



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Seventh Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamic Model

Robot Control



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Review of useful notions

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

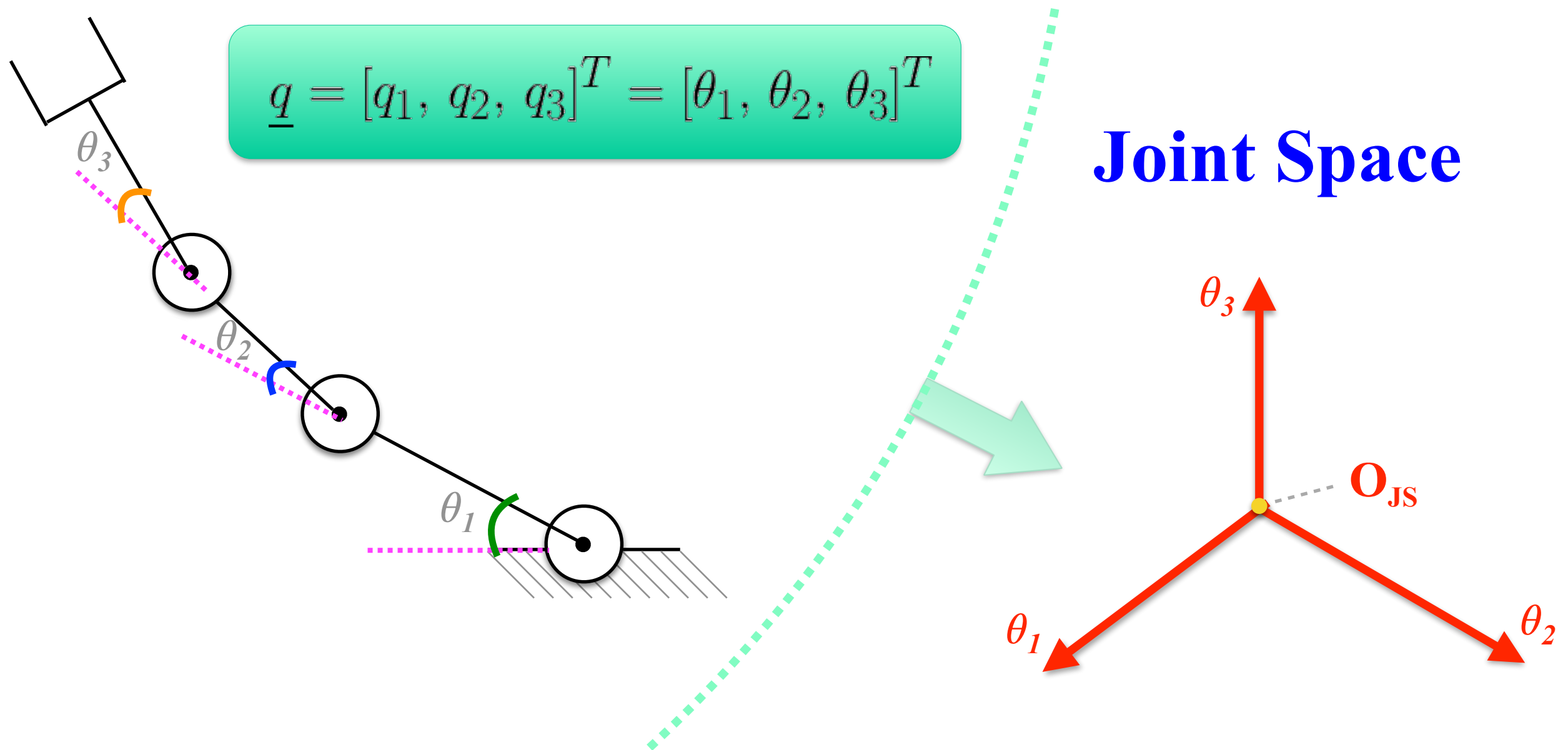
$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

$$B(\underline{q}) \ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} + F_v \dot{\underline{q}} + F_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - J^T(\underline{q}) \underline{h}_e$$

A Useful Definition: Joint Space

The space in which the $n \times 1$ vector of joint variables is defined is called “Joint Space” or “Configuration Space”



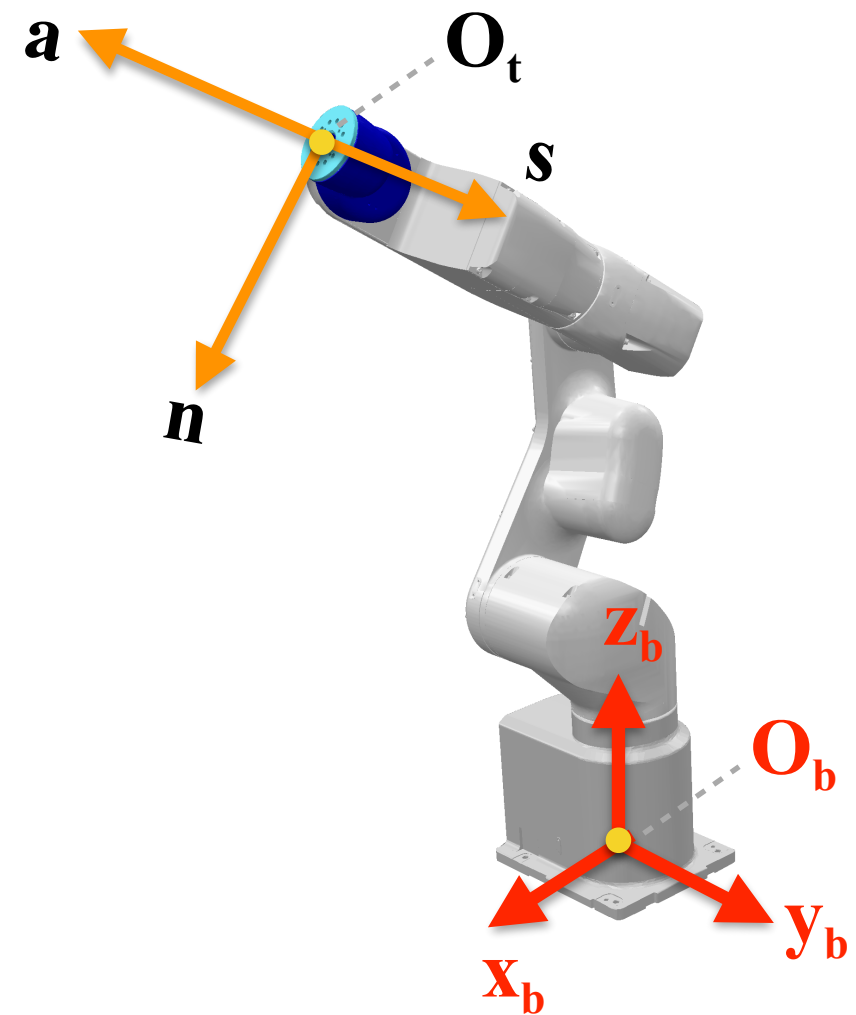
In a previous lecture..

A Useful Definition: Operational Space

Pose vector $\underline{x}(\underline{q}) = \begin{bmatrix} \underline{O}_t^b \\ \varphi \\ \vartheta \\ \psi \end{bmatrix}$

$m \times 1$

Operational Space



Operational Space Dimension : m



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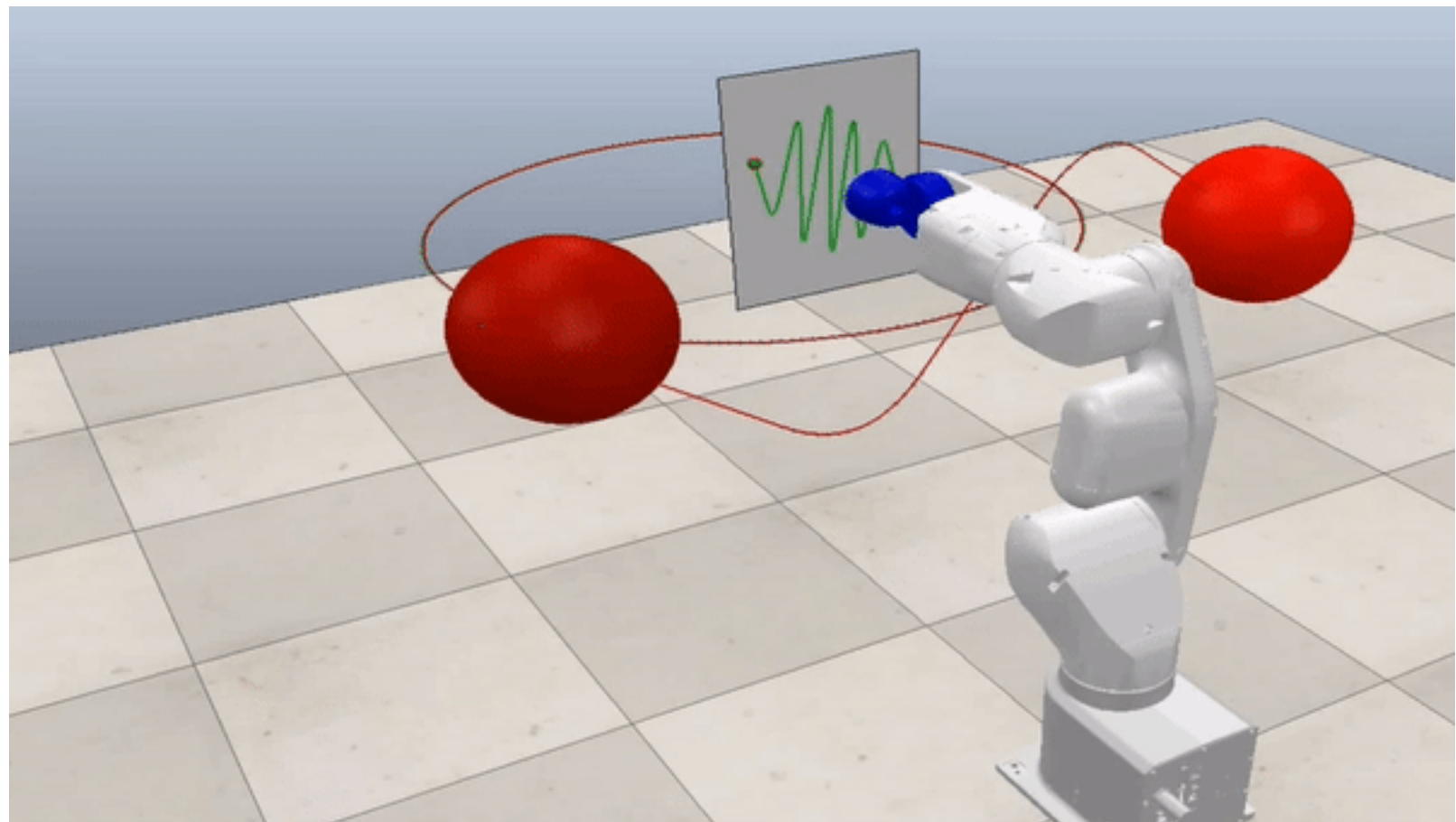
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Control of robot manipulators: introduction

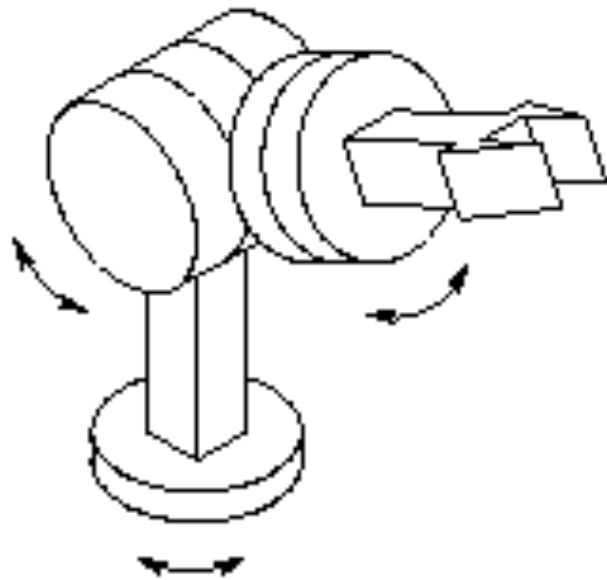
Control of Robot Manipulators

Control Problem: Determine the *generalized forces* (forces and torques) which the *actuators* need to apply to the joints so that the desired tasks are performed complying with the *transient and steady-state requirements*

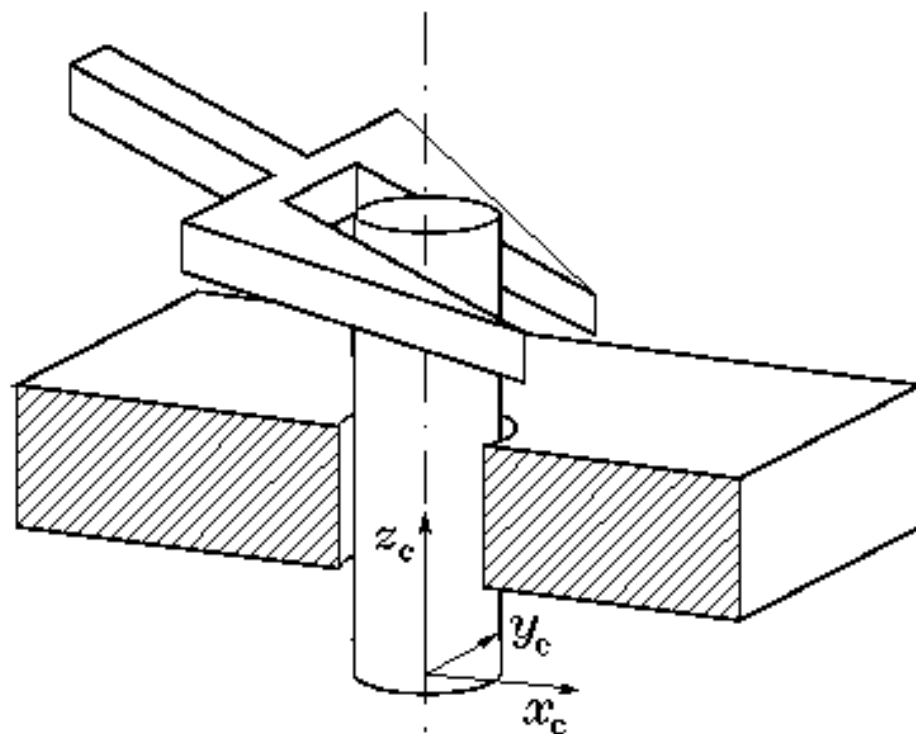


Control of Robot Manipulators

The types of tasks can require...



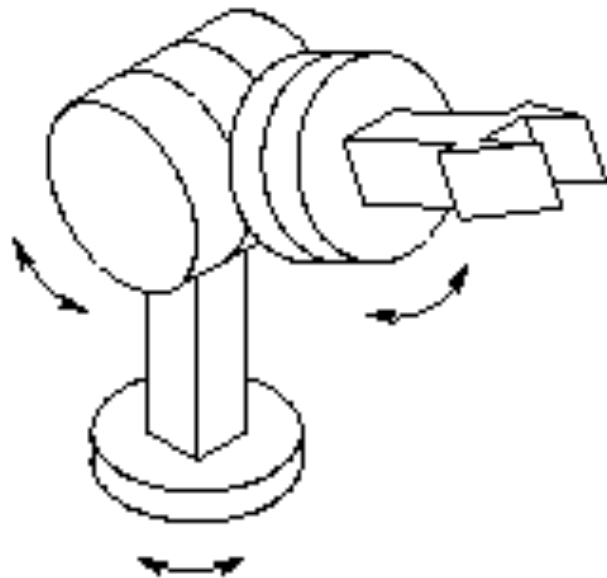
A motion of the robot in the working space without interaction with the environment:
MOTION CONTROL



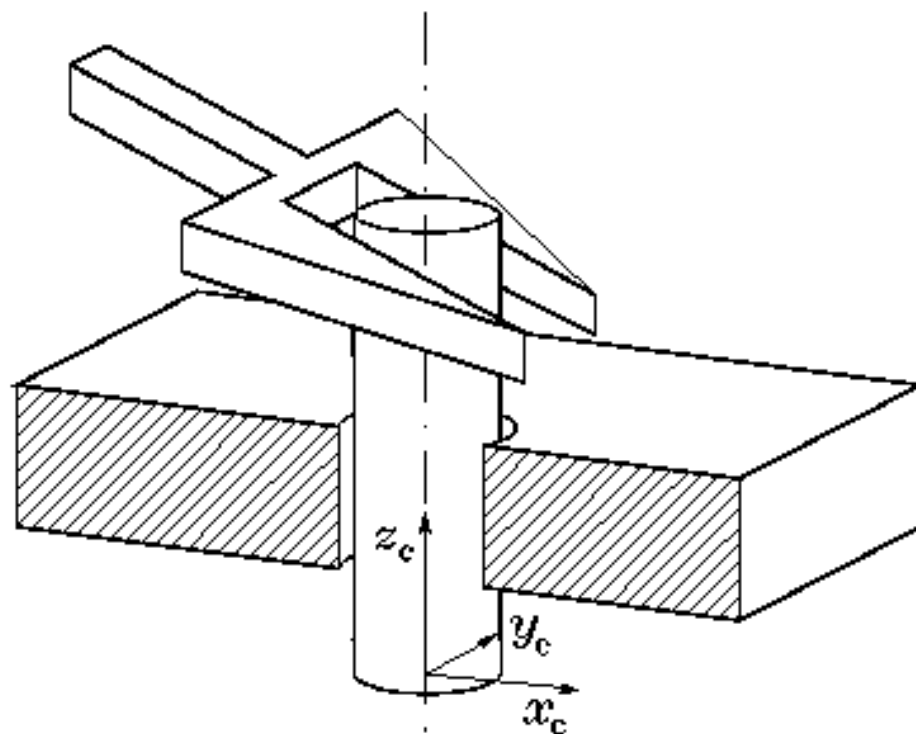
A motion of the robot in the working space with interaction with the environment:
INTERACTION CONTROL

Control of Robot Manipulators

The types of tasks can require...



A motion of the robot in the working space without interaction with the environment:
MOTION CONTROL



A motion of the robot in the working space with interaction with the environment:
INTERACTION CONTROL

Motion Control of Robot Manipulators

Two types of control schemes:

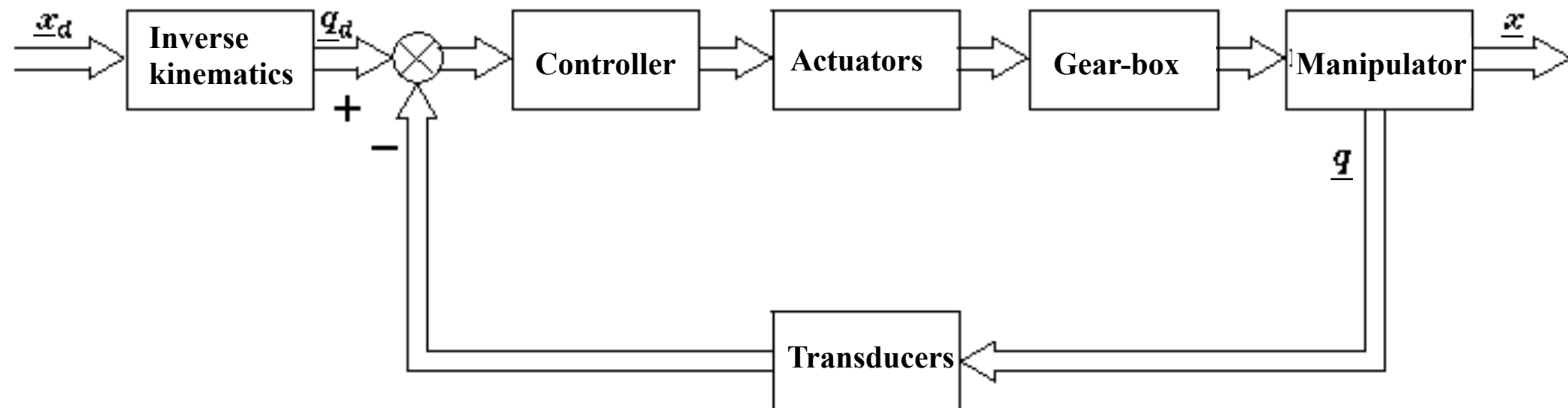
- Control in the Joint Space
- Control in the Operational Space

Two types of control approaches:

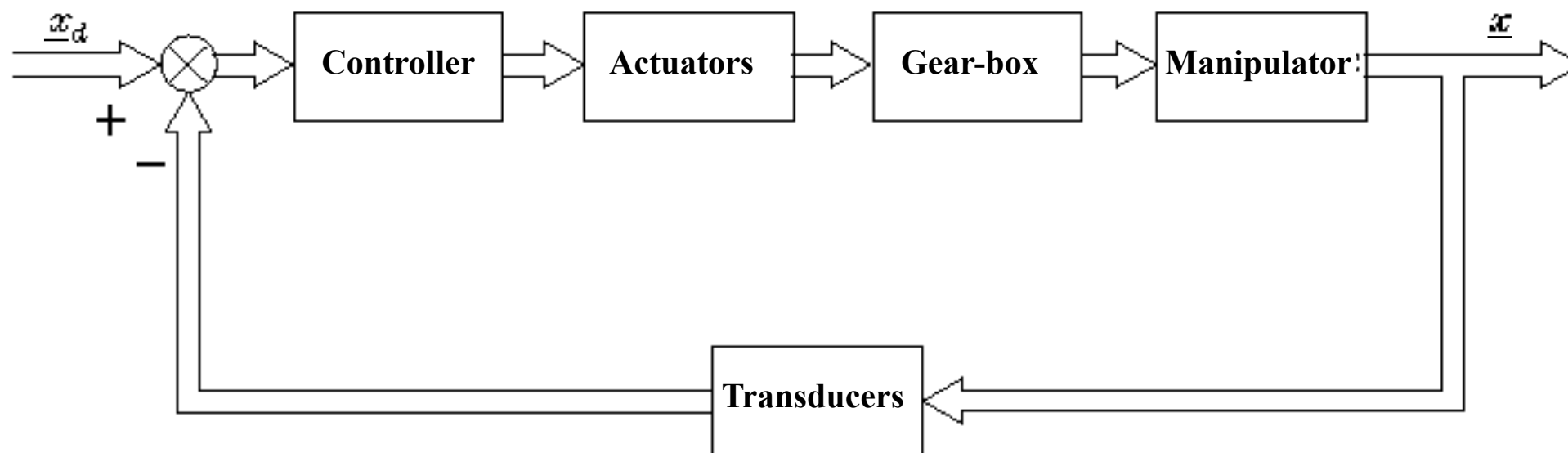
- decentralized
(each joint is controlled independently)
- centralized (the robot is regarded as a MIMO system)

Motion Control of Robot Manipulators

Control in the joint space

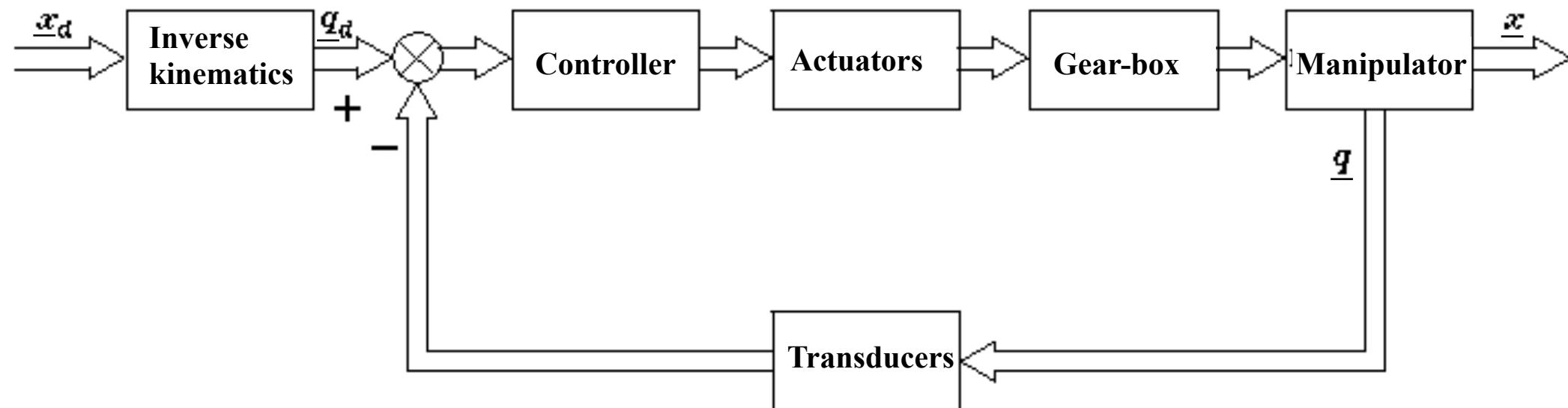


Control in the operational space

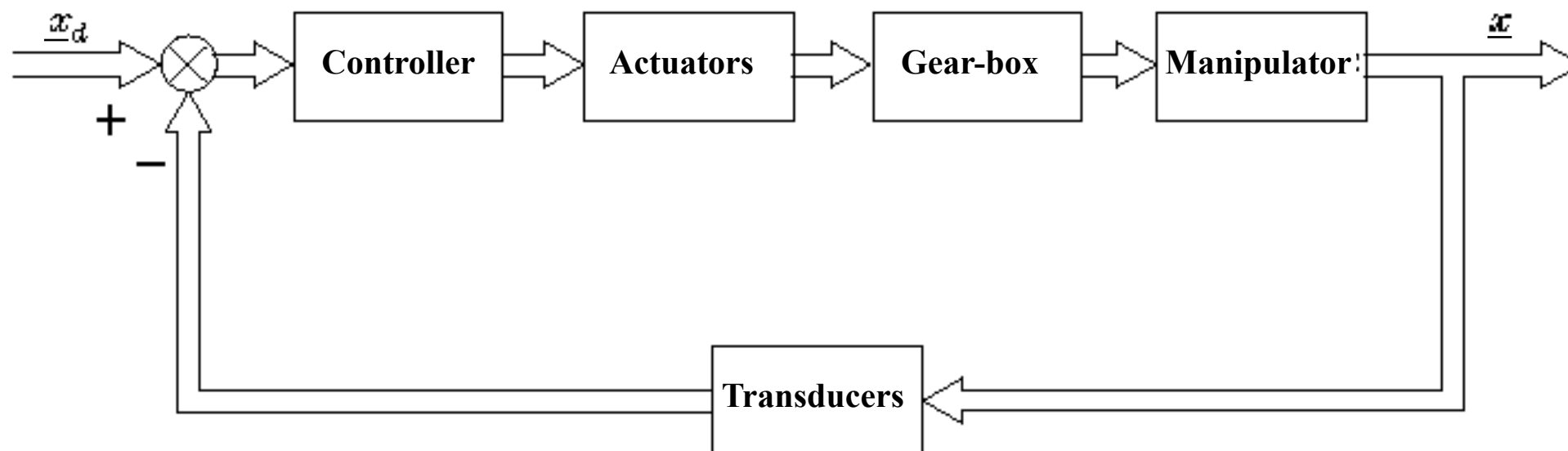


Motion Control of Robot Manipulators

Control in the joint space

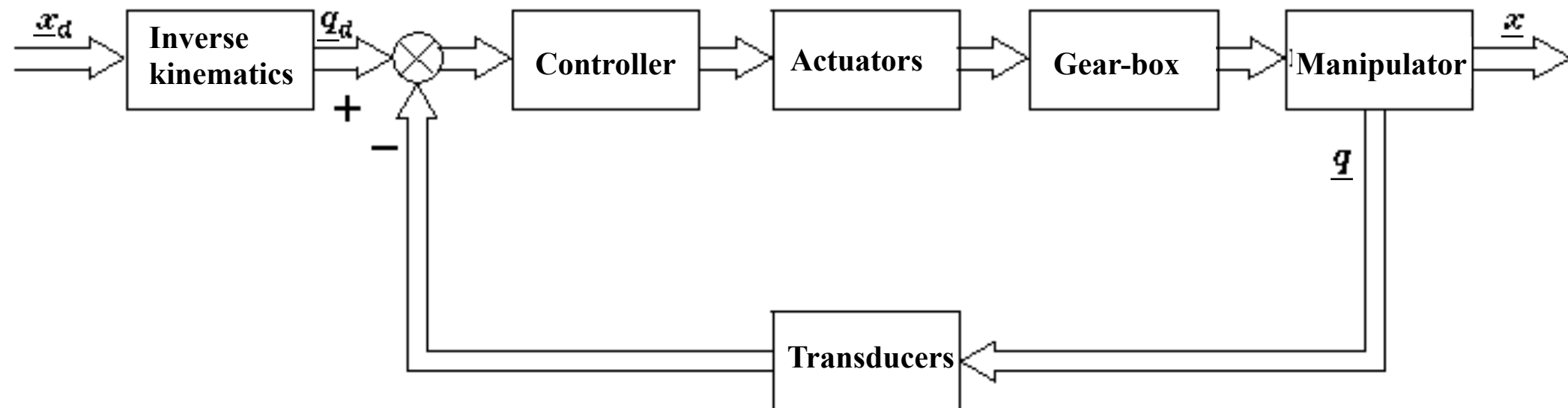


Control in the operational space

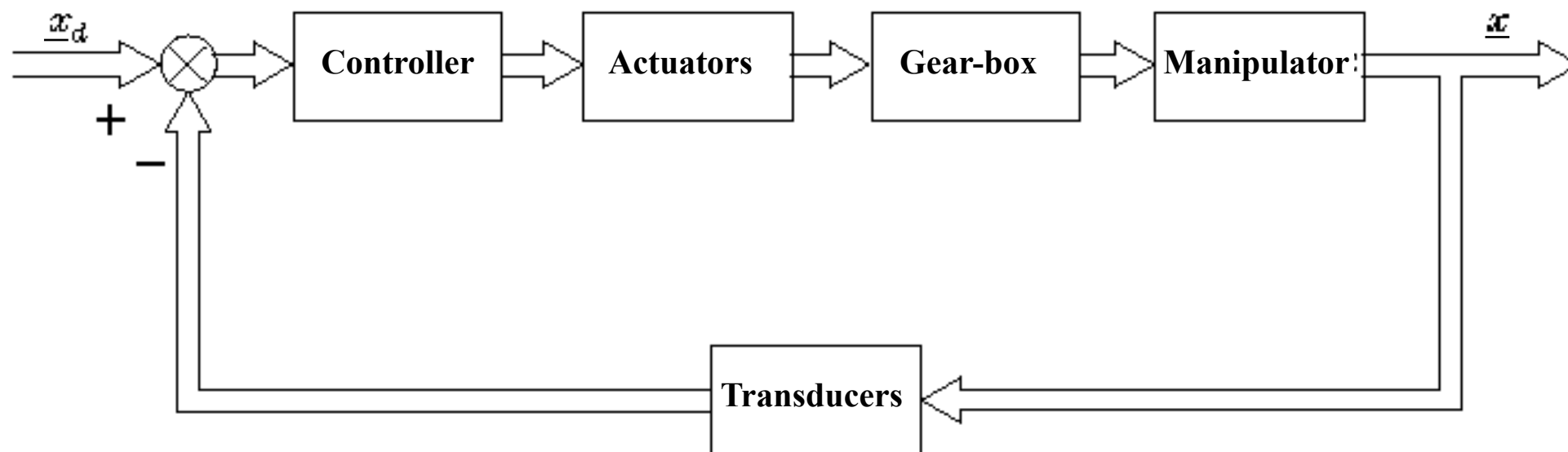


Motion Control of Robot Manipulators

Control in the joint space



Control in the operational space





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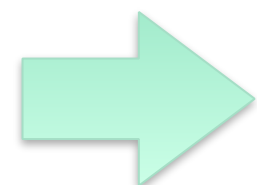
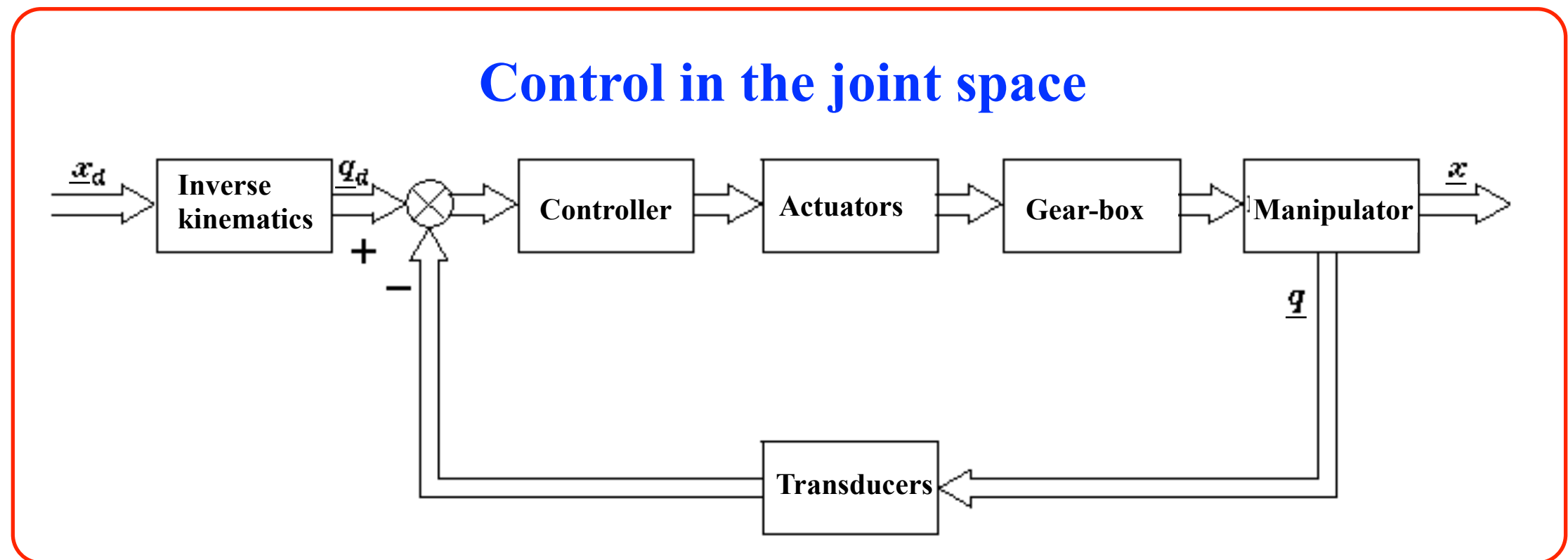
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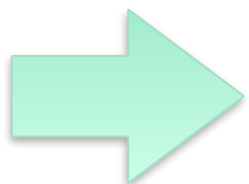
Motion Control: Control in the Joint Space

Motion Control: Control in the Joint Space



decentralized control (independent control of the joints)

- *decentralized control with decentralized feedforward compensation*
- *decentralized control with centralized computed torque based feedforward compensation*



centralized control of the joints

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v \dot{\underline{q}} + F_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - \underline{J}^T(\underline{q})\underline{h}_e$$

Robot Dynamic Model

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Robot Dynamic Model

Lagrange Equations

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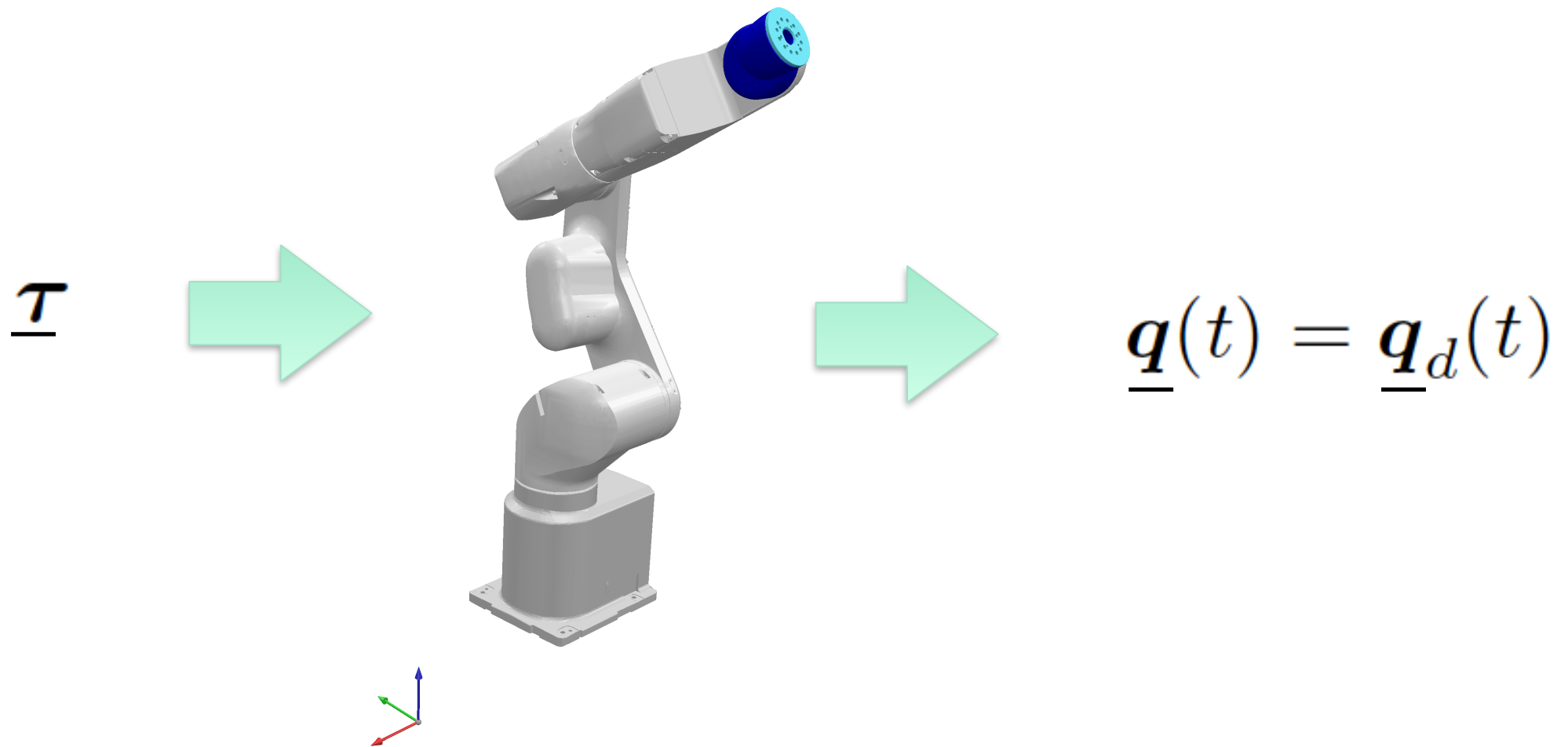
Robot Dynamic Model

$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

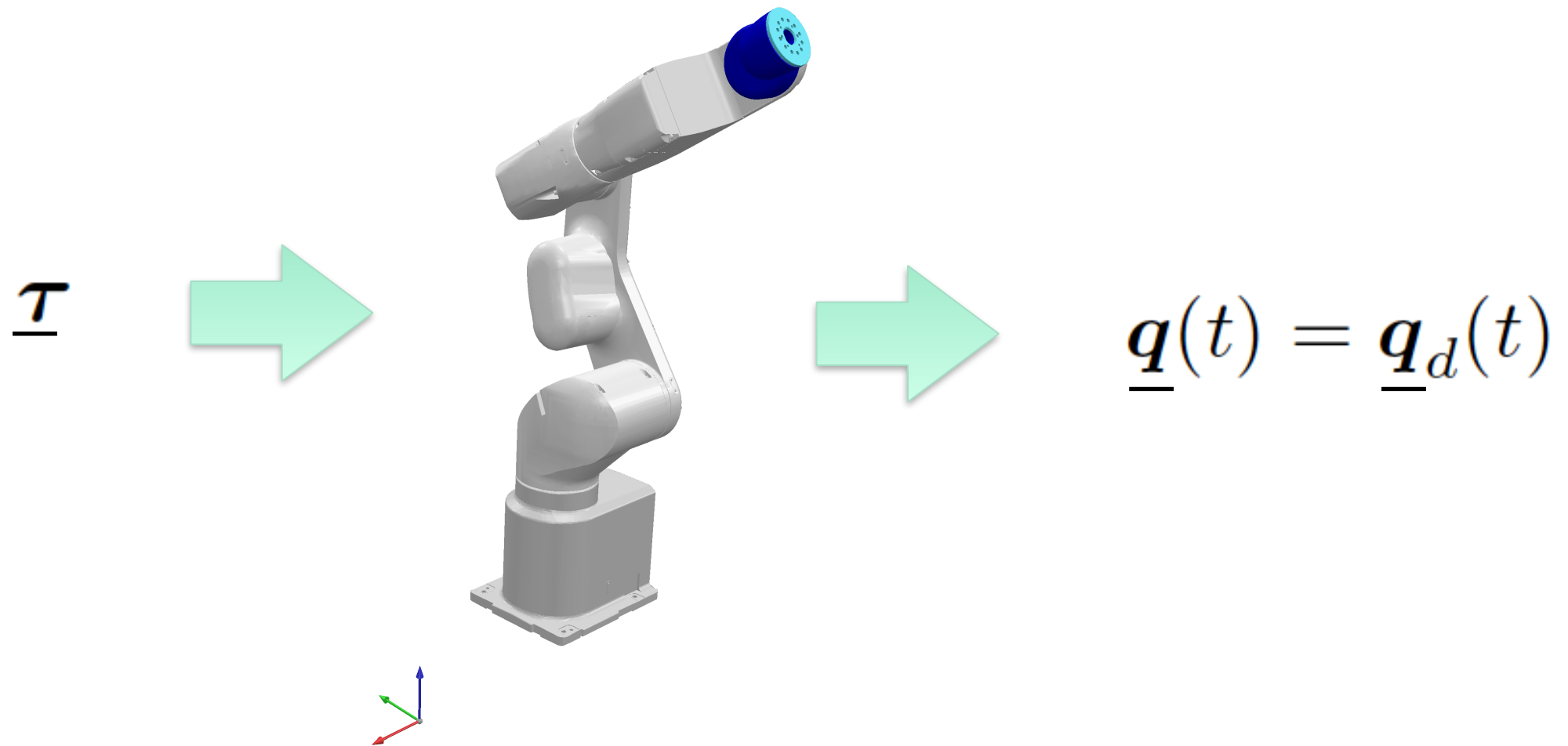
$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

$$\mathbf{B}(\underline{q}) \ddot{\underline{q}} + \mathbf{C}(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} + \mathbf{F}_v \dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{\tau}$$

Motion Control: Control in the Joint Space

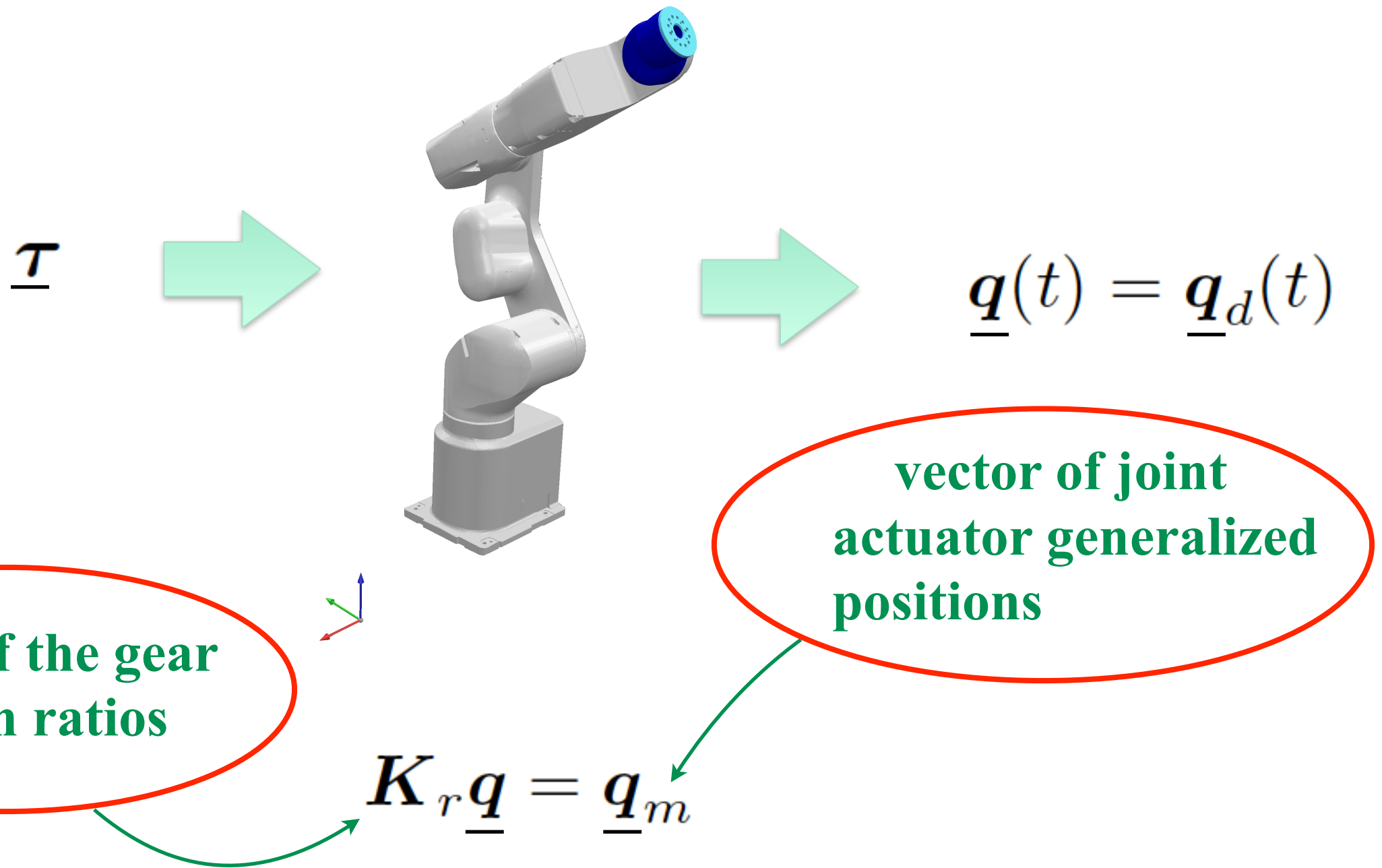


Motion Control: Control in the Joint Space

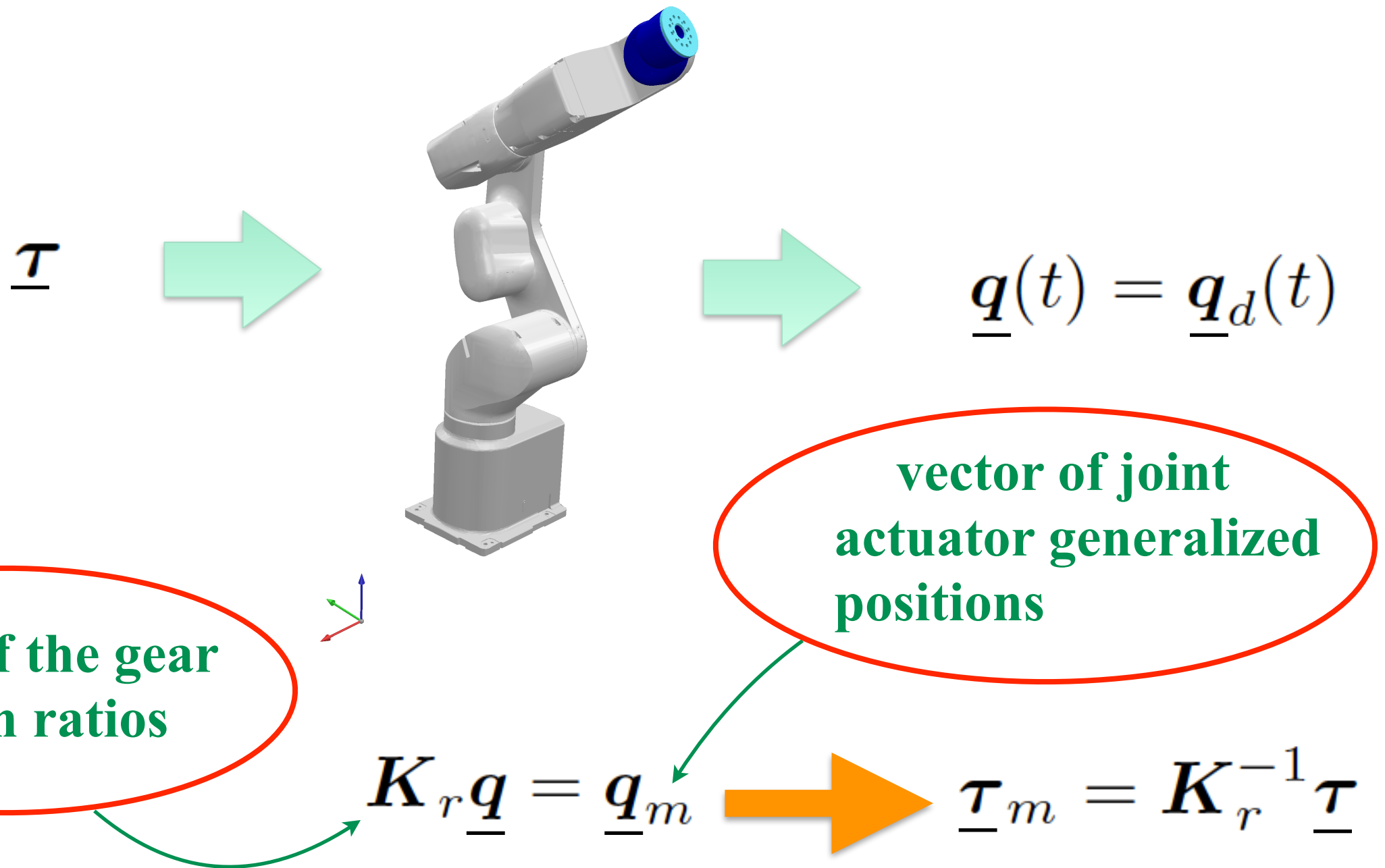


$$K_r \underline{q} = \underline{q}_m$$

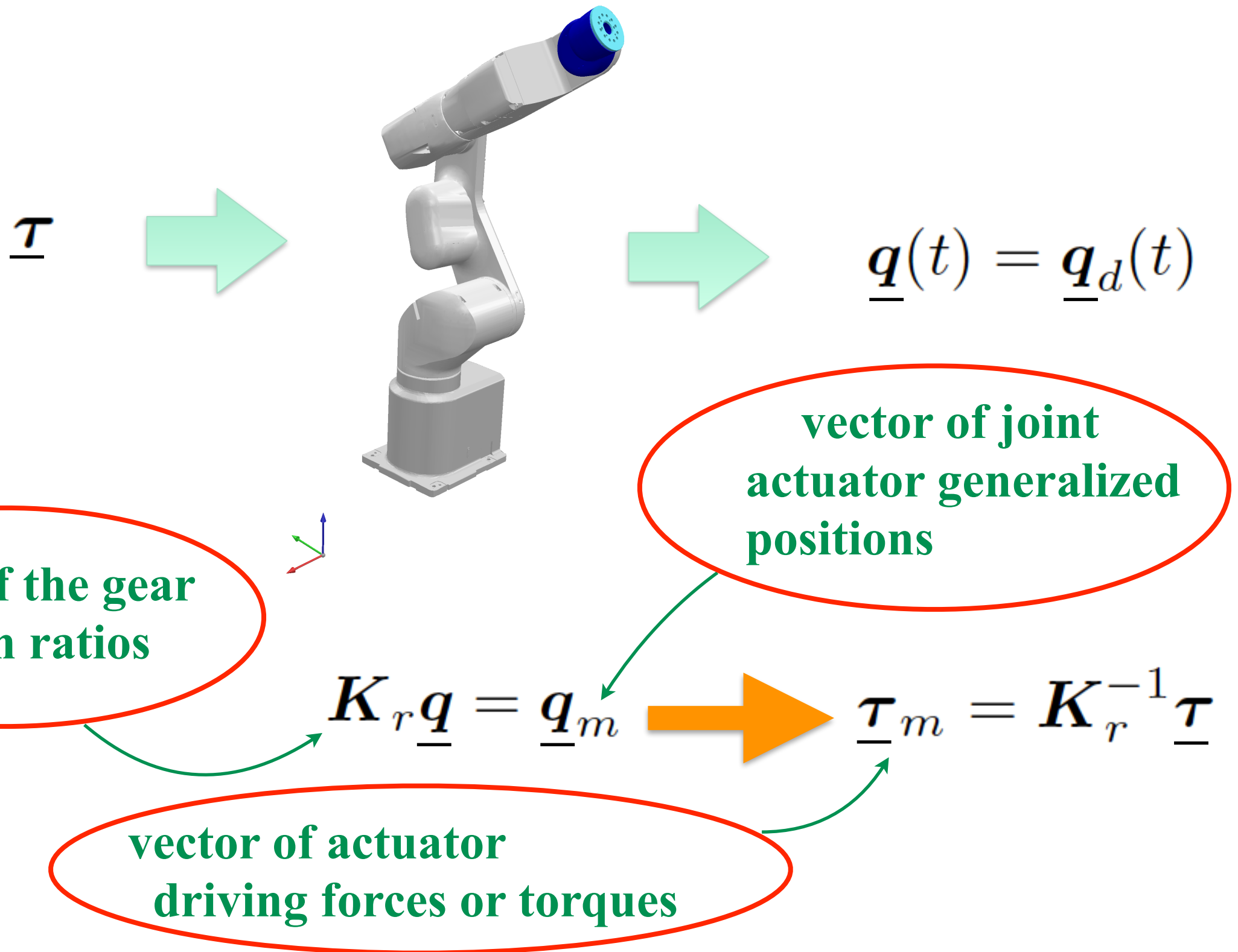
Motion Control: Control in the Joint Space



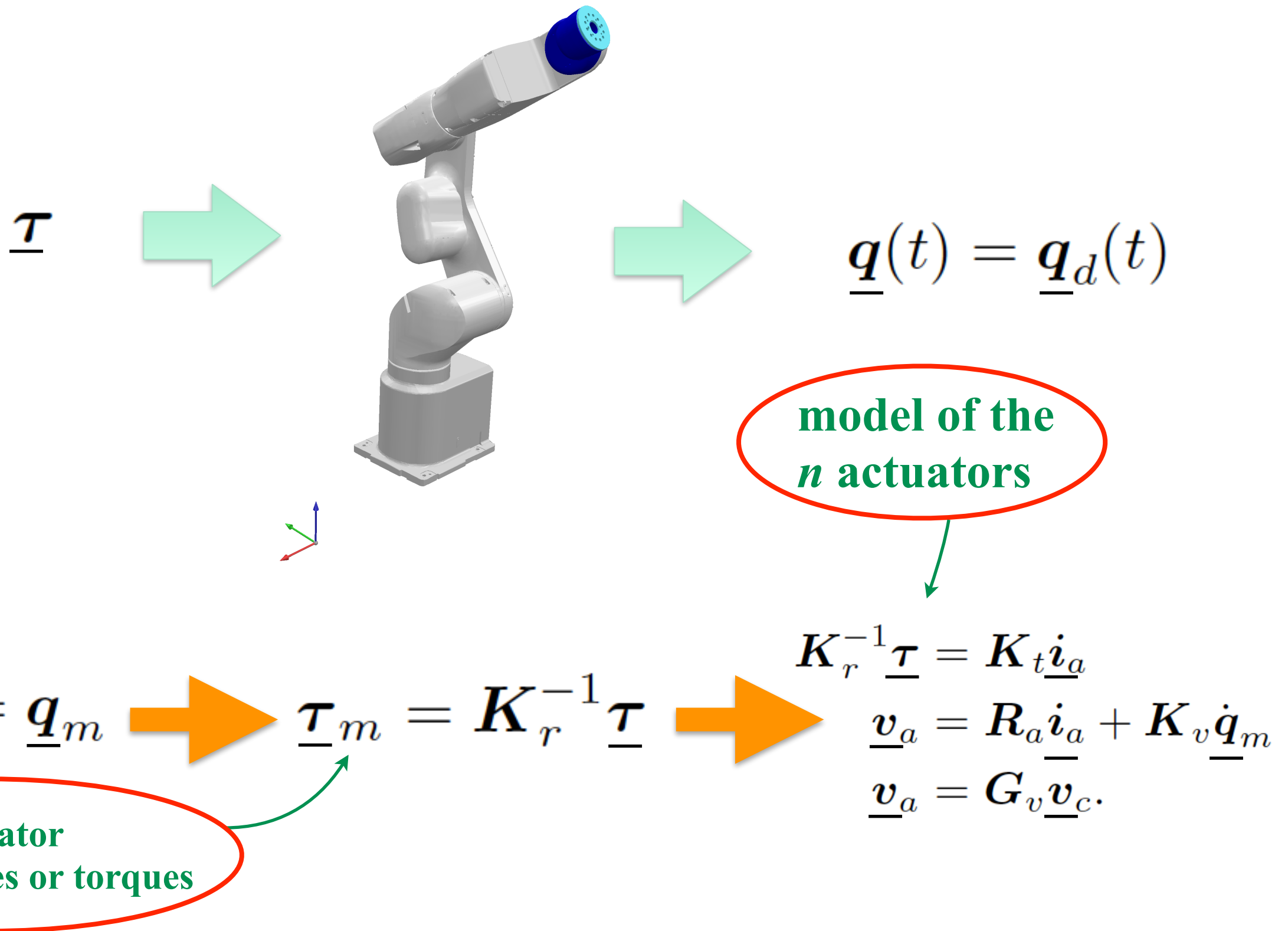
Motion Control: Control in the Joint Space



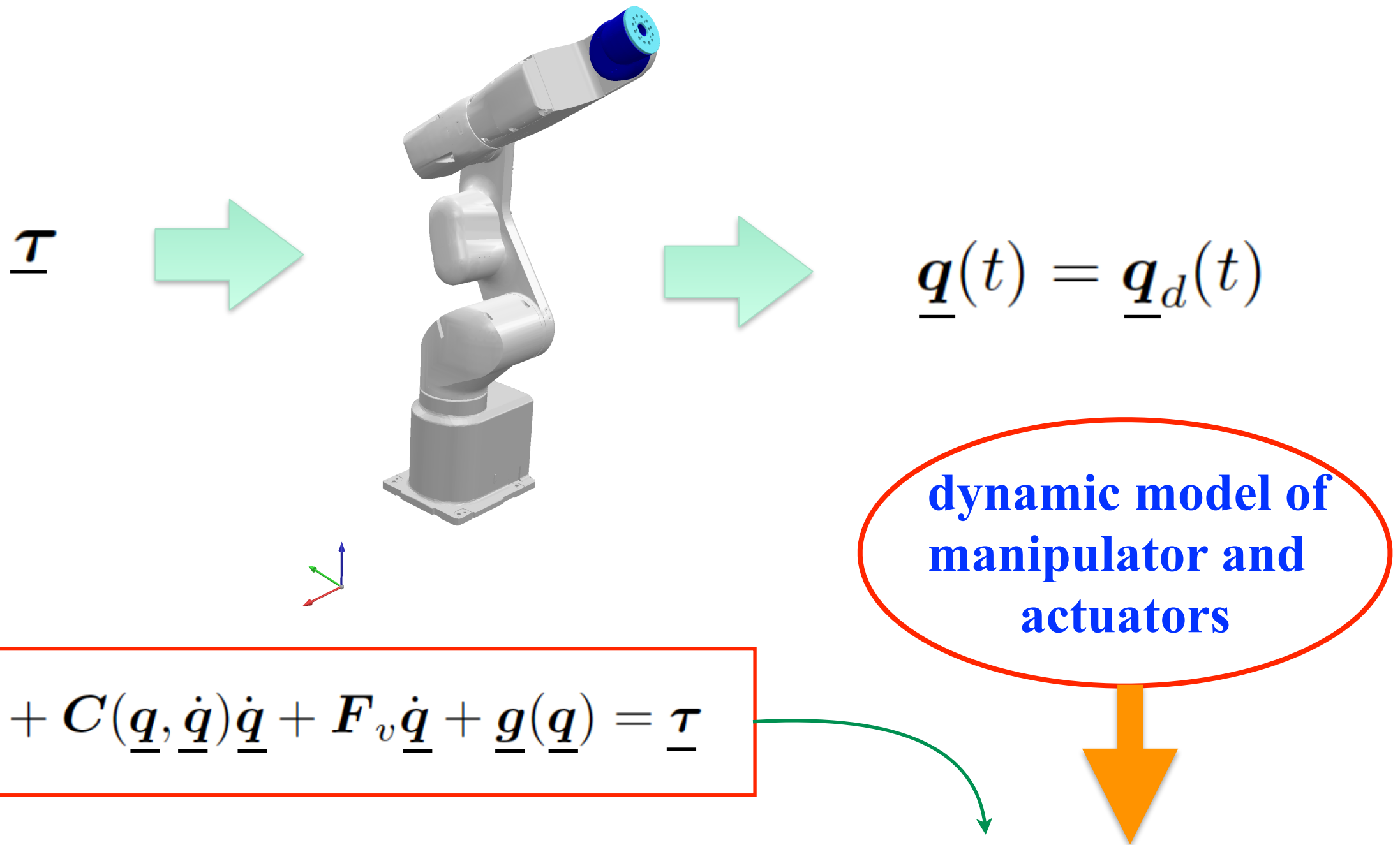
Motion Control: Control in the Joint Space



Motion Control: Control in the Joint Space



Motion Control: Control in the Joint Space

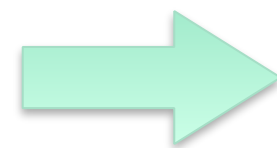


$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{\tau}$$

$$K_r^{-1}B(\underline{q})K_r^{-1}\ddot{\underline{q}}_m + K_r^{-1}C(\underline{q}, \dot{\underline{q}})K_r^{-1}\dot{\underline{q}}_m + K_r^{-1}F_vK_r^{-1}\dot{\underline{q}}_m + K_r^{-1}\underline{g}(\underline{q}) = \underline{\tau}_m$$

Motion Control: Control in the Joint Space

$$\mathbf{K}_r^{-1} \mathbf{B}(\underline{\mathbf{q}}) \mathbf{K}_r^{-1} \ddot{\underline{\mathbf{q}}}_m + \mathbf{K}_r^{-1} \mathbf{C}(\underline{\mathbf{q}}, \dot{\underline{\mathbf{q}}}) \mathbf{K}_r^{-1} \dot{\underline{\mathbf{q}}}_m + \mathbf{K}_r^{-1} \mathbf{F}_v \mathbf{K}_r^{-1} \dot{\underline{\mathbf{q}}}_m + \mathbf{K}_r^{-1} \mathbf{g}(\underline{\mathbf{q}}) = \underline{\boldsymbol{\tau}}_m$$



$$\mathbf{B}(\underline{\mathbf{q}}) = \bar{\mathbf{B}} + \Delta \mathbf{B}(\underline{\mathbf{q}})$$

resulting average
inertia at each joint



Motion Control: Control in the Joint Space

$$K_r^{-1} B(\underline{q}) K_r^{-1} \ddot{\underline{q}}_m + K_r^{-1} C(\underline{q}, \dot{\underline{q}}) K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} F_v K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} \underline{g}(\underline{q}) = \underline{\tau}_m$$



$$B(\underline{q}) = \bar{B} + \Delta B(\underline{q})$$

resulting average
inertia at each joint

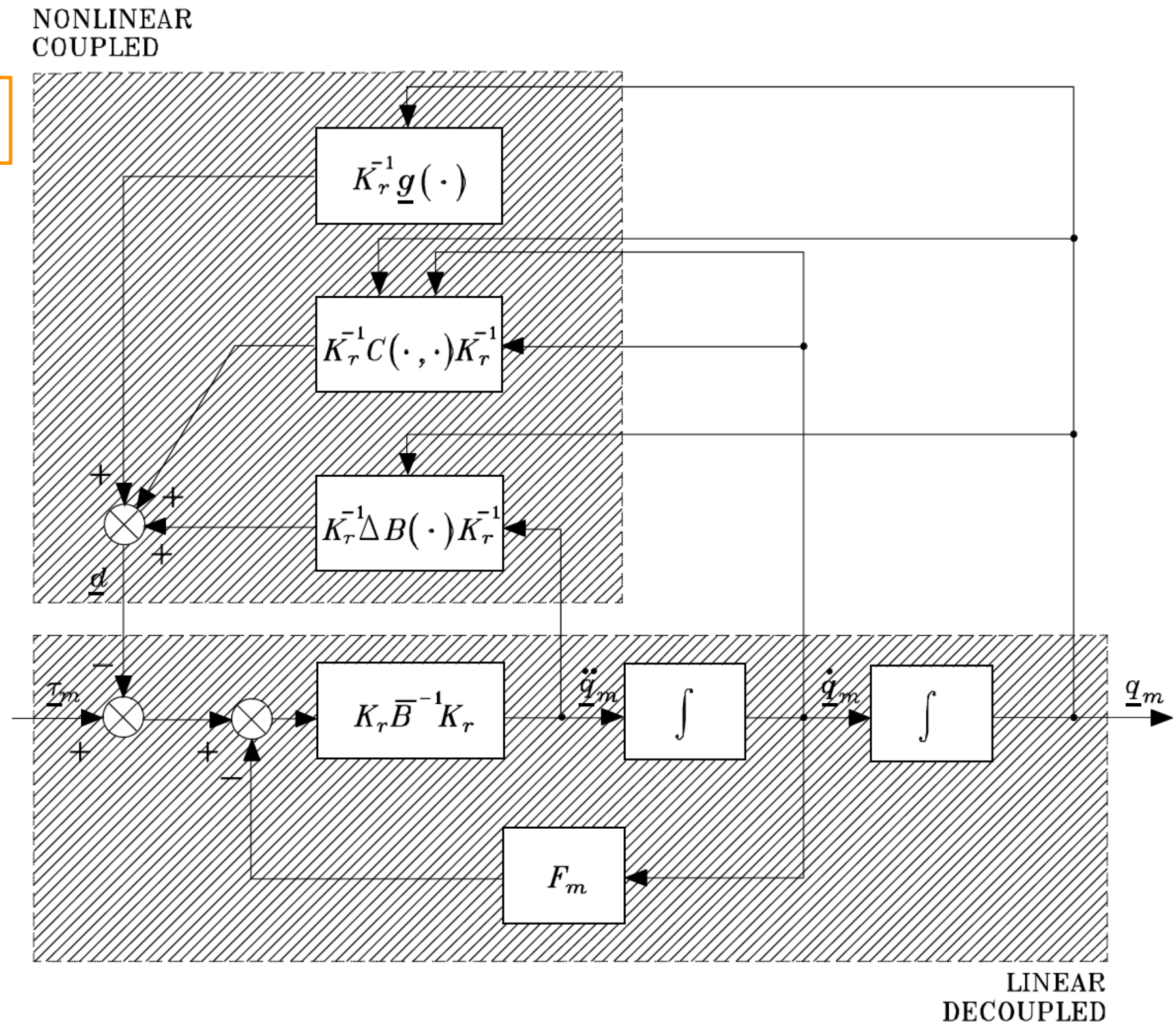
$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$F_m = K_r^{-1} F_v K_r^{-1}$$

$$\underline{d} = K_r^{-1} \Delta B(\underline{q}) K_r^{-1} \ddot{\underline{q}}_m + K_r^{-1} C(\underline{q}, \dot{\underline{q}}) K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} \underline{g}(\underline{q})$$

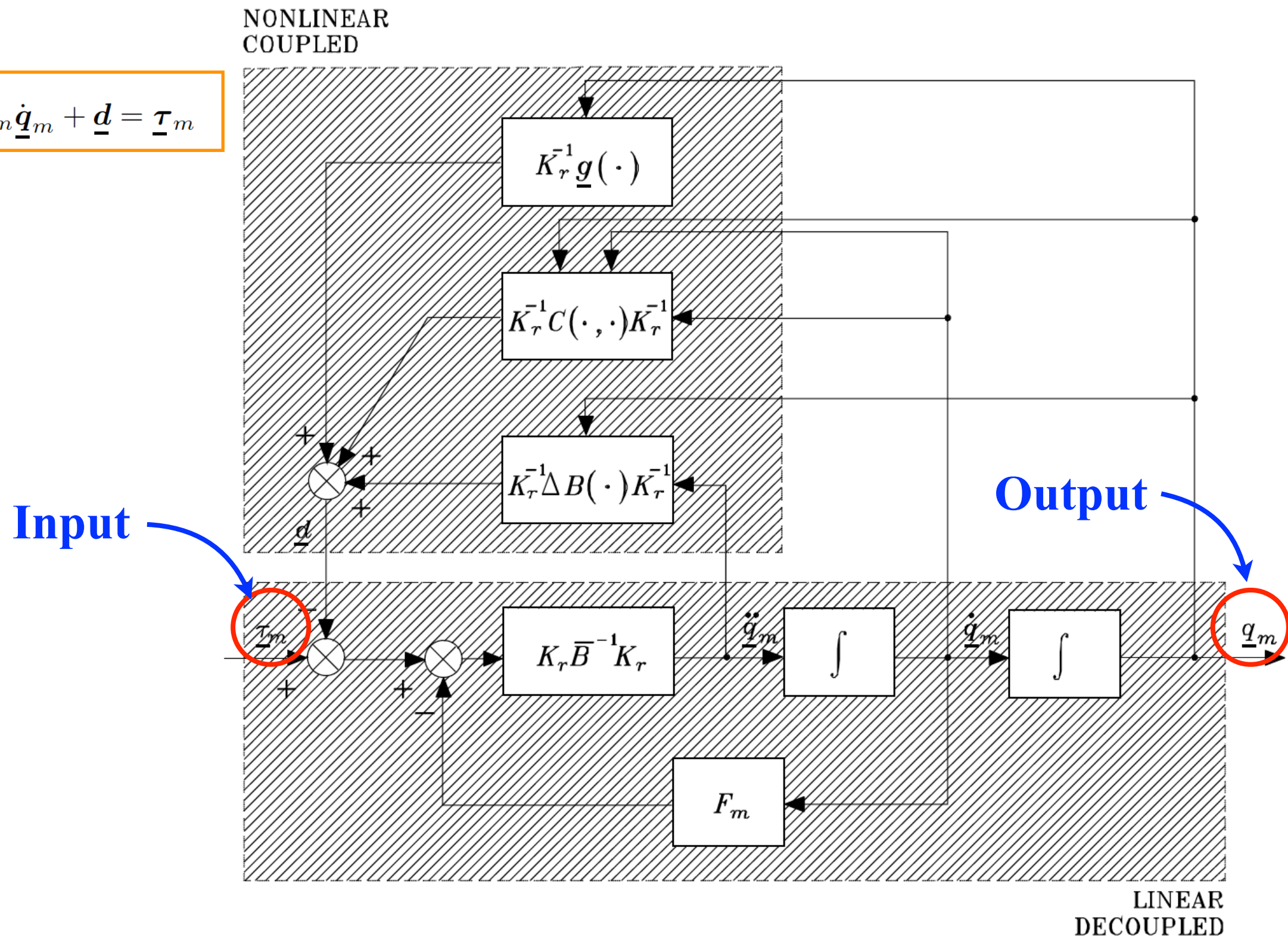
Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$



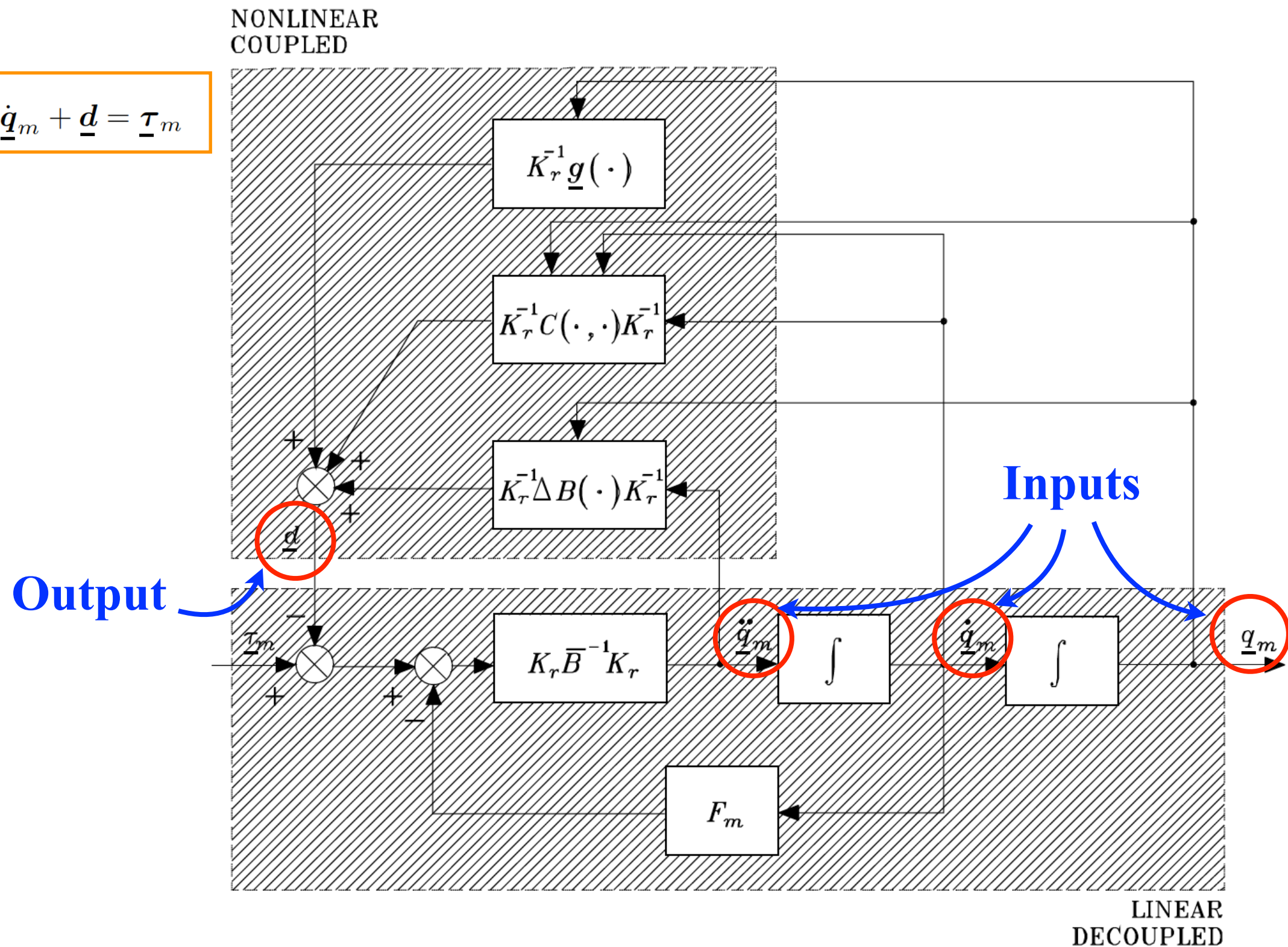
Motion Control: Control in the Joint Space

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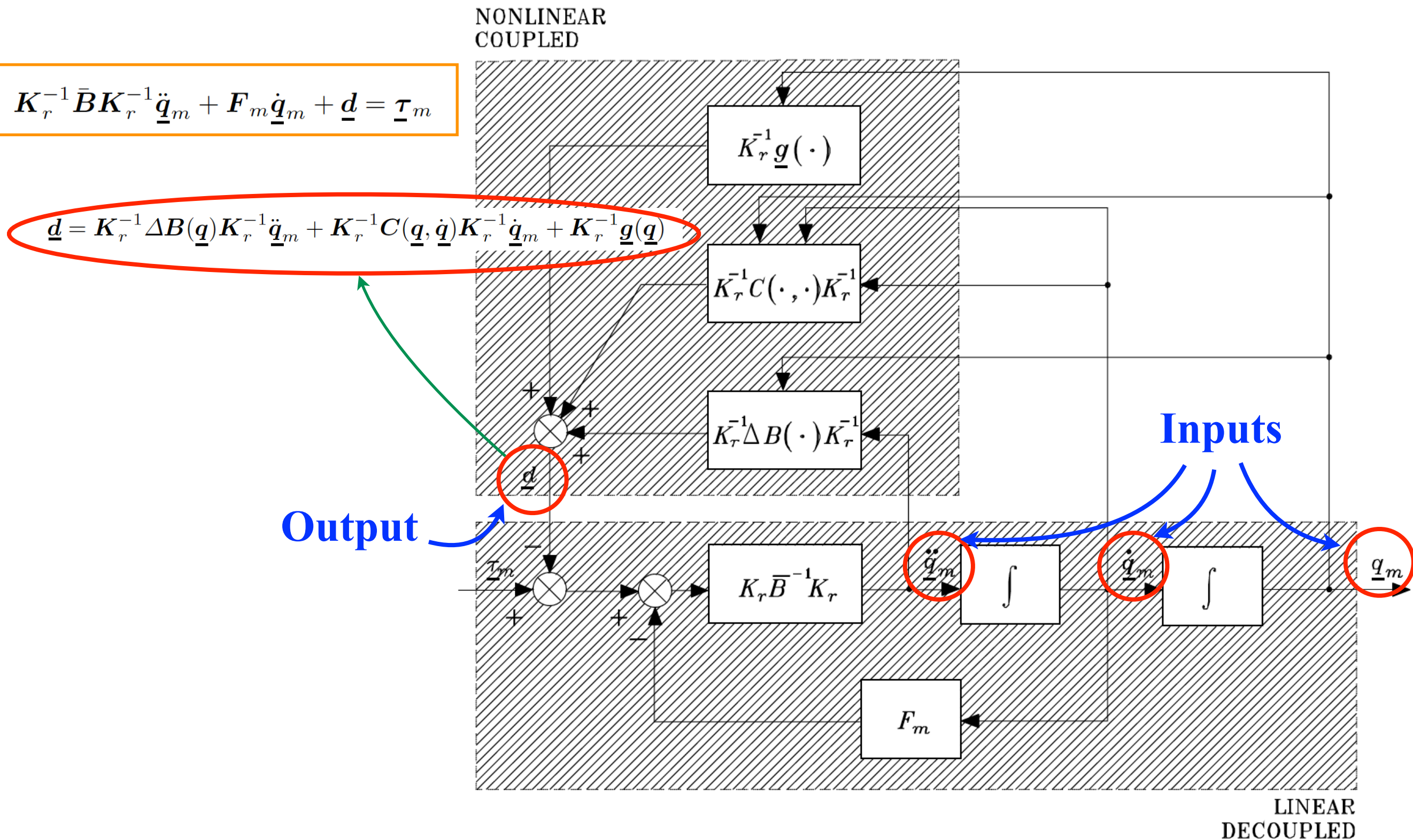


Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$



Motion Control: Control in the Joint Space



Motion Control: Control in the Joint Space

A first way to control the robot

- High transmission ratios
- No request of high performance in terms of velocity and acceleration

d (the terms accounting for nonlinearities and couplings): is regarded as an actual disturbance acting on the single joints

DECENTRALIZED CONTROL SCHEME

Each joint is controlled independently



Motion Control: Control in the Joint Space

Another way to control the robot

- No transmission (no gear box)
- Request of high performance in terms of velocity and acceleration

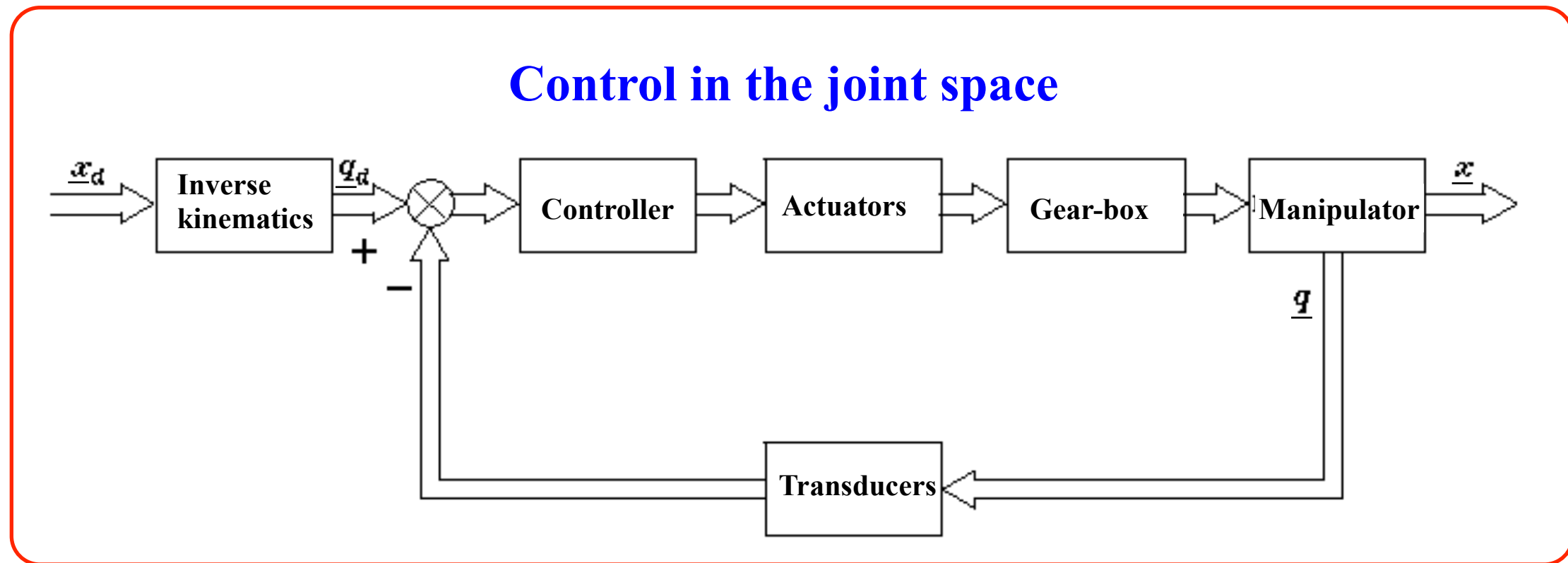
$\underline{d}$ (the terms accounting for nonlinearities and couplings): cannot be regarded as a disturbance acting on the joints. It needs to be explicitly considered in the design.



CENTRALIZED CONTROL SCHEME

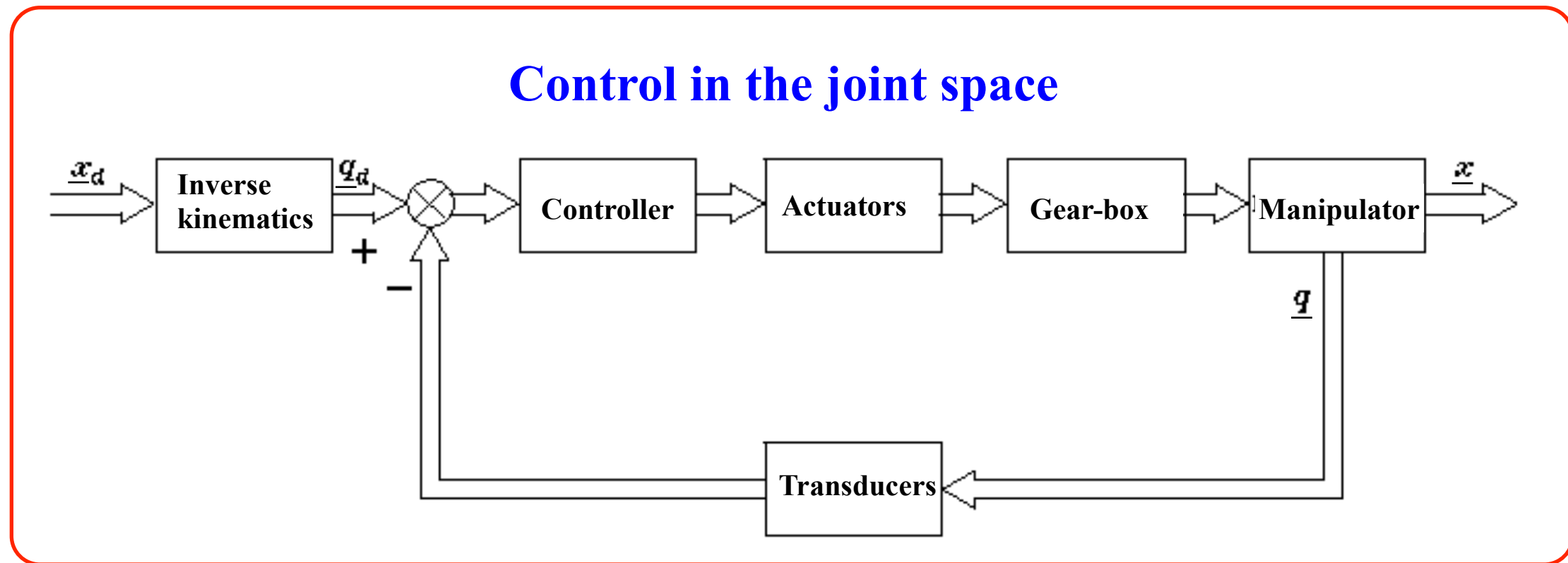
The robot is controlled as a MIMO system

Motion Control: Control in the Joint Space



- ➡ **decentralized control (independent control of the joints)**
- *decentralized control with decentralized feedforward compensation*
 - *decentralized control with centralized computed torque based feedforward compensation*
- ➡ **centralized control of the joints**

Motion Control: Control in the Joint Space

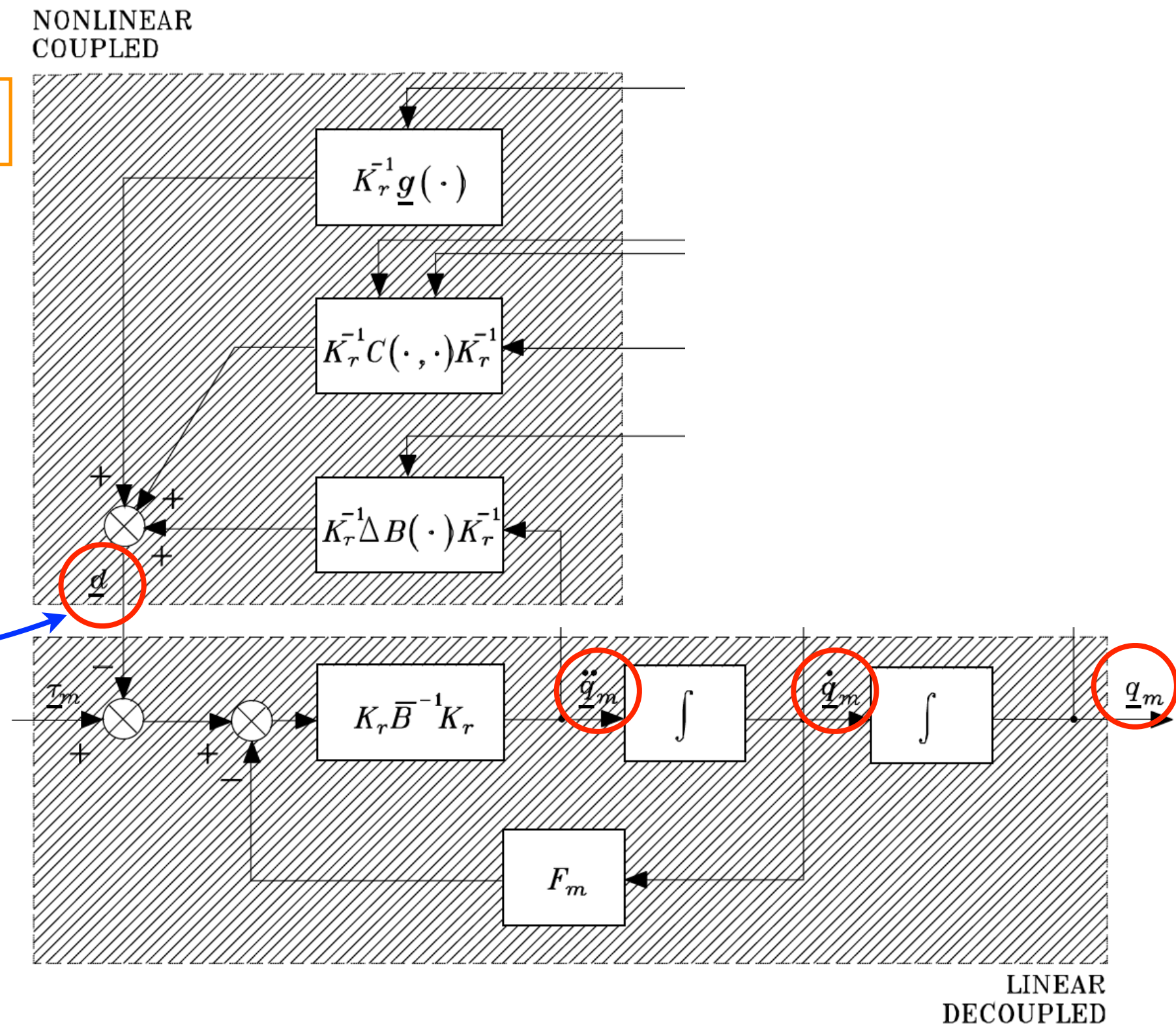


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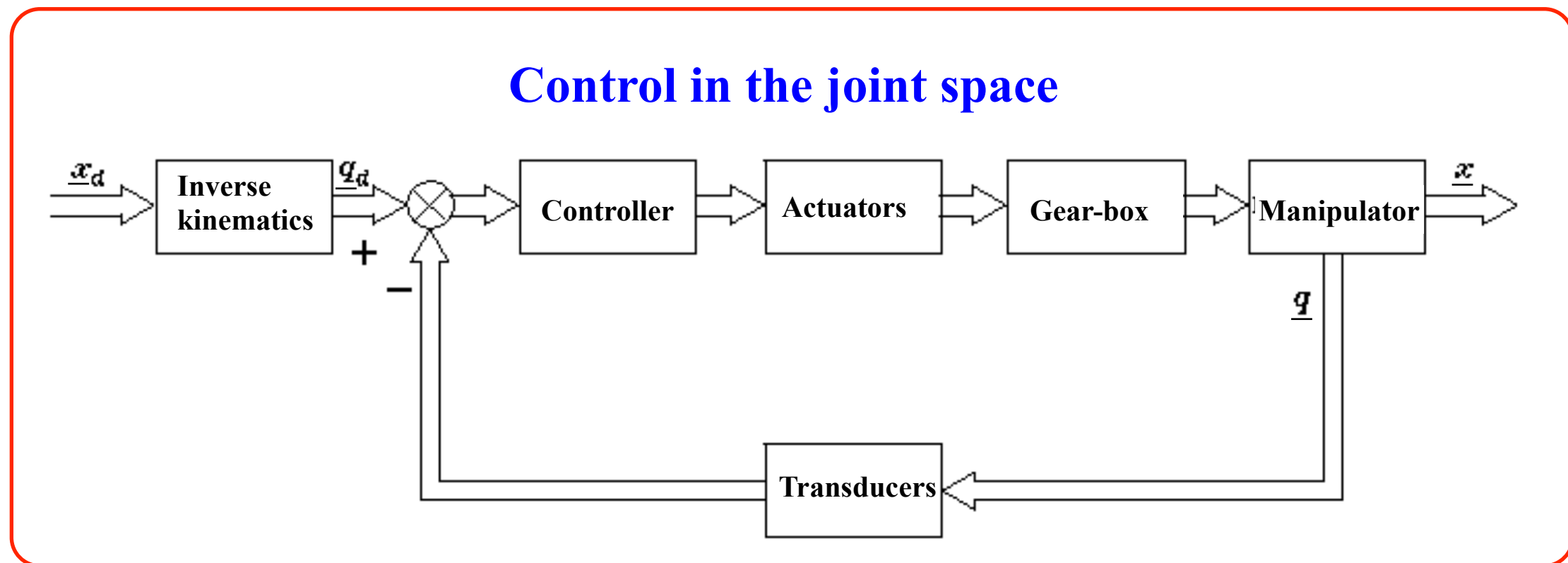
Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

It is a disturbance



Motion Control: Control in the Joint Space



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- ➡ **centralized control of the joints**

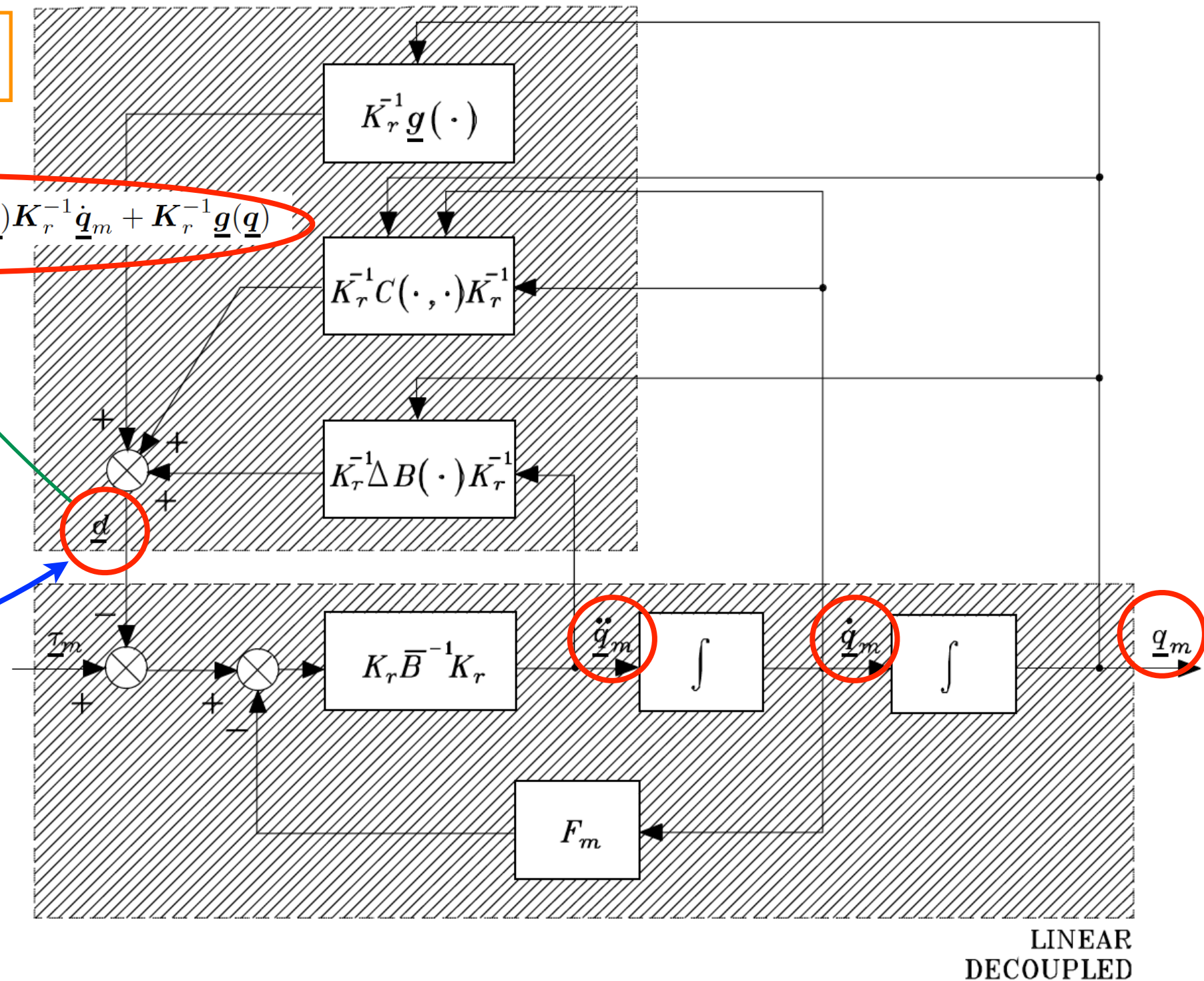
Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$\underline{d} = K_r^{-1} \Delta B(\underline{q}) K_r^{-1} \ddot{\underline{q}}_m + K_r^{-1} C(\underline{q}, \dot{\underline{q}}) K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} \underline{g}(\underline{q})$$

NONLINEAR
COUPLED

It is not a disturbance





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Motion Control: Decentralized Control in the Joint Space

Motion Control: Control in the Joint Space

A first way to control the robot

- High transmission ratios
- No request of high performance in terms of velocity and acceleration

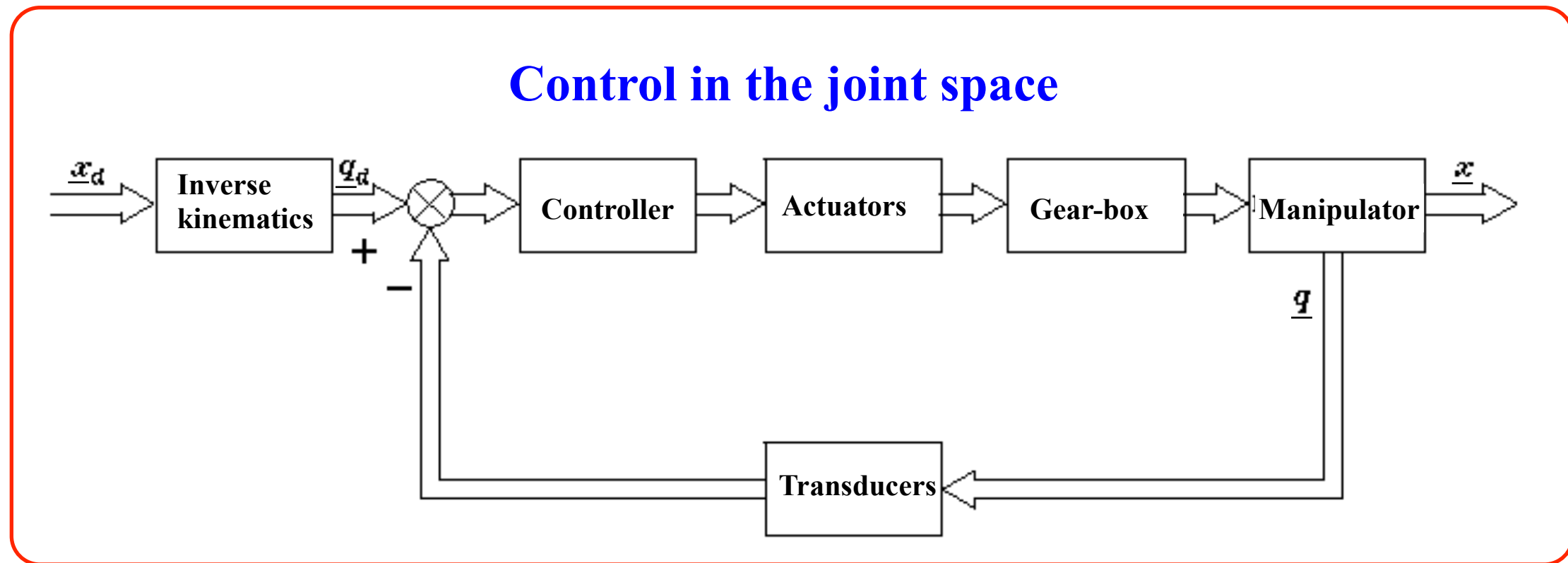
d (the terms accounting for nonlinearities and couplings): is regarded as an actual disturbance acting on the single joints

DECENTRALIZED CONTROL SCHEME

Each joint is controlled independently



Motion Control: Control in the Joint Space

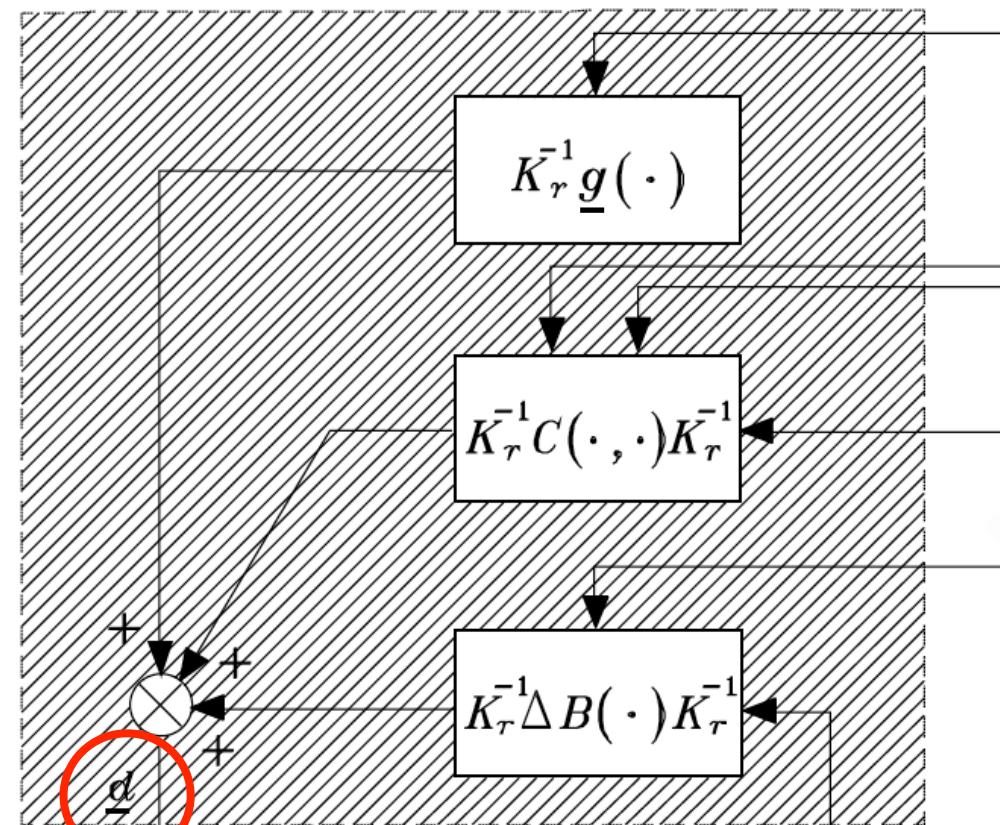


- ➔ **decentralized control (independent control of the joints)**
- *decentralized control with decentralized feedforward compensation*
 - *decentralized control with centralized computed torque based feedforward compensation*
- ➔ **centralized control of the joints**

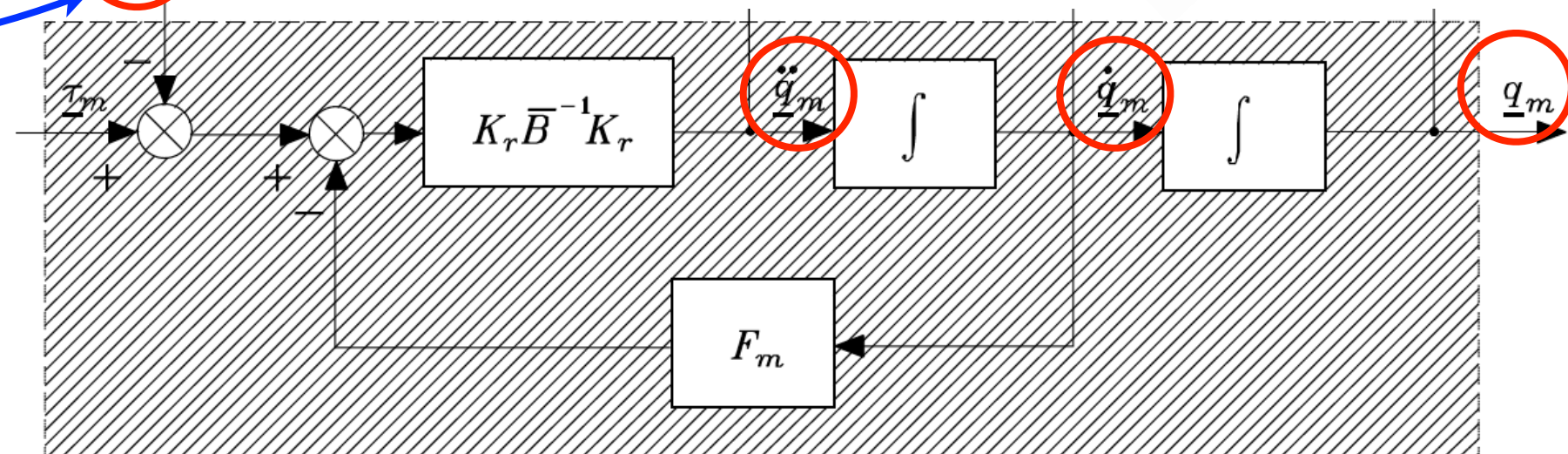
Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

NONLINEAR
COUPLED



It is a disturbance

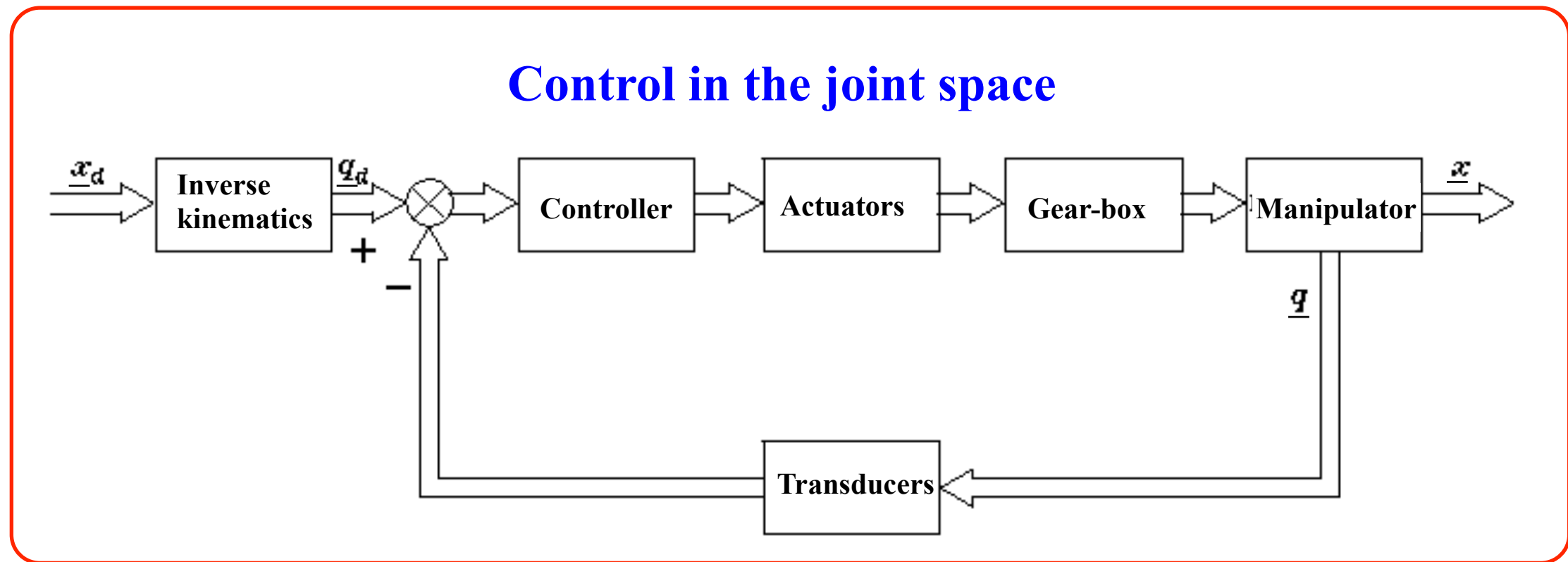


LINEAR
DECOUPLED



Motion Control: Control in the Joint Space

Decentralized Approach



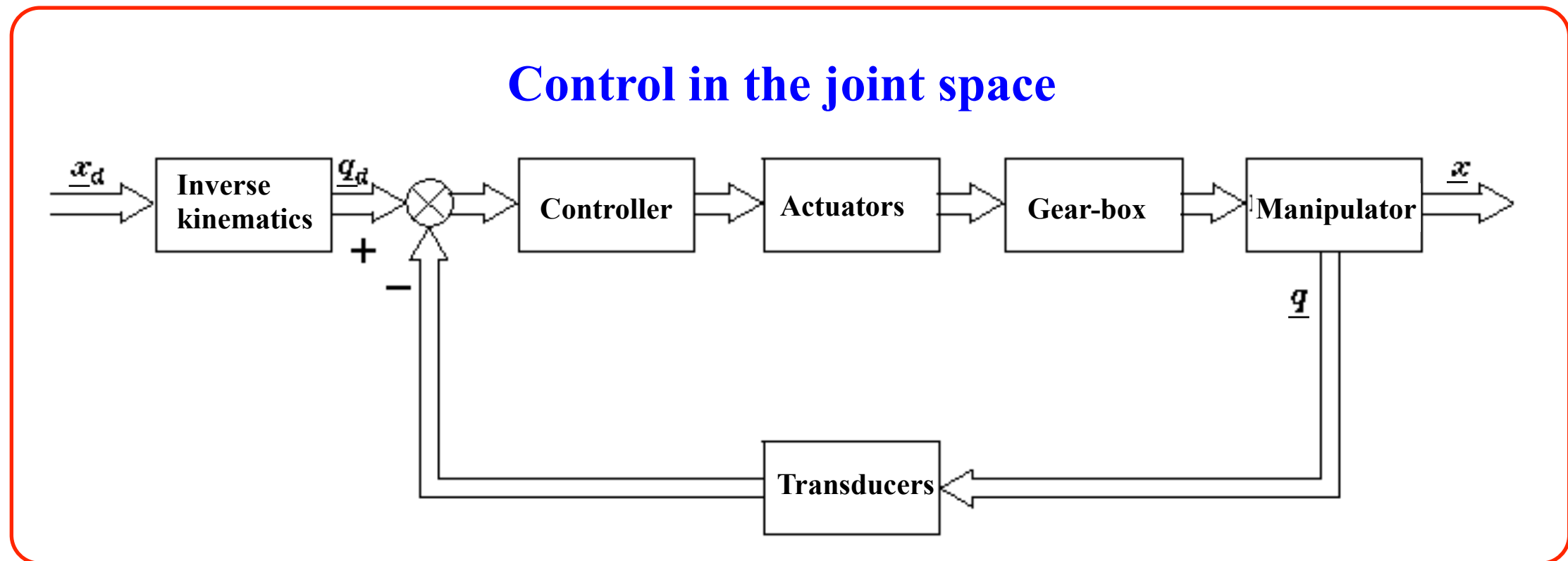
$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$F_m = K_r^{-1} F_v K_r^{-1}$$

generalized force disturbance

Motion Control: Control in the Joint Space

Decentralized Approach



diagonal matrices

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$F_m = K_r^{-1} F_v K_r^{-1}$$

generalized force disturbance

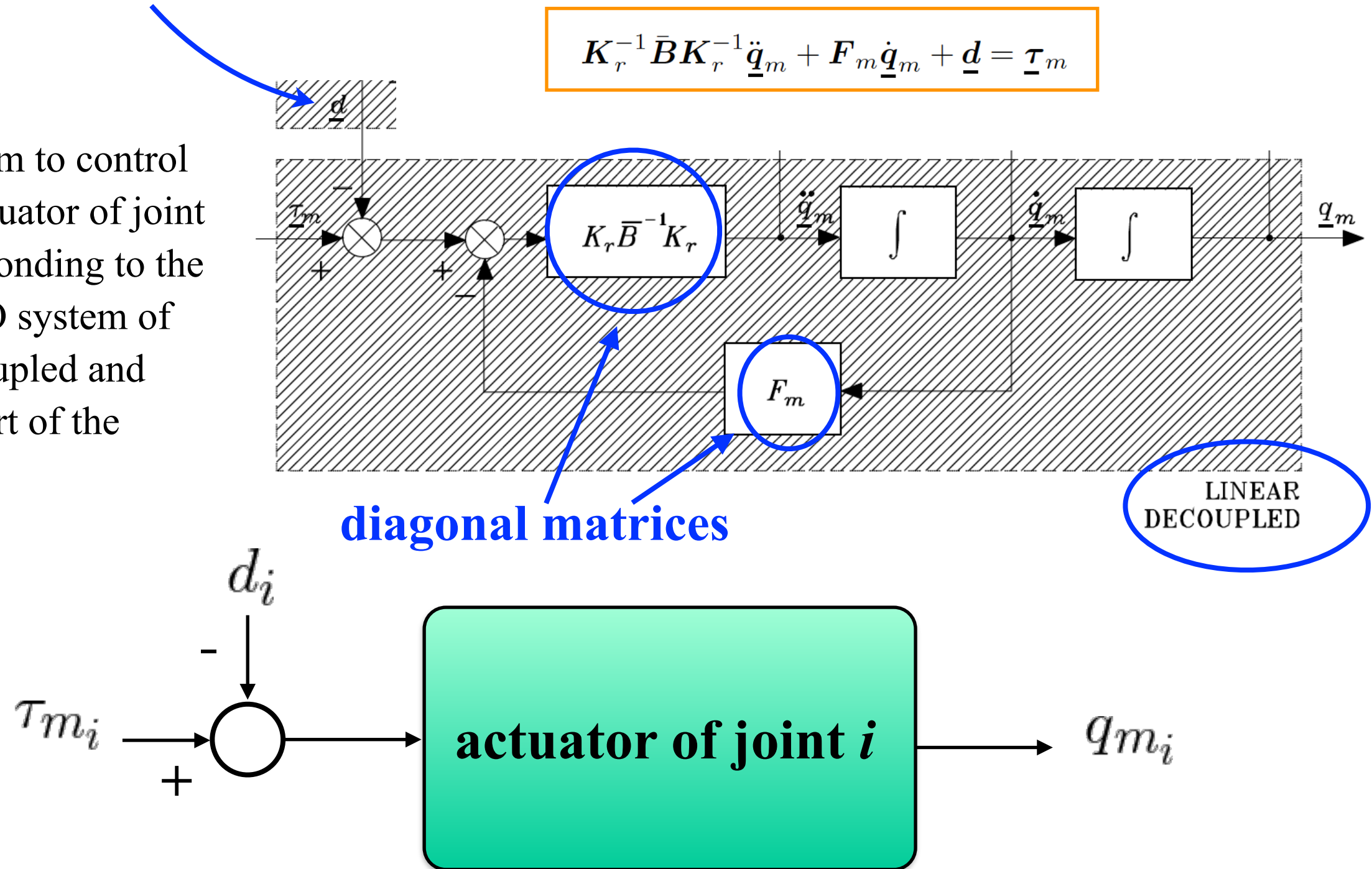
collection of n independent differential equation one for any joint actuator

Motion Control: Control in the Joint Space

Decentralized Approach

It is a disturbance

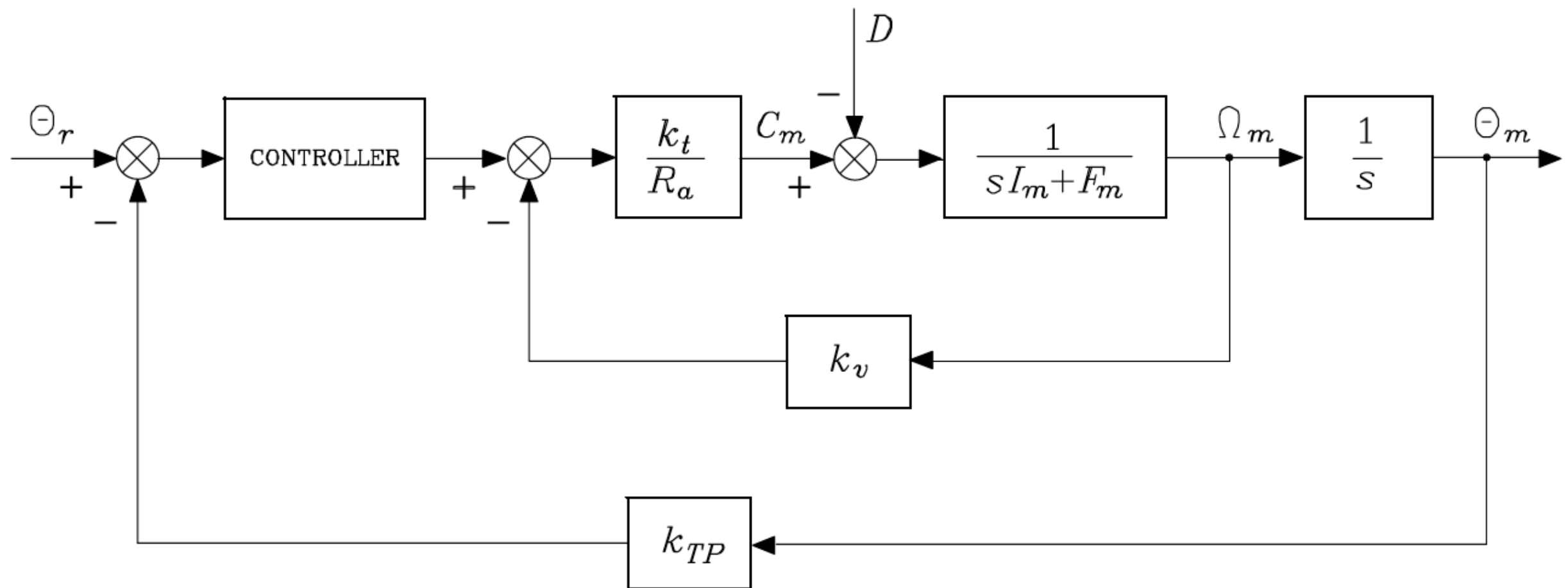
the system to control is the actuator of joint i corresponding to the i -th SISO system of the decoupled and linear part of the scheme



Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor

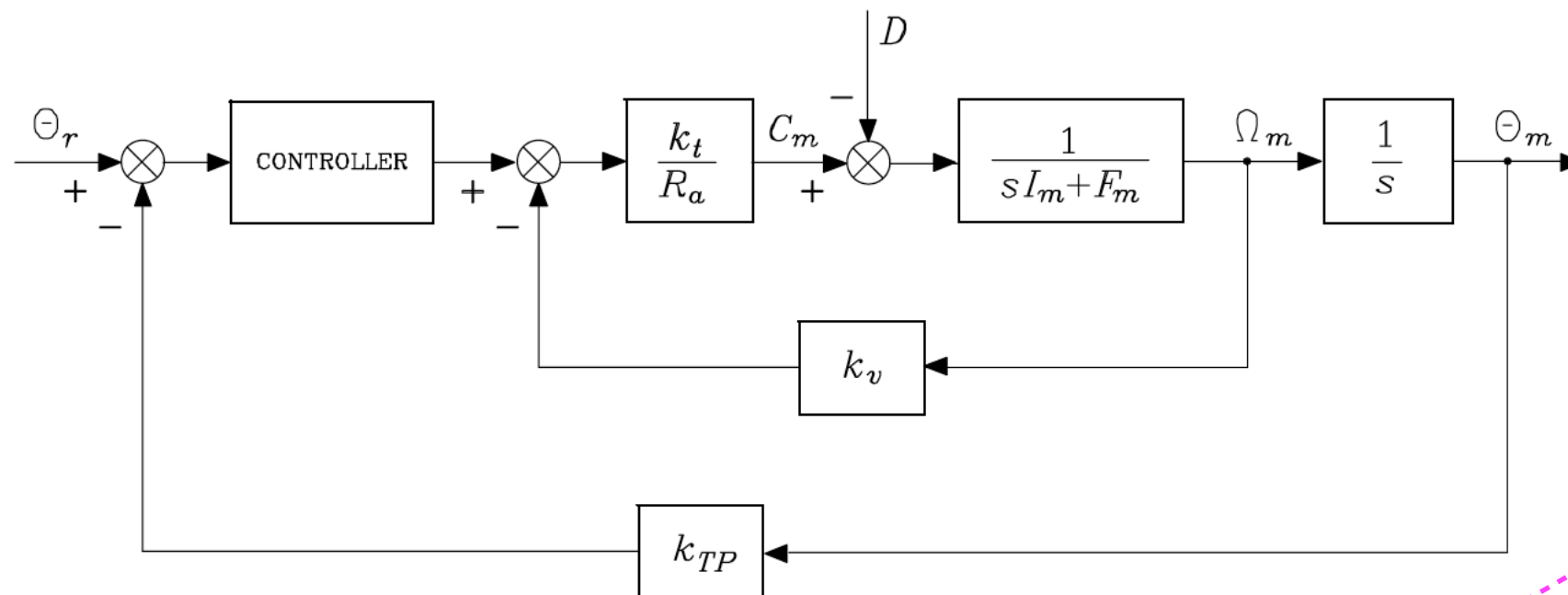


Note that to avoid to have to two subscripts we omit the subscript i

Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor



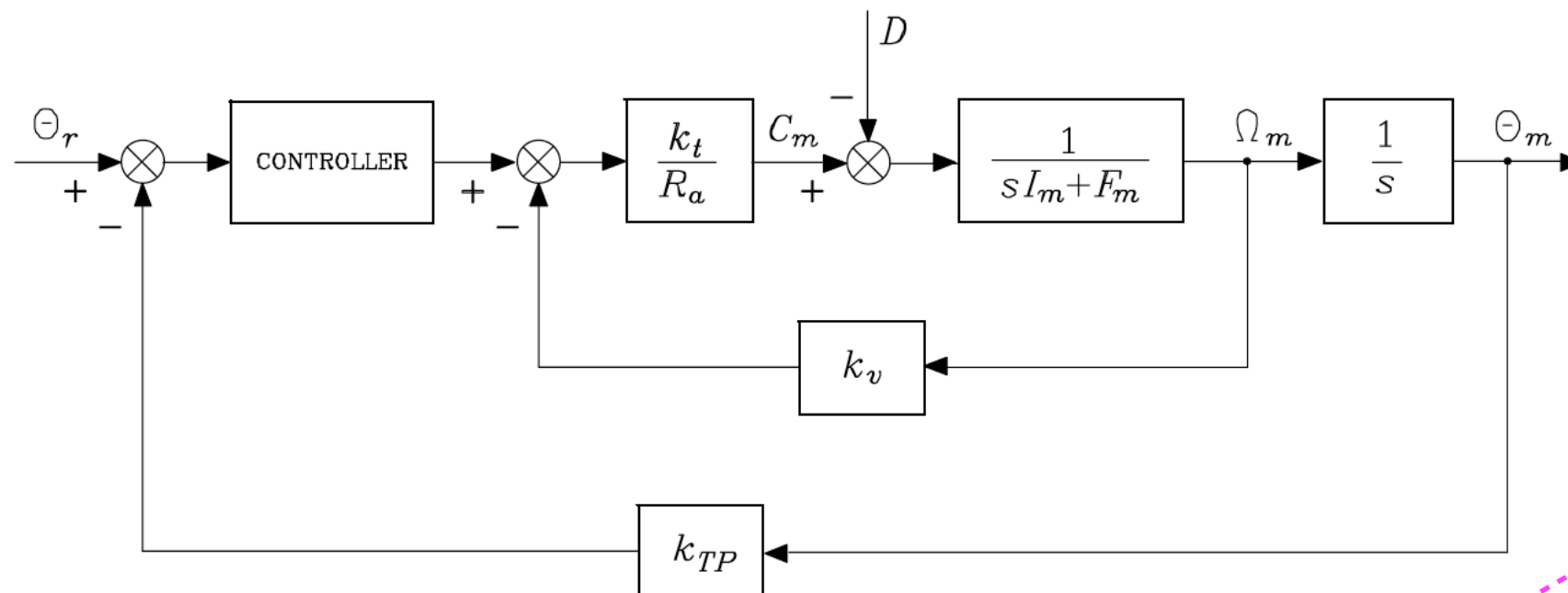
Disturbance rejection

- a **large value of the amplifier gain** before the point of intervention of the disturbance
- the presence of an **integral action** in the controller so as to cancel the effect of the gravitational component on the output at steady state

Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor



Disturbance rejection

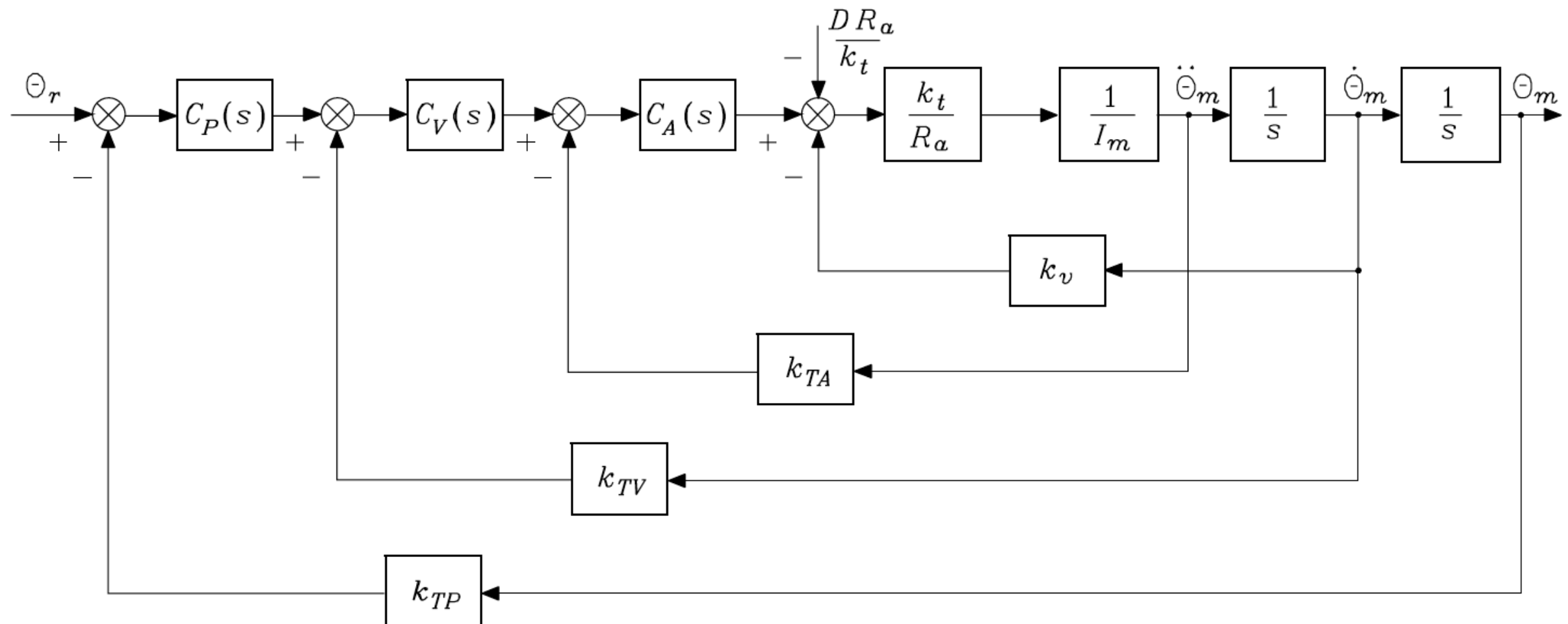
PI Controller

$$C(s) = K_c \frac{1 + sT_c}{s}$$

Motion Control: Control in the Joint Space

Decentralized Approach

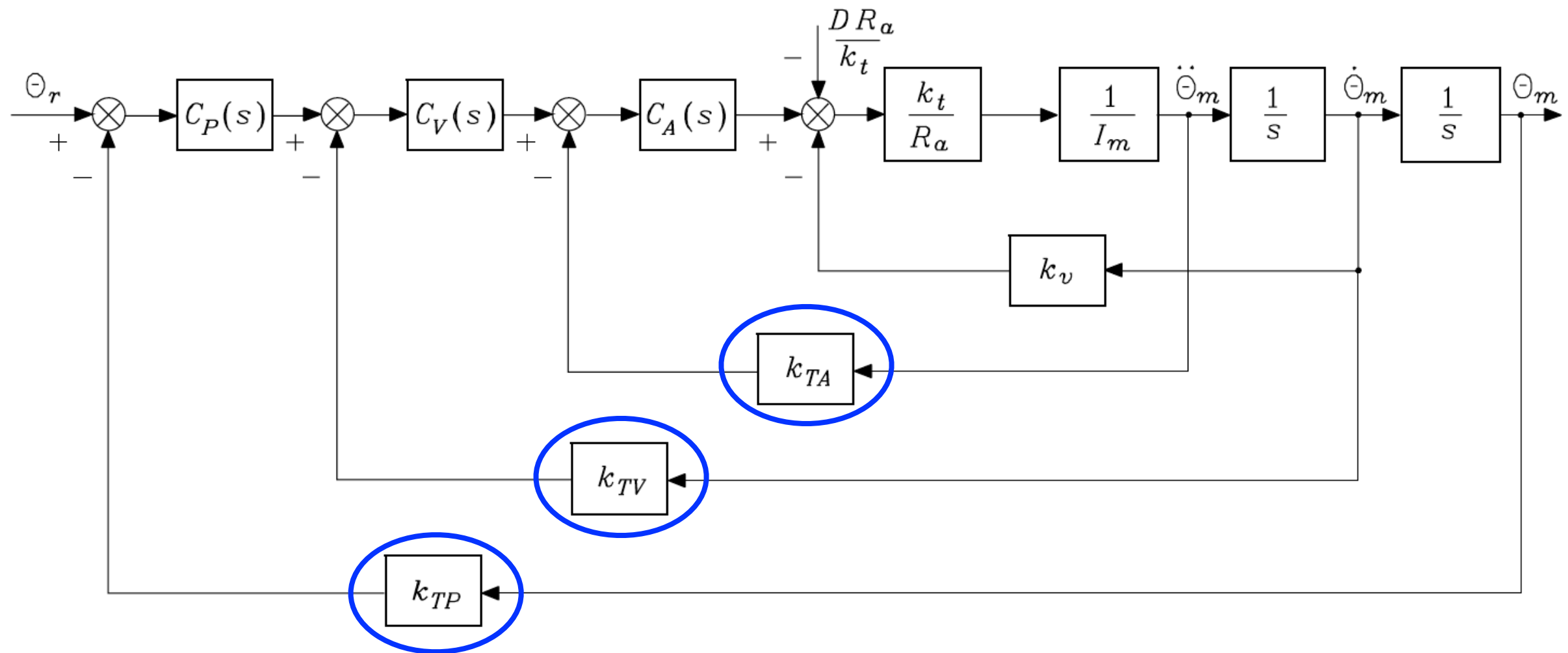
Example: the actuator of joint i is a rotary electric DC motor



Motion Control: Control in the Joint Space

Decentralized Approach

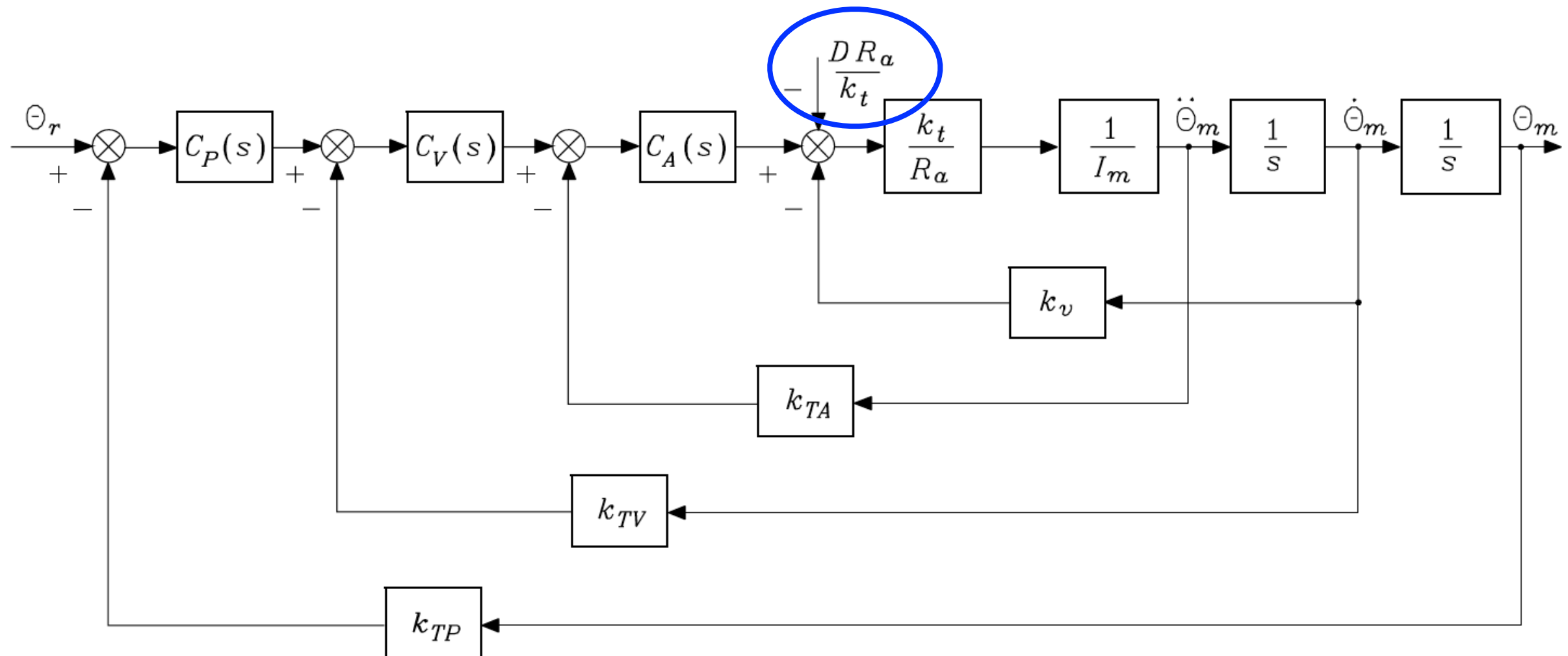
Example: the actuator of joint i is a rotary electric DC motor



Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor

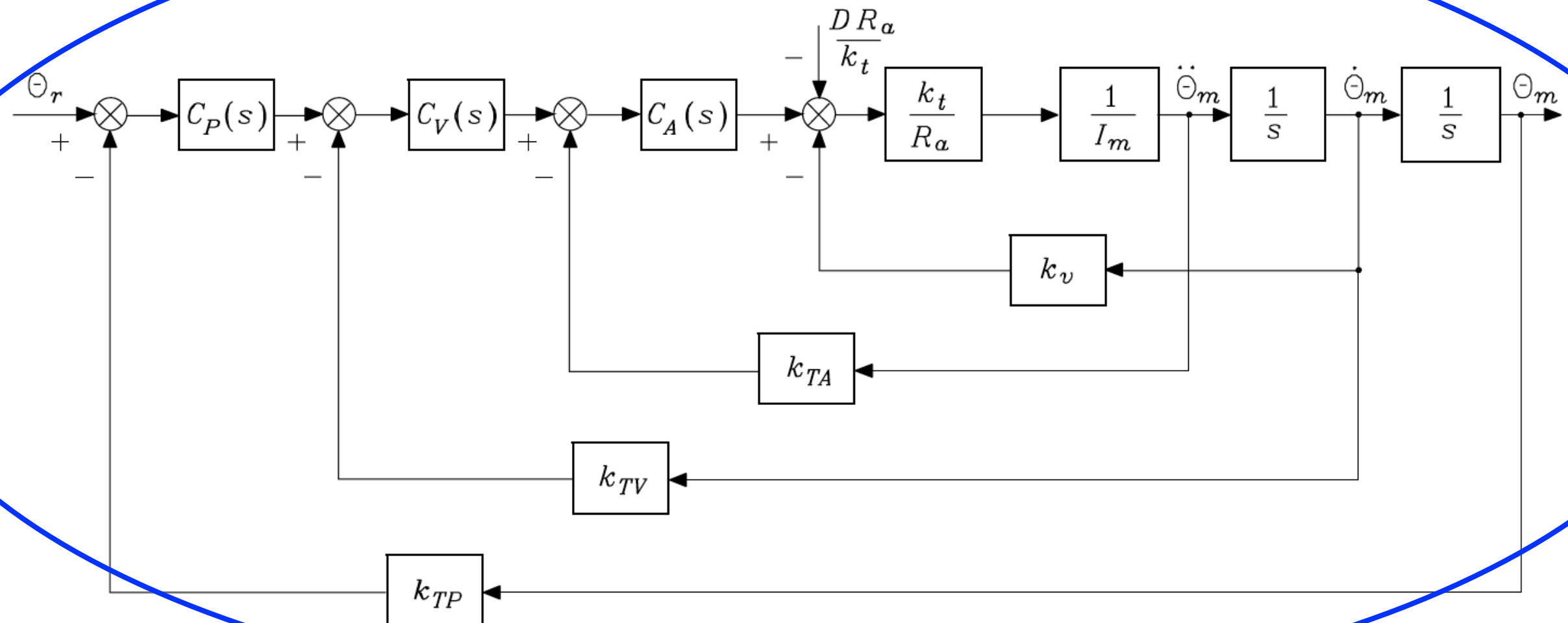


D has been suitably transformed into a voltage

Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor



control scheme for the actuator of joint i

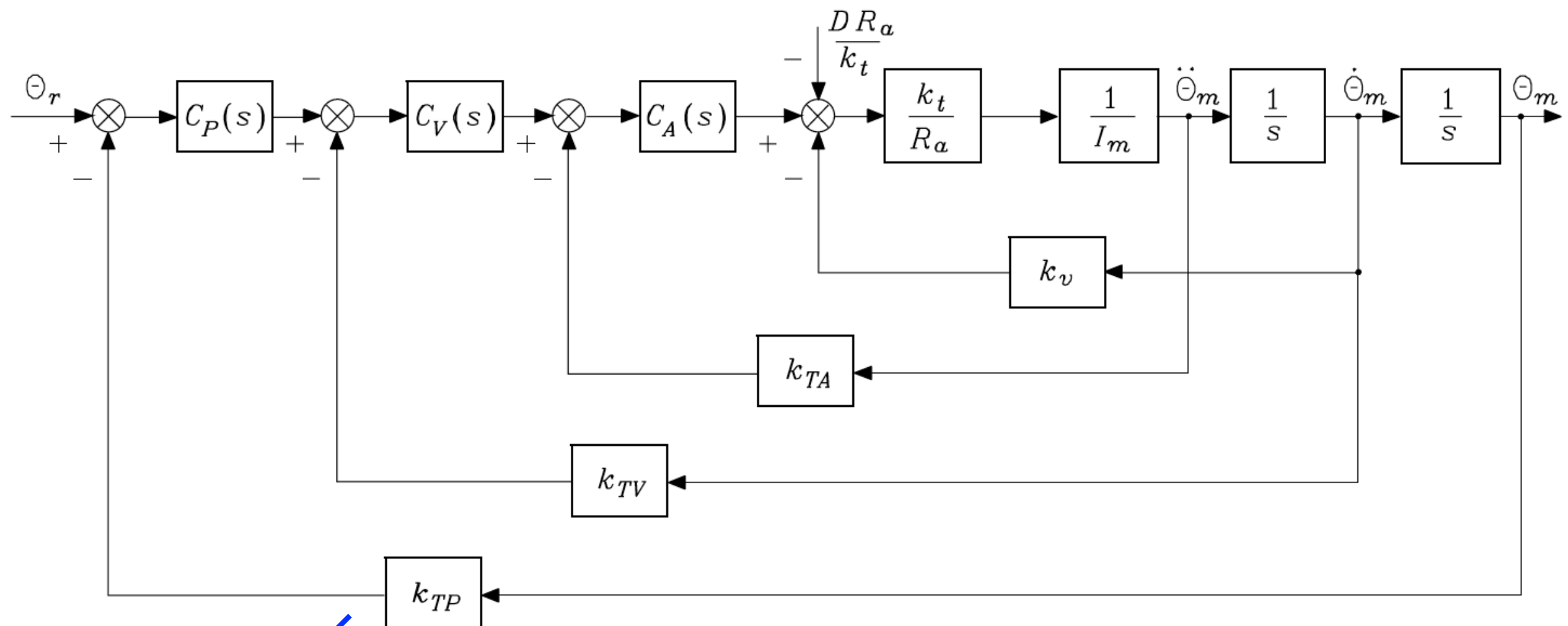
final controller:
 n of these schemes

Motion Control: Control in the Joint Space

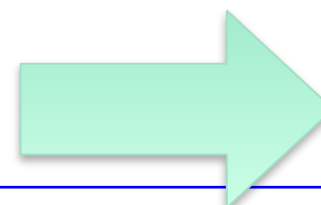
Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor

$$C_P(s) = K_P \quad C_V(s) = K_V \quad C_A(s) = K_A \frac{1 + sT_A}{s}$$



control scheme for the actuator of joint i

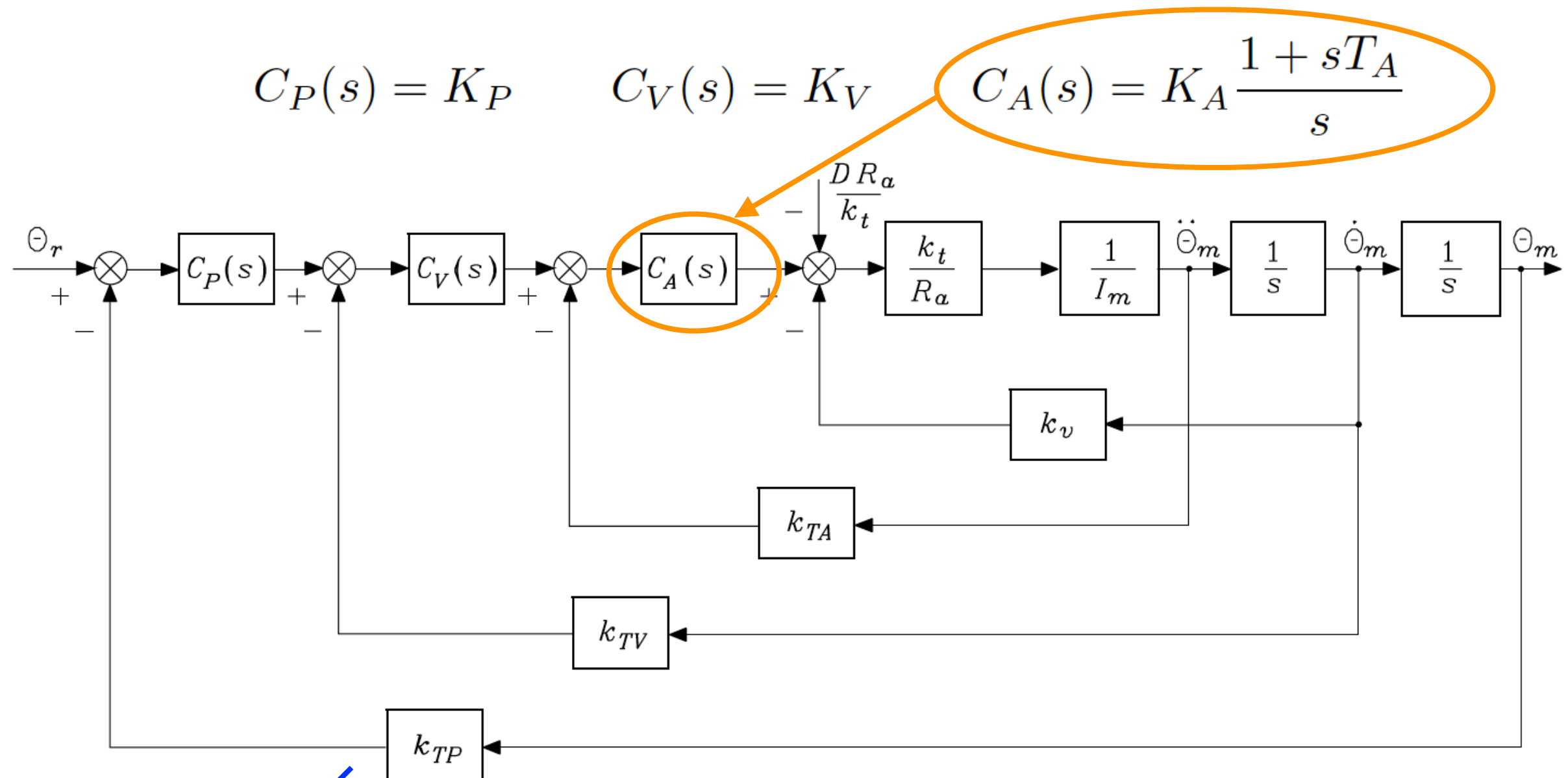


final controller:
 n of these schemes

Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor



control scheme for the actuator of joint i

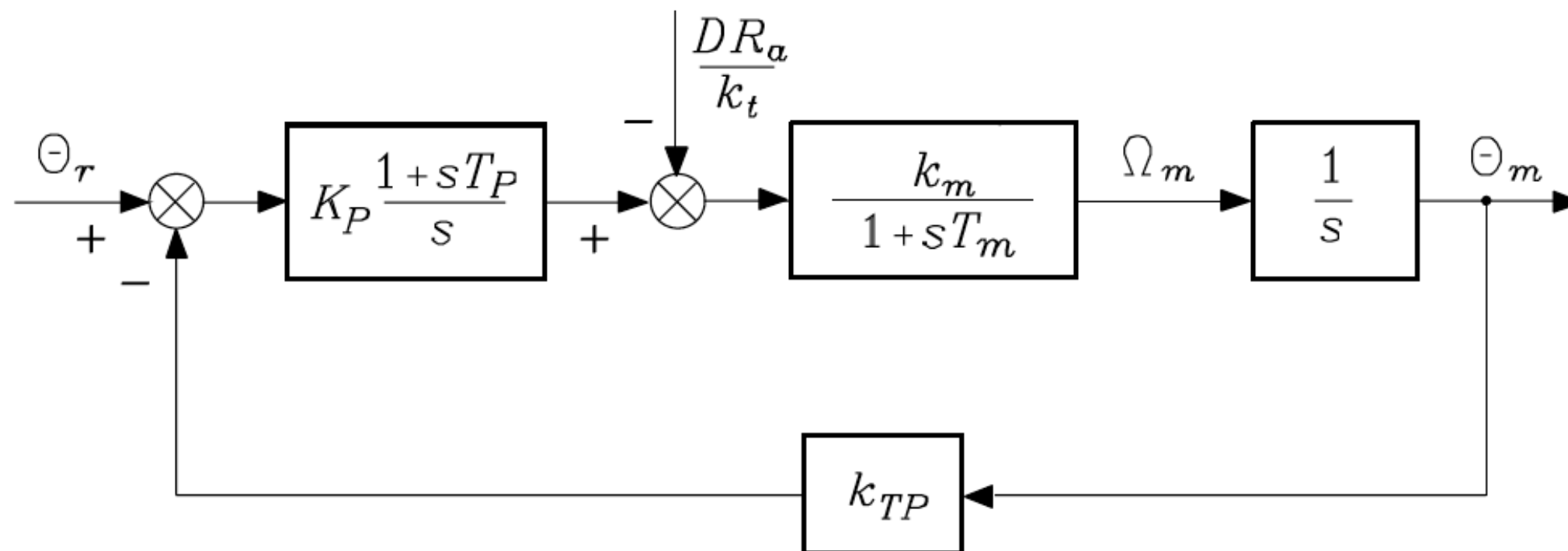
final controller:
 n of these schemes

Motion Control: Control in the Joint Space

Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor

POSITION FEEDBACK $C_P(s) = K_P \frac{1 + sT_P}{s}$ $C_V(s) = 1$ $C_A(s) = 1$
 $k_{TV} = k_{TA} = 0.$



control scheme for the actuator of joint i

final controller:
 n of these schemes

Motion Control: Control in the Joint Space

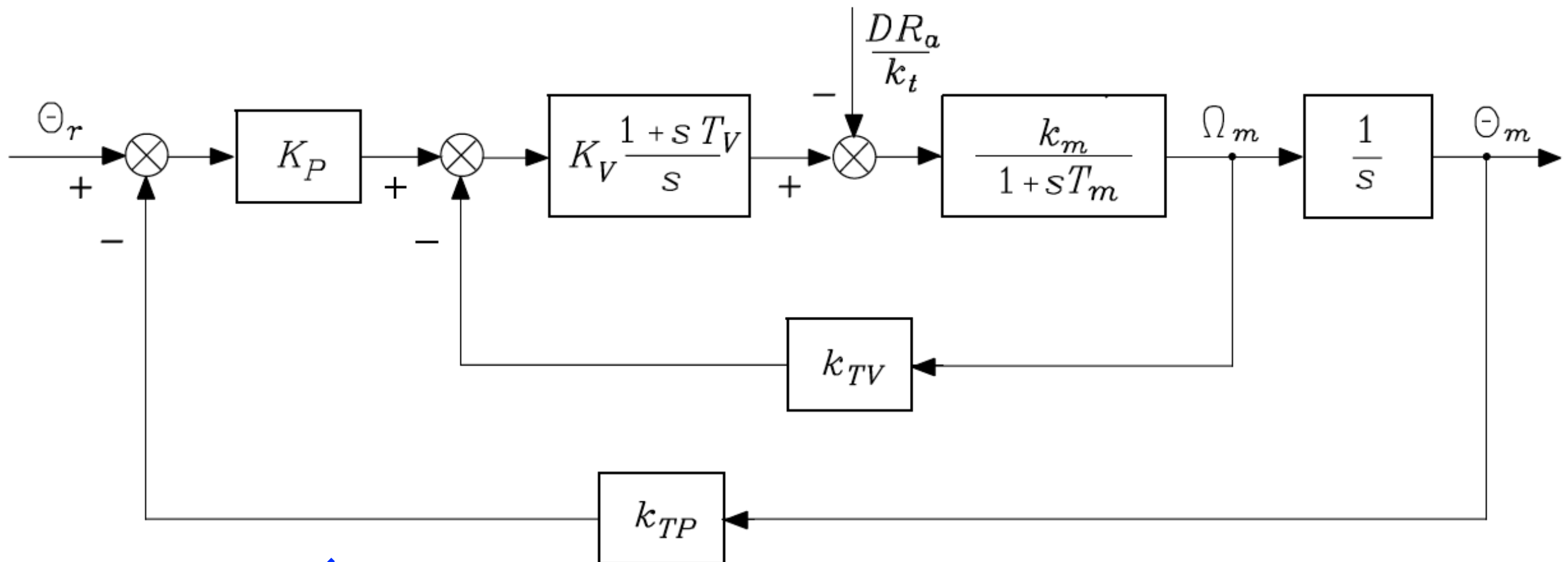
Decentralized Approach

Example: the actuator of joint i is a rotary electric DC motor

**POSITION AND
VELOCITY**

$$C_P(s) = K_P \quad C_V(s) = K_V \frac{1 + sT_V}{s} \quad C_A(s) = 1$$

$$k_{TA} = 0;$$



control scheme for the actuator of joint i

final controller:
 n of these schemes



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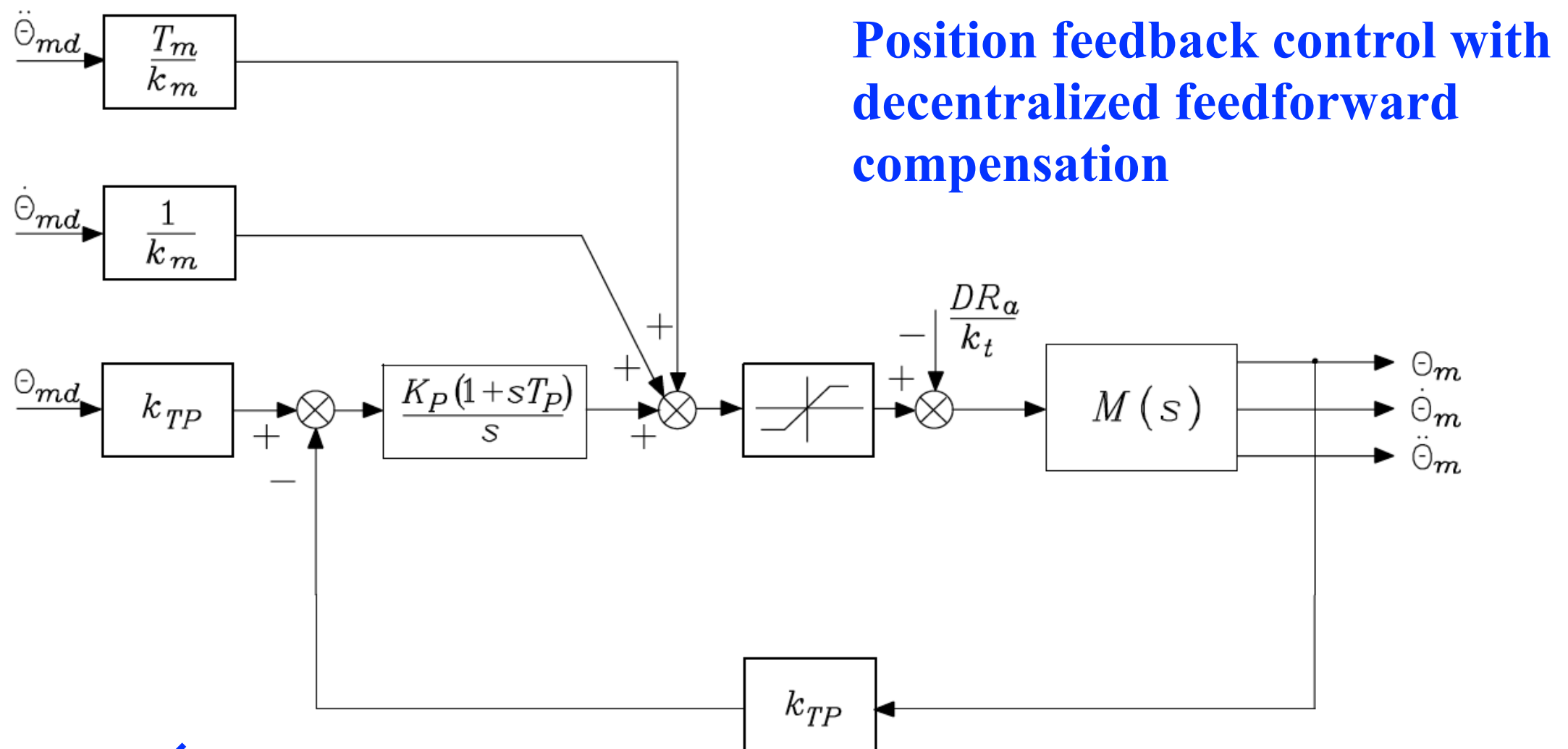


**Motion Control: Decentralized Control in the Joint Space
with Decentralized Feedforward Compensation**

Motion Control: Control in the Joint Space

Decentralized Approach and Decentralized Feedforward Compensation

Example: the actuator of joint i is a rotary electric DC motor



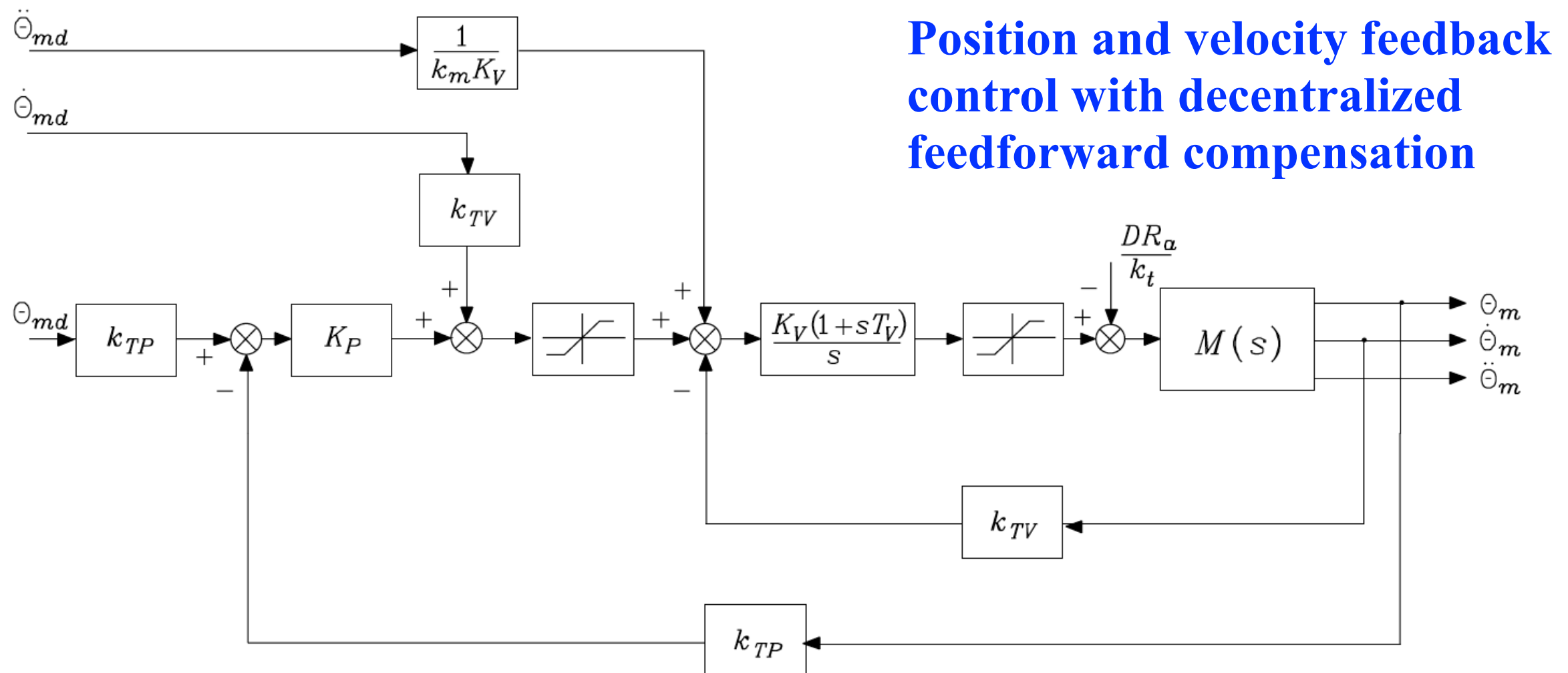
control scheme for the actuator of joint i

final controller:
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Motion Control: Control in the Joint Space

Decentralized Approach and Decentralized Feedforward Compensation

Example: the actuator of joint i is a rotary electric DC motor



control scheme for the actuator of joint i

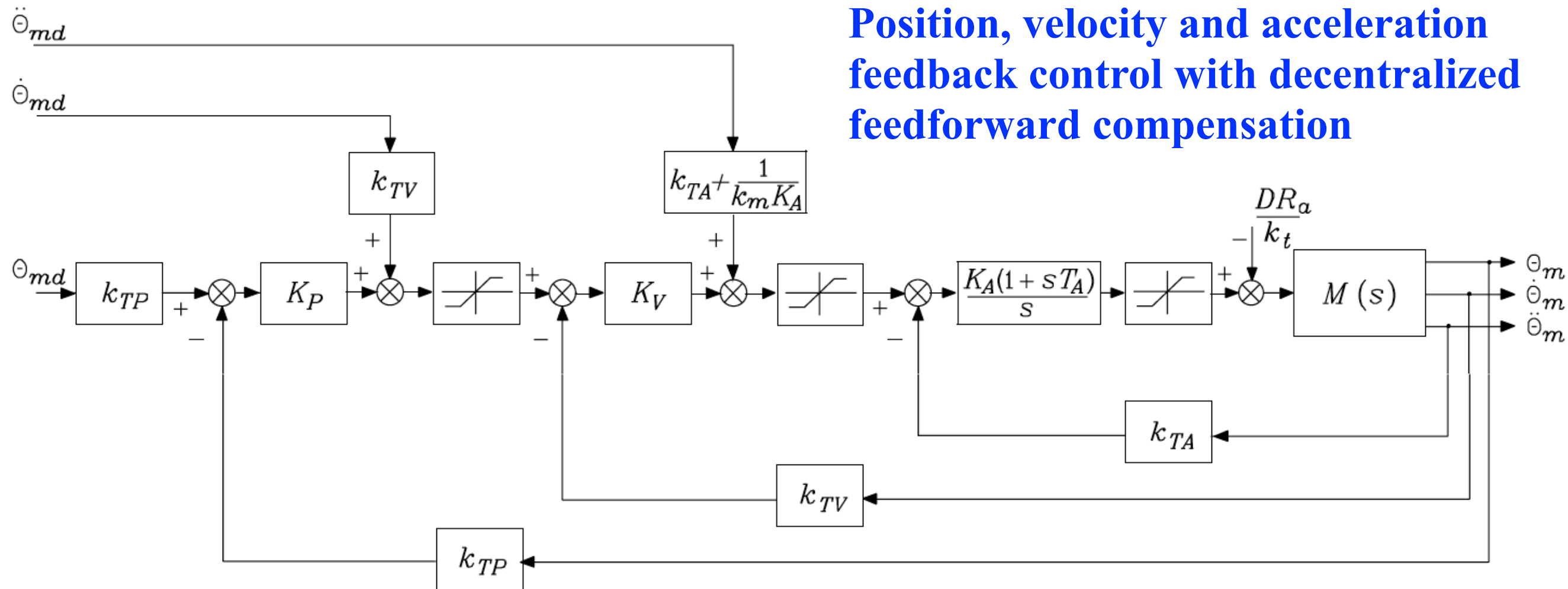
final controller:
 n of these schemes

Motion Control: Control in the Joint Space

Decentralized Approach and Decentralized Feedforward Compensation

Example: the actuator of joint i is a rotary electric DC motor

Position, velocity and acceleration feedback control with decentralized feedforward compensation



control scheme for the actuator of joint i

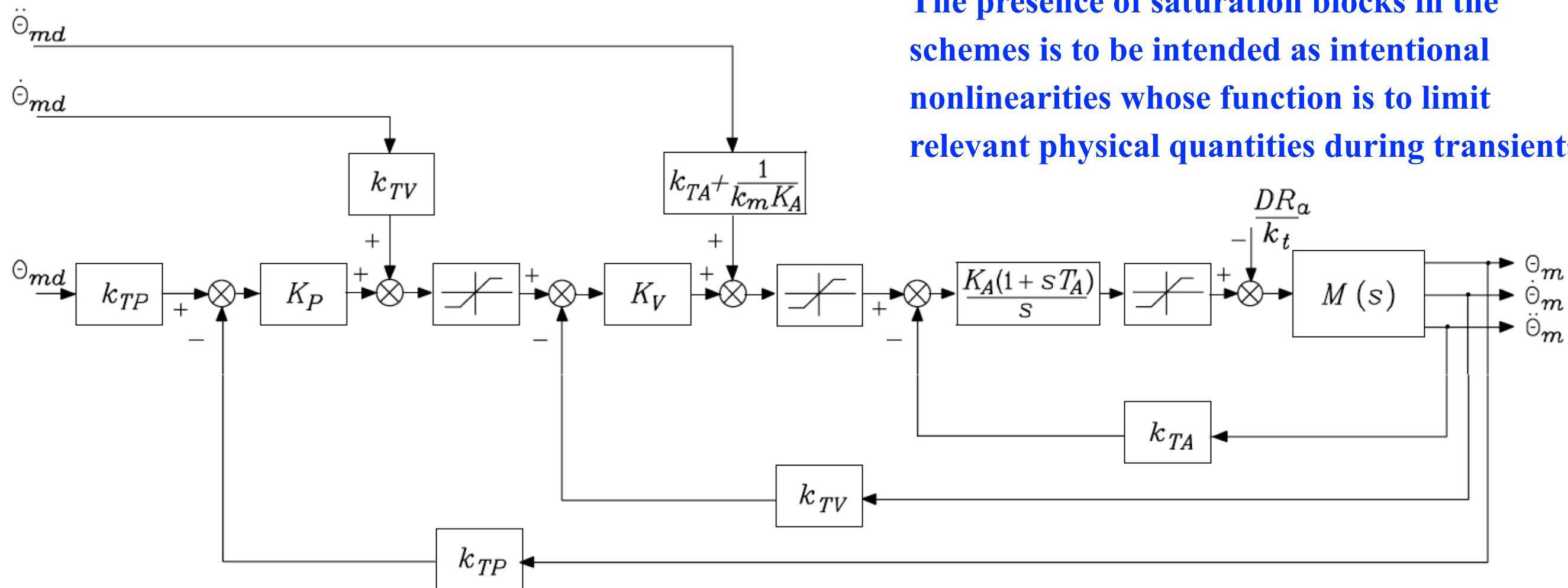
final controller:
 n of these schemes

Motion Control: Control in the Joint Space

Decentralized Approach and Decentralized Feedforward Compensation

Example: the actuator of joint i is a rotary electric DC motor

The presence of saturation blocks in the schemes is to be intended as intentional nonlinearities whose function is to limit relevant physical quantities during transients



control scheme for the actuator of joint i

final controller:
 n of these schemes



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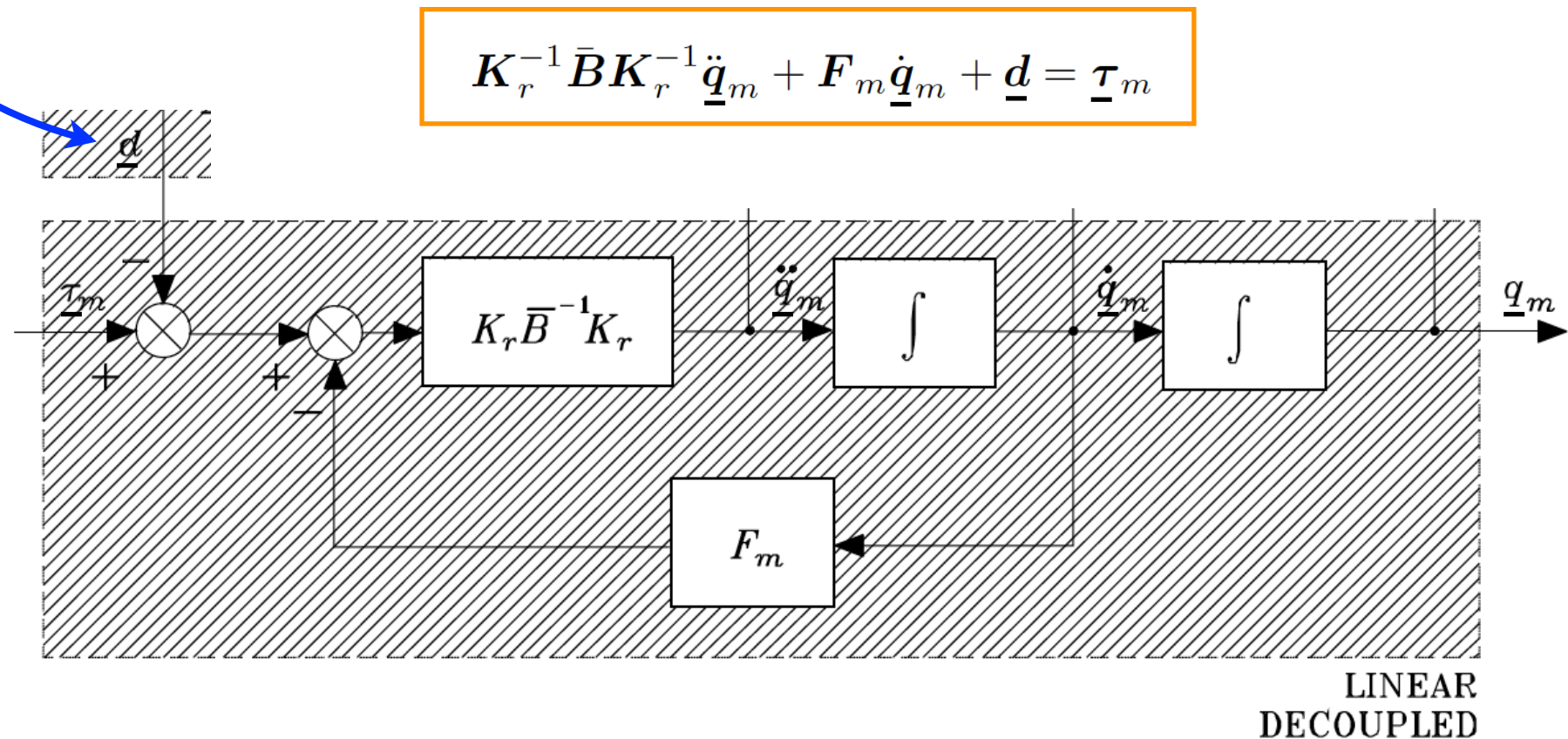


**Motion Control: Decentralized Control in the Joint Space
with Centralized Feedforward Compensation**

Motion Control: Control in the Joint Space

Decentralized Approach and Centralized Feedforward Compensation

It is a disturbance

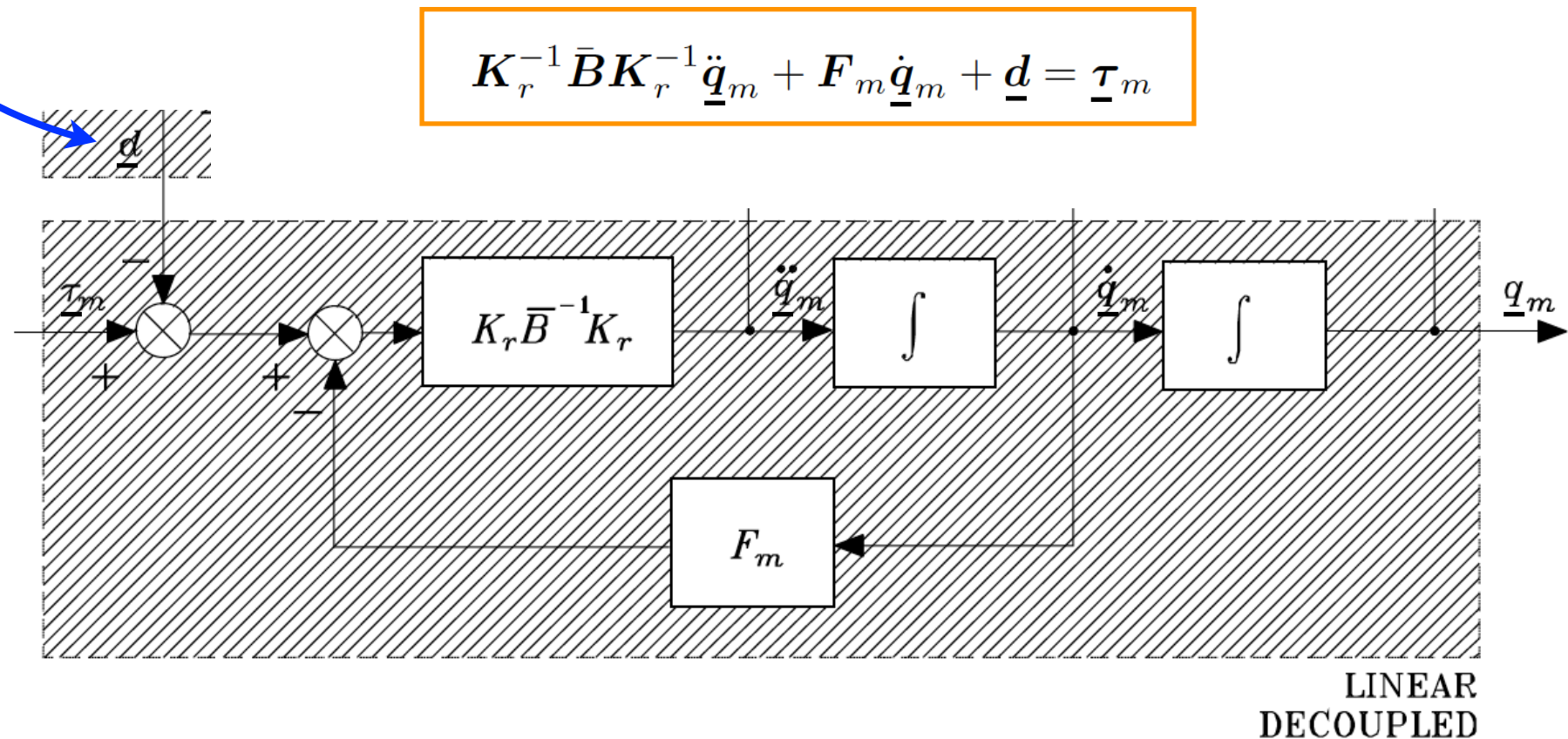


$$\underline{d} = K_r^{-1} \Delta B(\underline{q}) K_r^{-1} \ddot{\underline{q}}_m + K_r^{-1} C(\underline{q}, \dot{\underline{q}}) K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} \underline{g}(\underline{q})$$

Motion Control: Control in the Joint Space

Decentralized Approach and Centralized Feedforward Compensation

It is a disturbance



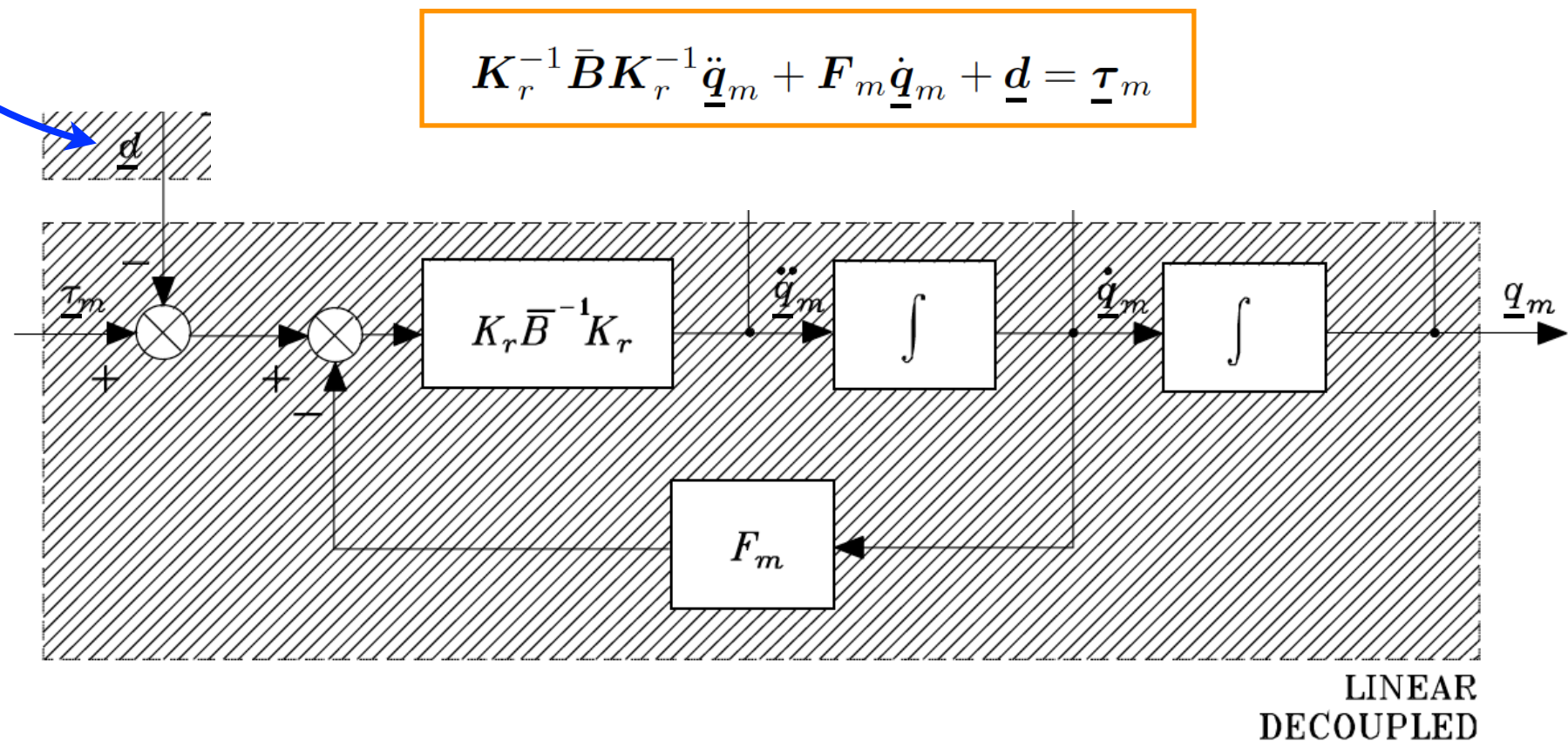
$$\underline{d} = K_r^{-1} \Delta B(\underline{q}) K_r^{-1} \ddot{\underline{q}}_m + K_r^{-1} C(\underline{q}, \dot{\underline{q}}) K_r^{-1} \dot{\underline{q}}_m + K_r^{-1} \underline{g}(\underline{q})$$

$$\underline{d}_d = K_r^{-1} \Delta B(\underline{q}_d) K_r^{-1} \ddot{\underline{q}}_{md} + K_r^{-1} C(\underline{q}_d, \dot{\underline{q}}_d) K_r^{-1} \dot{\underline{q}}_{md} + K_r^{-1} \underline{g}(\underline{q}_d)$$

Motion Control: Control in the Joint Space

Decentralized Approach and Centralized Feedforward Compensation

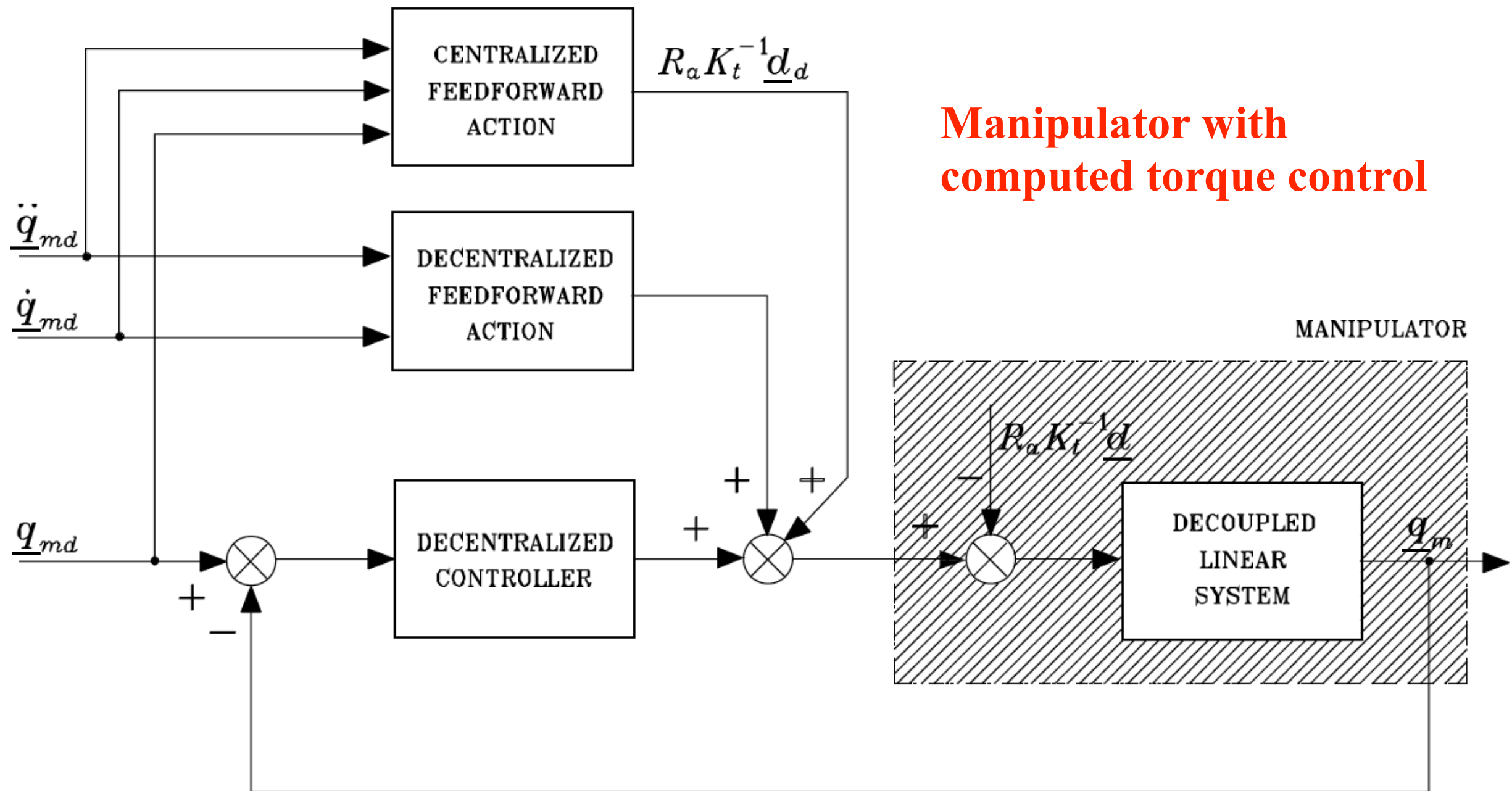
It is a disturbance



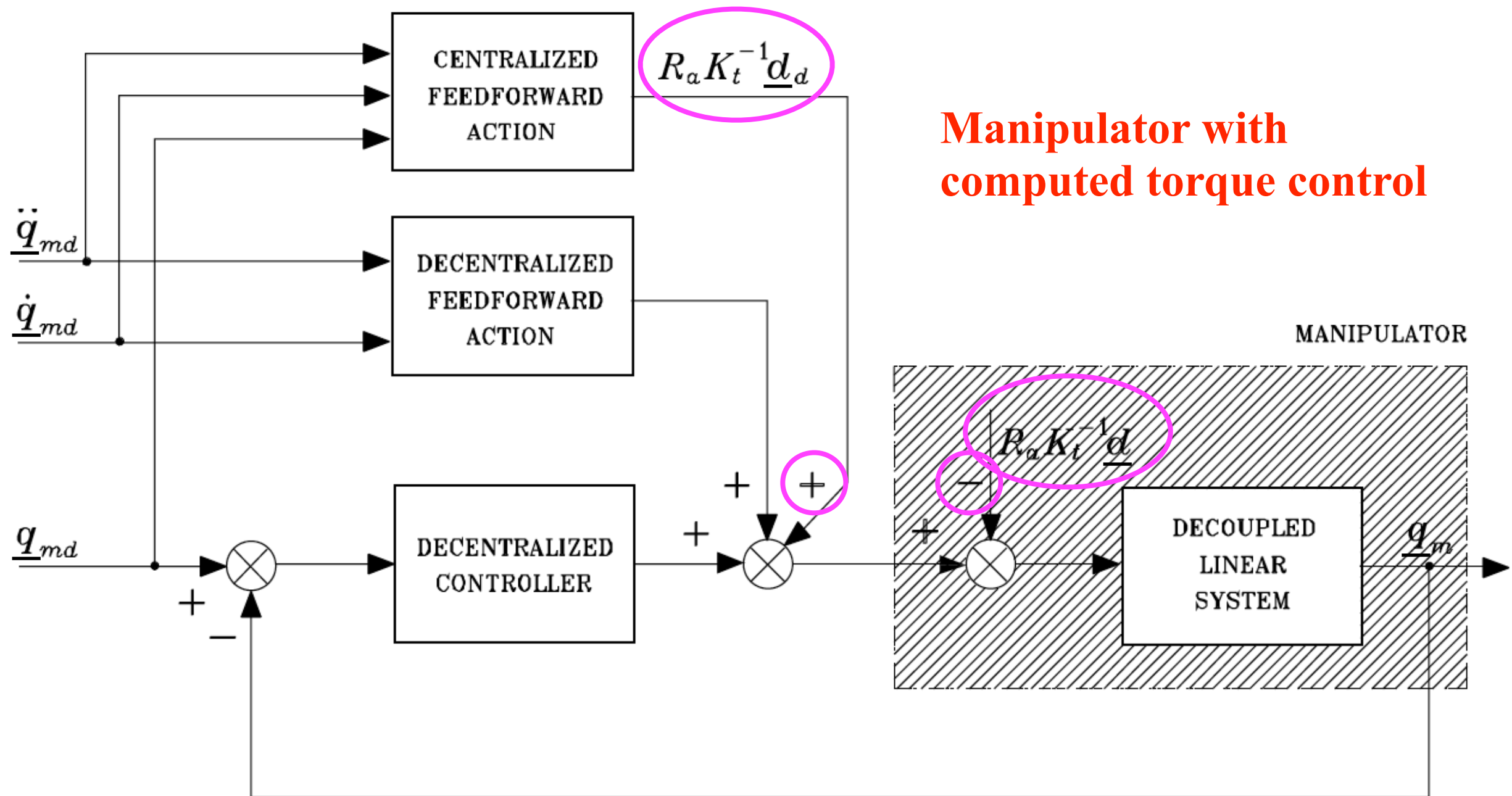
$$\underline{\underline{d}} = \mathbf{K}_r^{-1} \Delta \mathbf{B}(\underline{q}) \mathbf{K}_r^{-1} \ddot{\underline{q}}_m + \mathbf{K}_r^{-1} \mathbf{C}(\underline{q}, \dot{\underline{q}}) \mathbf{K}_r^{-1} \dot{\underline{q}}_m + \mathbf{K}_r^{-1} \underline{g}(\underline{q})$$

$$\underline{\underline{d}}_d = \mathbf{K}_r^{-1} \Delta \mathbf{B}(\underline{\underline{q}}_d) \mathbf{K}_r^{-1} \ddot{\underline{\underline{q}}}_{md} + \mathbf{K}_r^{-1} \mathbf{C}(\underline{\underline{q}}_d, \dot{\underline{\underline{q}}}_d) \mathbf{K}_r^{-1} \dot{\underline{\underline{q}}}_{md} + \mathbf{K}_r^{-1} \underline{\underline{g}}(\underline{\underline{q}}_d).$$

Overall MIMO Decentralized Motion Control Scheme in the Joint Space with Decentralized and Centralized Feedforward Compensation



Overall MIMO Decentralized Motion Control Scheme in the Joint Space with Decentralized and Centralized Feedforward Compensation





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Motion Control: Centralized Control in the Joint Space

Motion Control: Control in the Joint Space

Another way to control the robot

- No transmission (no gear box)
- Request of high performance in terms of velocity and acceleration

$\underline{d}$ (the terms accounting for nonlinearities and couplings): cannot be regarded as a disturbance acting on the joints. It needs to be explicitly considered in the design.



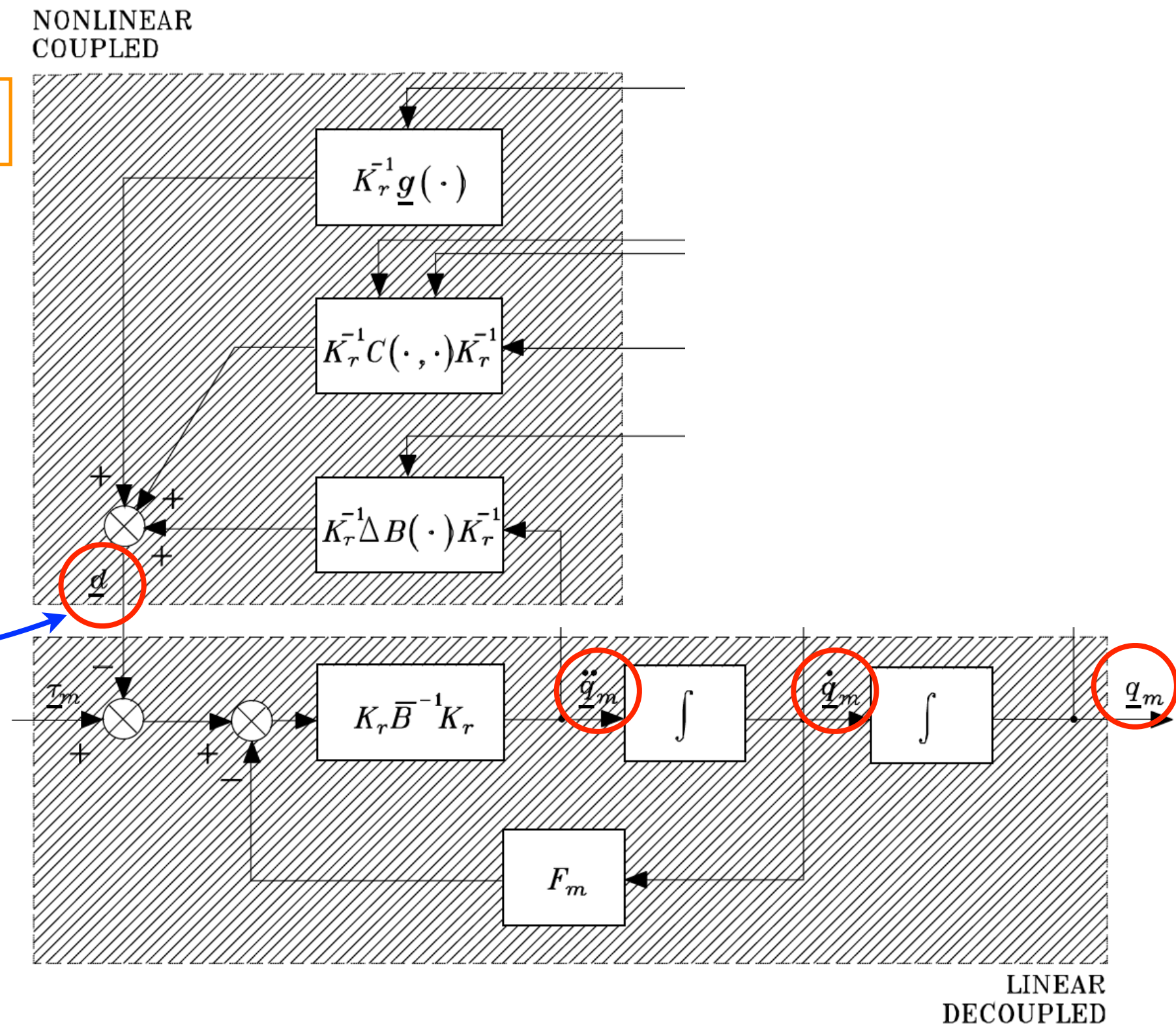
CENTRALIZED CONTROL SCHEME

The robot is controlled as a MIMO system

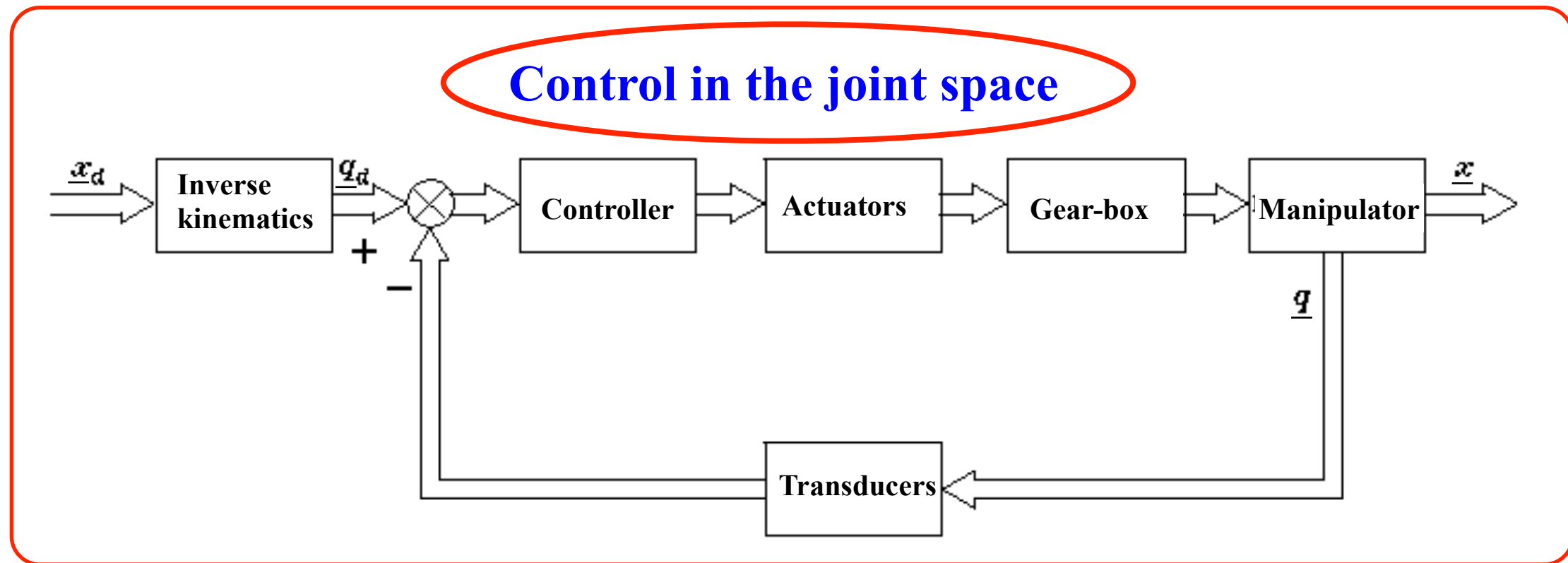
Motion Control: Control in the Joint Space

$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

It is a disturbance



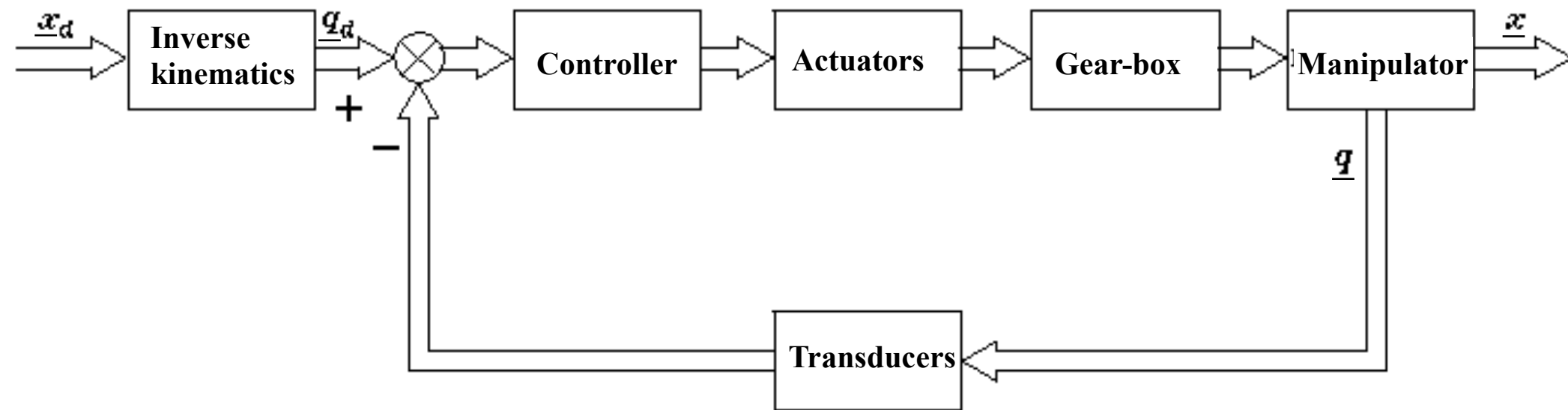
Motion Control: Control in the Joint Space



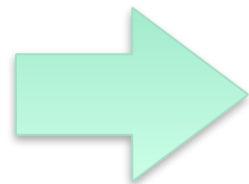
- ➡ **decentralized control (independent control of the joints)**
 - *decentralized control with decentralized feedforward compensation*
 - *decentralized control with centralized computed torque based feedforward compensation*
- ➡ **centralized control of the joints**

Motion Control: Control in the Joint Space

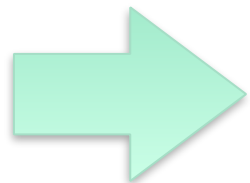
In this Lecture... **Control in the joint space**



Centralized Control Approach in the Joint Space:



P and PD Control with Gravity Compensation



Inverse Dynamics Control



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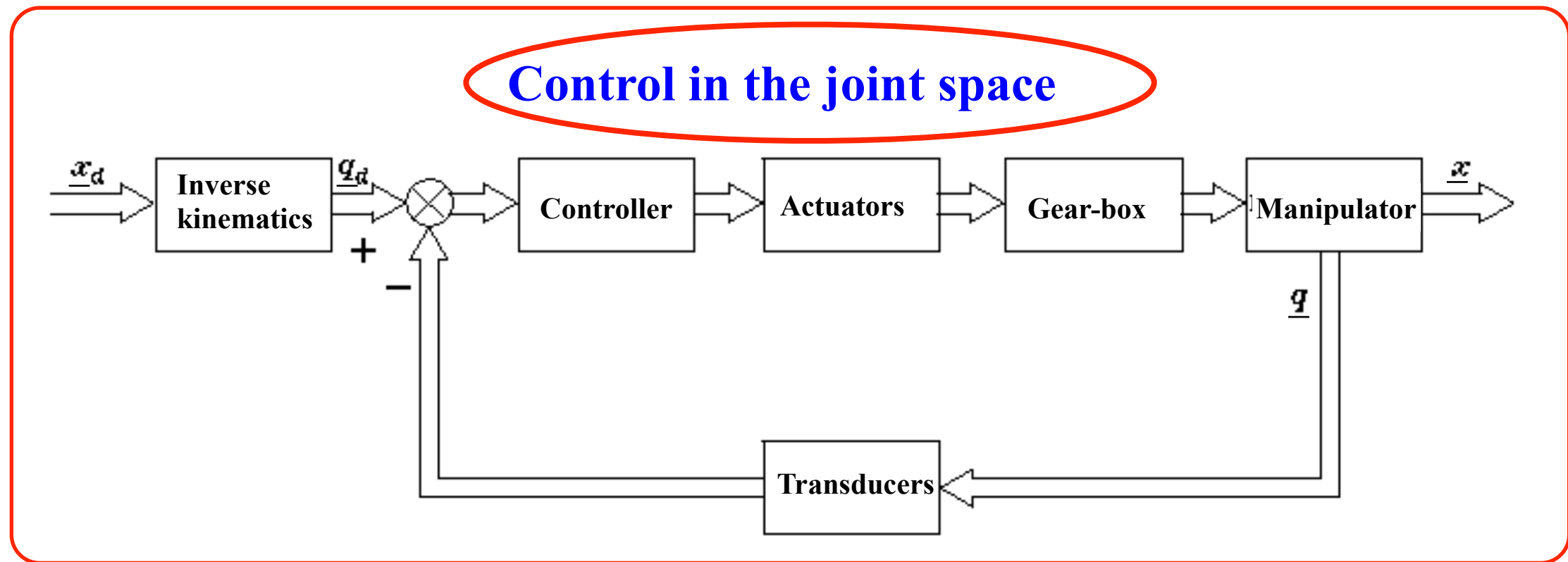
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Model reformulation for centralized control in joint space

Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation

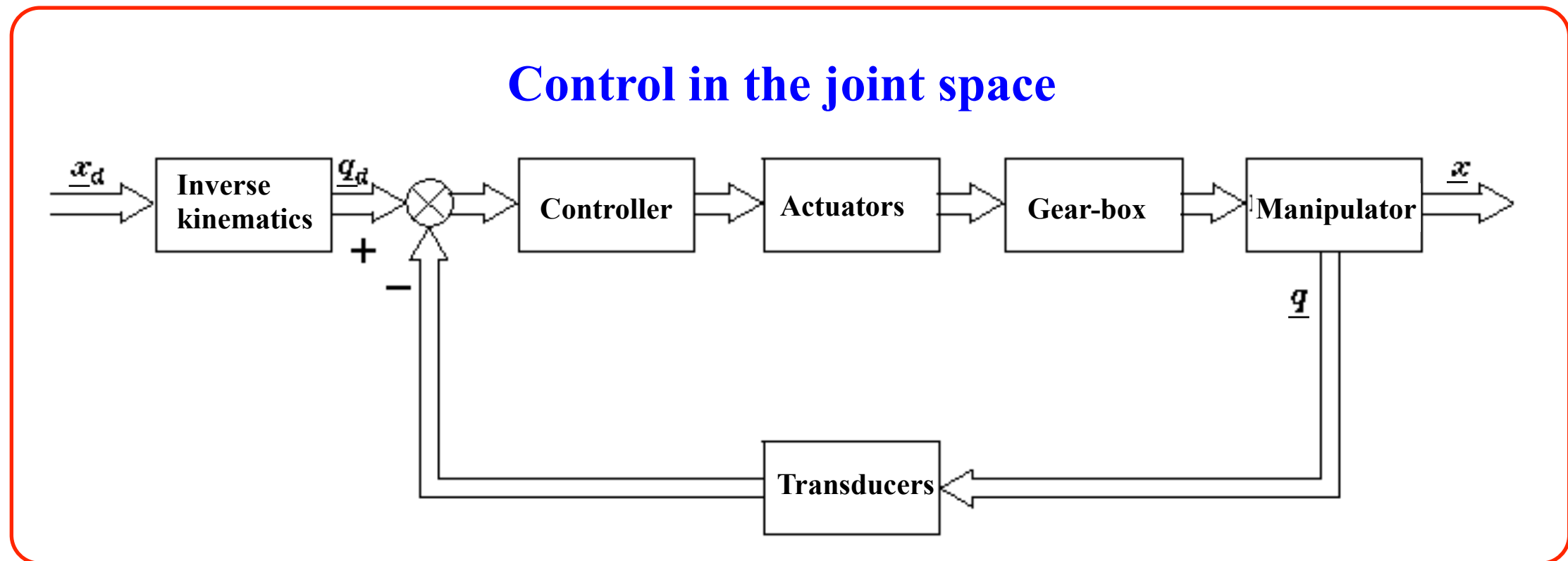


$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$F_m = K_r^{-1} F_v K_r^{-1}$$

Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation



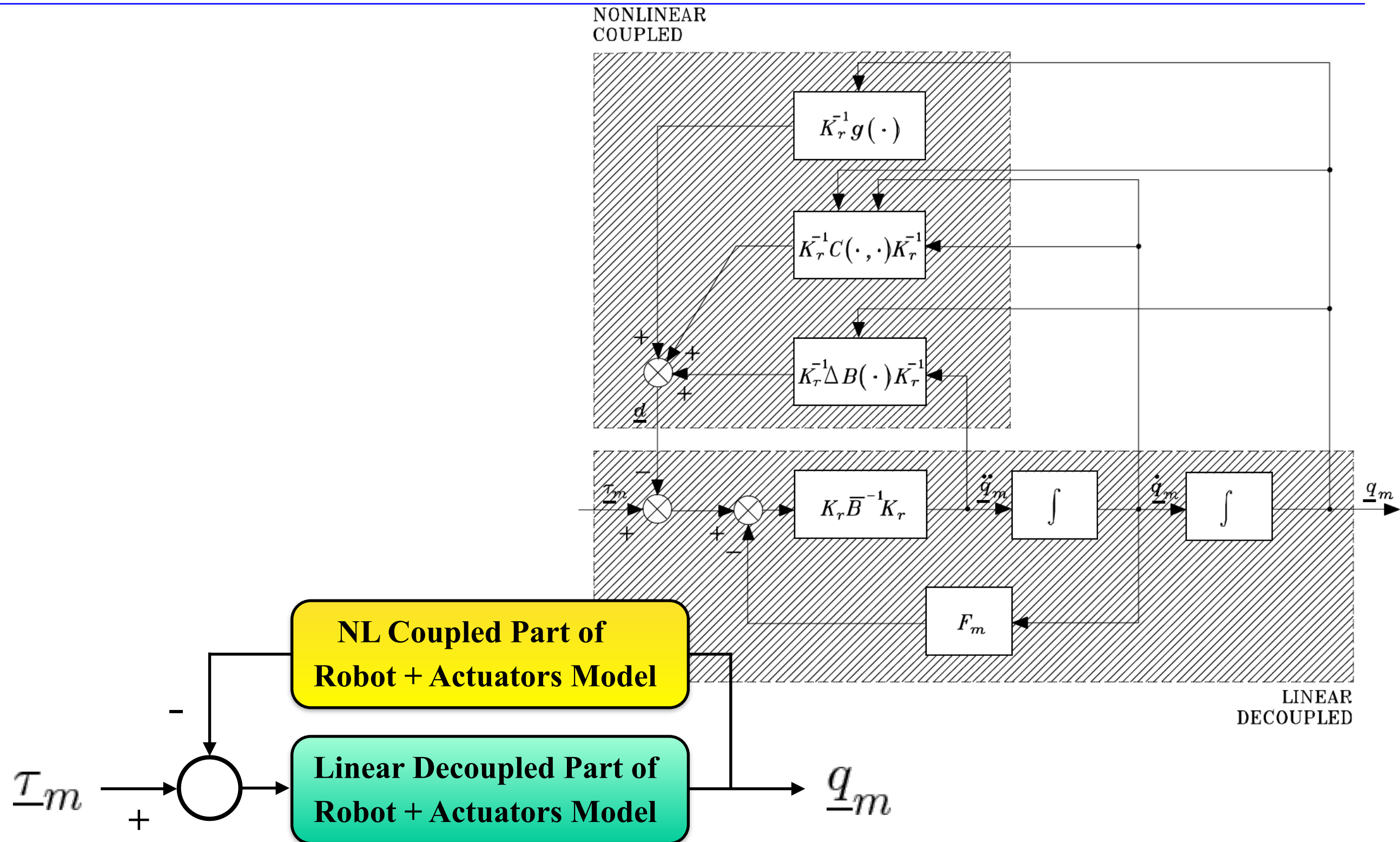
$$K_r^{-1} \bar{B} K_r^{-1} \ddot{\underline{q}}_m + F_m \dot{\underline{q}}_m + \underline{d} = \underline{\tau}_m$$

$$F_m = K_r^{-1} F_v K_r^{-1}$$

It cannot be regarded as a
generalized force disturbance

Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation



Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation

MIMO Dynamic Model of Robot Manipulator and Actuators



$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{\tau}$$

$$K_r \underline{q} = \underline{q}_m \quad \underline{\tau}_m = K_r^{-1} \underline{\tau} \quad \text{transmissions}$$

$$K_r^{-1} \underline{\tau} = K_t \dot{\underline{a}}$$

$$\underline{v}_a = R_a \dot{\underline{a}} + K_v \dot{\underline{q}}_m \quad n \text{ actuators}$$

$$\underline{v}_a = G_v \underline{v}_c.$$

to design control schemes of centralized type, it is convenient to go back to the dynamic model formulation expressed in terms of the joint variables vector and its derivative

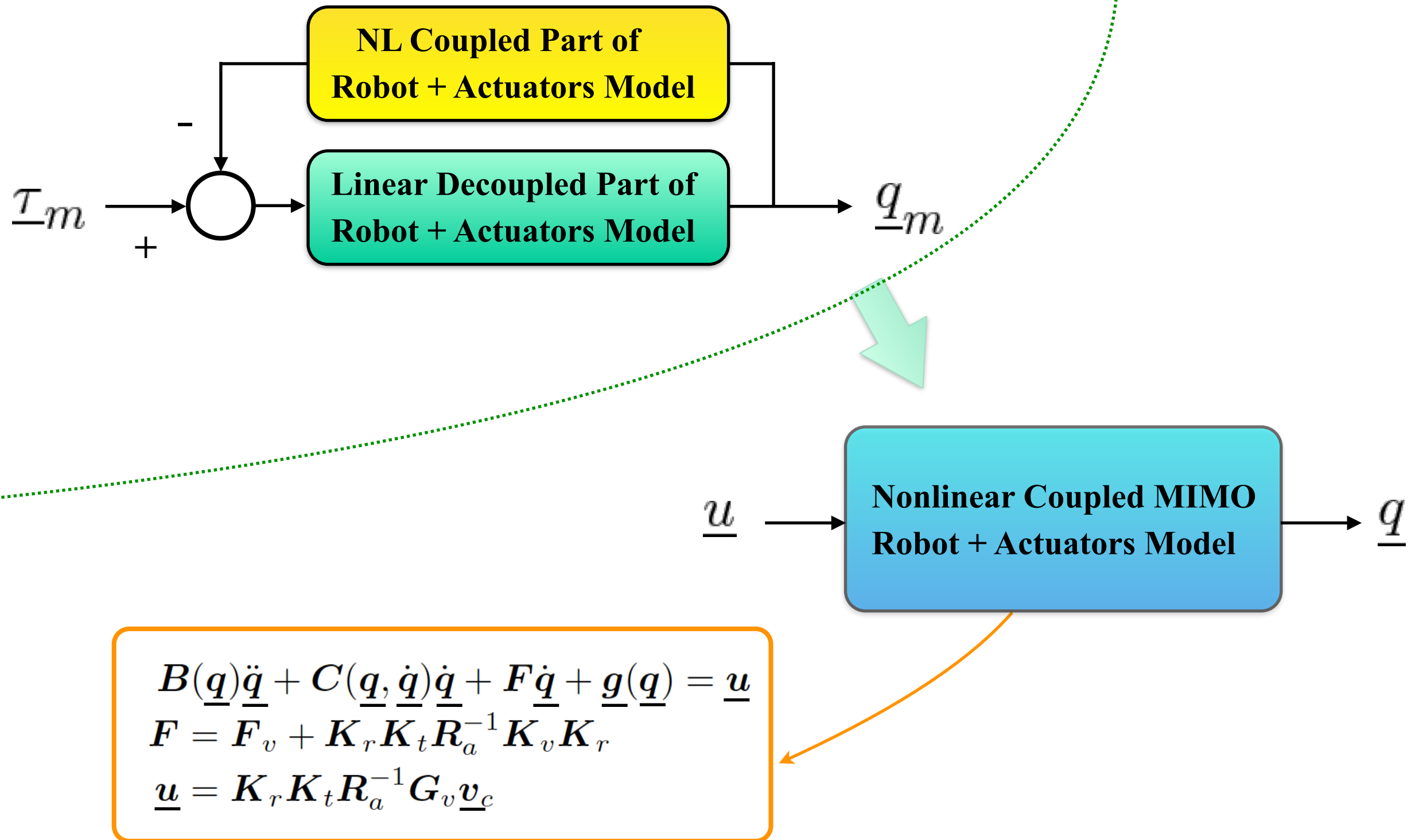
$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation





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P and PD Control with Gravity Compensation

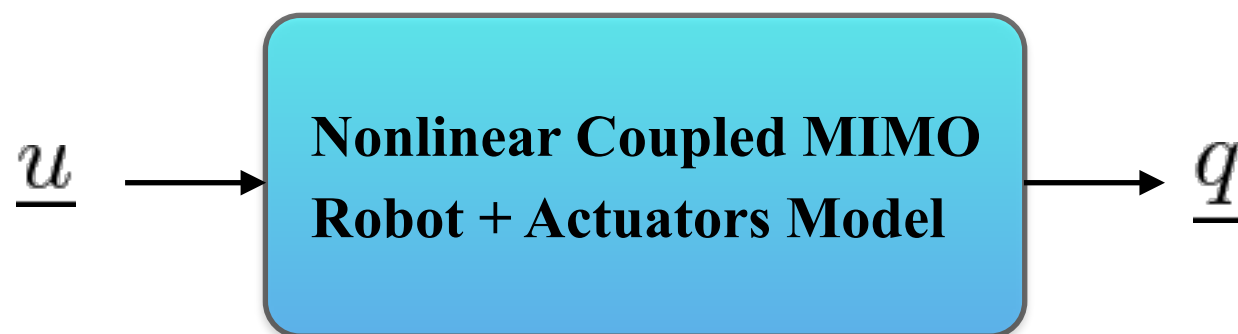
Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

System state: $[\underline{\tilde{q}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{q}} = \underline{q_d} - \underline{q}$

$\underline{q_d}$ is a **constant equilibrium configuration**



Motion Control: Control in the Joint Space

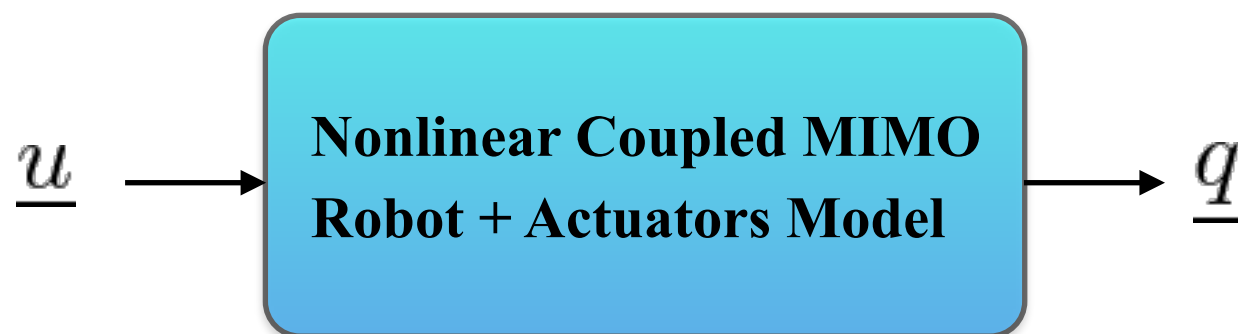
Centralized Approach

P and PD Control with Gravity Compensation

System state: $[\underline{\tilde{q}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{q}} = \underline{q_d} - \underline{q}$

$\underline{q_d}$ is a **constant equilibrium configuration**

PROBLEM: find the structure of the controller which guarantees the global asymptotic stability of the equilibrium configuration



Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

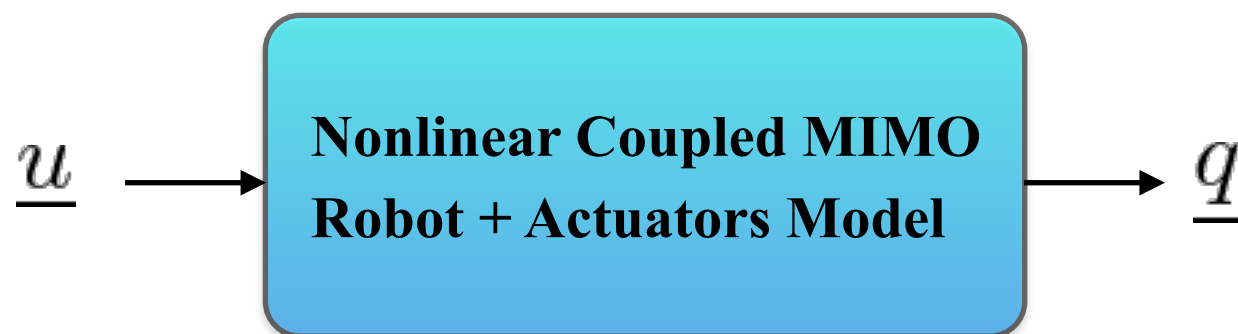
System state: $[\underline{\tilde{q}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{q}} = \underline{q_d} - \underline{q}$

constant equilibrium configuration

Lyapunov function candidate:

(n x n) symmetric positive definite matrix

$$V(\underline{\dot{q}}, \underline{\tilde{q}}) = \frac{1}{2} \underline{\dot{q}}^T B(\underline{q}) \underline{\dot{q}} + \frac{1}{2} \underline{\tilde{q}}^T K_P \underline{\tilde{q}} > 0 \quad \forall \underline{\dot{q}}, \underline{\tilde{q}} \neq 0$$



Motion Control: Control in the Joint Space

Centralized Approach

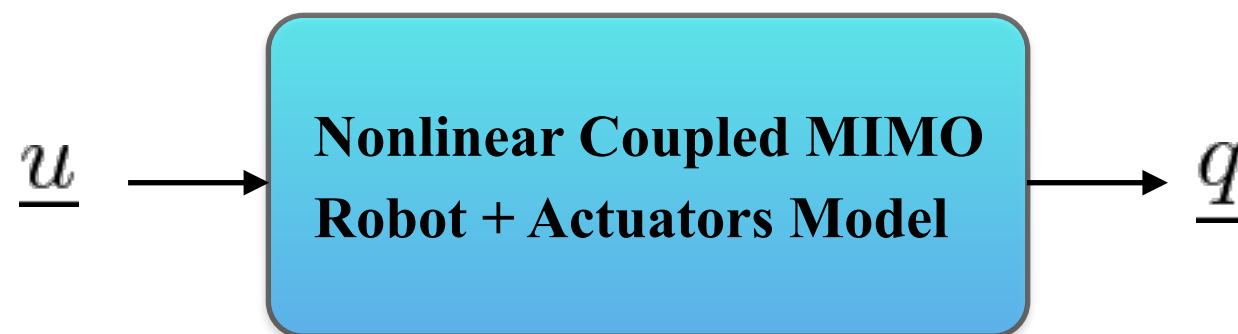
P and PD Control with Gravity Compensation

System state: $[\underline{\tilde{q}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{q}} = \underline{q_d} - \underline{q}$ constant equilibrium configuration

Lyapunov function candidate:

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$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T K_P \underline{\tilde{q}}$$



Motion Control: Control in the Joint Space

Centralized Approach

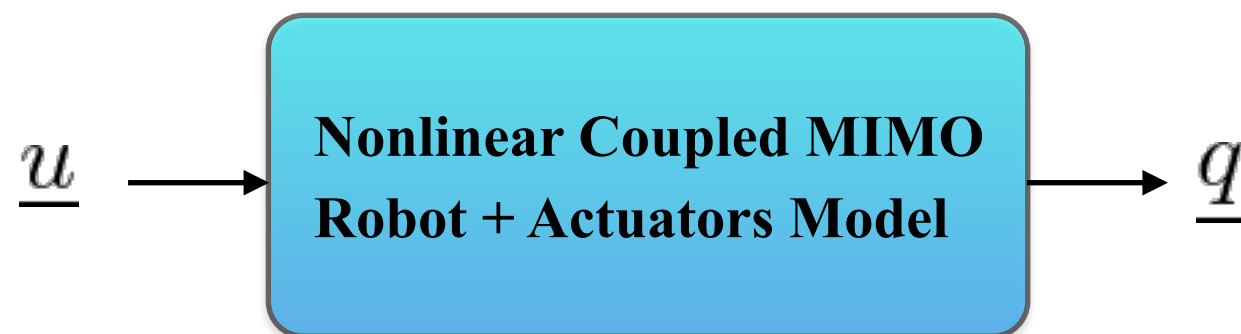
P and PD Control with Gravity Compensation

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$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} - \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T K_P \underline{\tilde{q}}$$



Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

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Skew symmetry property:

$$N(\underline{q}, \underline{\dot{q}}) = \dot{B}(\underline{q}) - 2C(\underline{q}, \underline{\dot{q}}) \quad \text{s.t.} \quad \underline{w}^T N(\underline{q}, \underline{\dot{q}}) \underline{w} = 0$$

Direct implication: $\underline{w} = \underline{\dot{q}} \longrightarrow \underline{\dot{q}}^T N(\underline{q}, \underline{\dot{q}}) \underline{\dot{q}} = 0$

Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

$$\begin{aligned} B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) &= \underline{u} \\ F &= F_v + K_r K_t R_a^{-1} K_v K_r \\ \underline{u} &= K_r K_t R_a^{-1} G_v \underline{v}_c \end{aligned}$$

$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} - \frac{1}{2}\dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

Skew symmetry property:

$$N(\underline{q}, \dot{\underline{q}}) = \dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}) \quad \text{s.t.} \quad \underline{w}^T N(\underline{q}, \dot{\underline{q}}) \underline{w} = 0$$

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Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

$$\begin{aligned}
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 F &= F_v + K_r K_t R_a^{-1} K_v K_r \\
 \underline{u} &= K_r K_t R_a^{-1} G_v \underline{v}_c
 \end{aligned}$$

$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} - \frac{1}{2} \dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

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Motion Control: Control in the Joint Space

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$$\dot{V} = \frac{1}{2}\dot{\underline{q}}^T (\dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}))\dot{\underline{q}} - \dot{\underline{q}}^T F\dot{\underline{q}} + \dot{\underline{q}}^T (\underline{u} - \underline{g}(\underline{q}) - K_P \tilde{\underline{q}})$$

Skew symmetry property:

$$N(\underline{q}, \dot{\underline{q}}) = \dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}) \quad \text{s.t.} \quad \underline{w}^T N(\underline{q}, \dot{\underline{q}}) \underline{w} = 0$$

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Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

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$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} + \frac{1}{2}\dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

$$\dot{V} = \frac{1}{2}\dot{\underline{q}}^T (\dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}))\dot{\underline{q}} - \dot{\underline{q}}^T F\dot{\underline{q}} + \dot{\underline{q}}^T (\underline{u} - \underline{g}(\underline{q}) - K_P \tilde{\underline{q}})$$

Skew symmetry property:

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Direct implication:

$$\underline{w} = \dot{\underline{q}}$$



$$\dot{\underline{q}}^T N(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} = 0$$

Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} + \frac{1}{2}\dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

The second term is negative definite

$$\dot{V} = \frac{1}{2}\dot{\underline{q}}^T (\dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}))\dot{\underline{q}} - \dot{\underline{q}}^T F\dot{\underline{q}} + \dot{\underline{q}}^T (\underline{u} - \underline{g}(\underline{q}) - K_P \tilde{\underline{q}})$$

Skew symmetry property:

$$N(\underline{q}, \dot{\underline{q}}) = \dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}) \quad \text{s.t.} \quad \underline{w}^T N(\underline{q}, \dot{\underline{q}}) \underline{w} = 0$$

Direct implication: $\underline{w} = \dot{\underline{q}} \longrightarrow \dot{\underline{q}}^T N(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} = 0$

Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

$$\begin{aligned} B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) &= \underline{u} \\ F &= F_v + K_r K_t R_a^{-1} K_v K_r \\ \underline{u} &= K_r K_t R_a^{-1} G_v \underline{v}_c \end{aligned}$$

$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} + \frac{1}{2}\dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

$$\dot{V} = \frac{1}{2}\dot{\underline{q}}^T (\dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}))\dot{\underline{q}} - \dot{\underline{q}}^T F\dot{\underline{q}} + \dot{\underline{q}}^T (\underline{u} - \underline{g}(\underline{q}) - K_P \tilde{\underline{q}})$$

a control law with compensation of gravitational terms and a proportional action, leads to a negative semi-definite $\dot{V}$


$$\underline{u} = \underline{g}(\underline{q}) + K_P \tilde{\underline{q}}$$

Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

$$\dot{V} = \dot{\underline{q}}^T B(\underline{q})\ddot{\underline{q}} + \frac{1}{2}\dot{\underline{q}}^T \dot{B}(\underline{q})\dot{\underline{q}} - \dot{\underline{q}}^T K_P \tilde{\underline{q}}$$

$$\dot{V} = \frac{1}{2}\dot{\underline{q}}^T (\dot{B}(\underline{q}) - 2C(\underline{q}, \dot{\underline{q}}))\dot{\underline{q}} - \dot{\underline{q}}^T F\dot{\underline{q}} + \dot{\underline{q}}^T (\underline{u} - \underline{g}(\underline{q}) - K_P \tilde{\underline{q}})$$

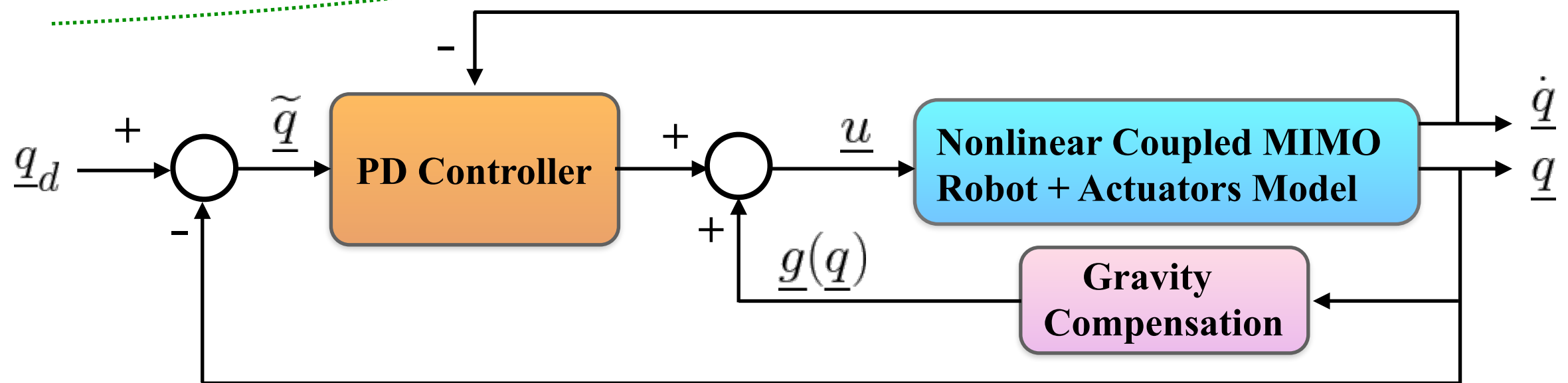
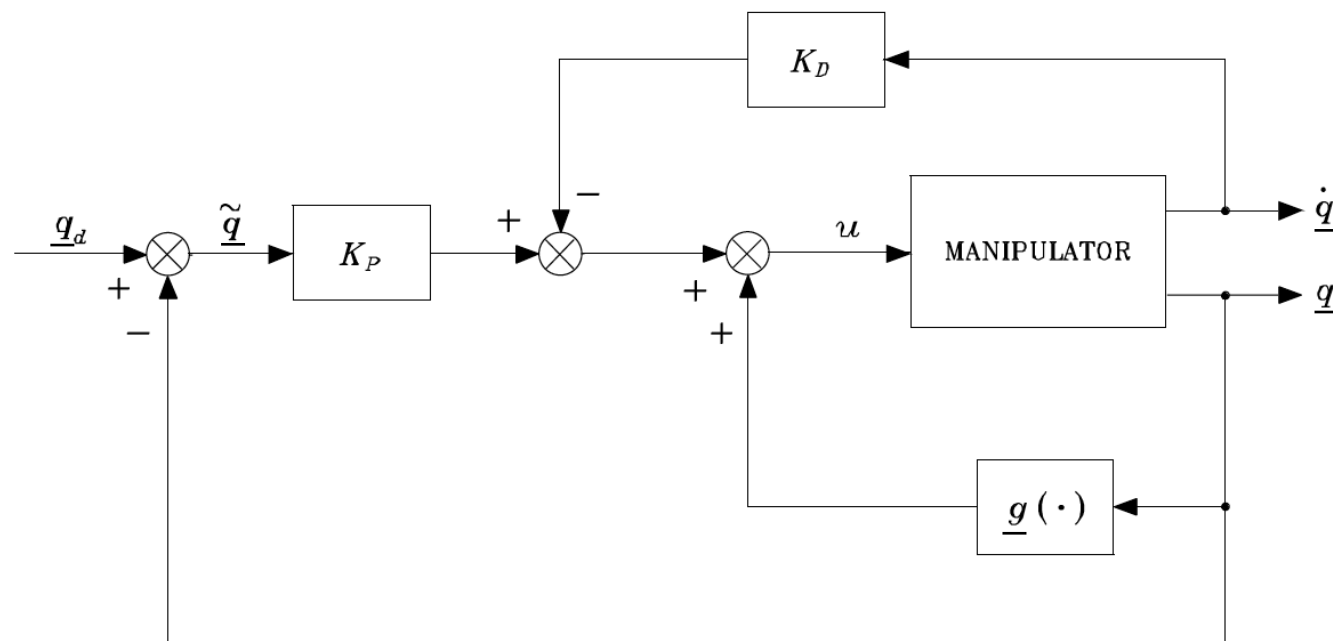
in alternative, a control law with compensation of gravitational terms and a proportional-derivative action, leads to the same result

$$\underline{u} = \underline{g}(\underline{q}) + K_P \tilde{\underline{q}} - K_D \dot{\underline{q}}$$

Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation



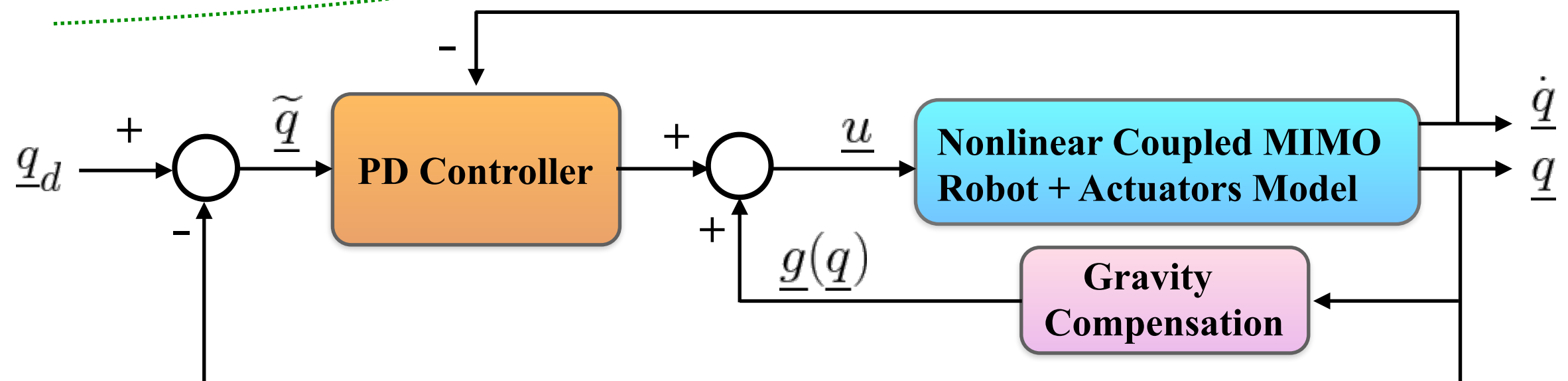
Motion Control: Control in the Joint Space

Centralized Approach

P and PD Control with Gravity Compensation

It can be shown that that any manipulator equilibrium configuration is *globally asymptotically stable* under a controller with a PD linear action and a nonlinear gravity compensating action.

The global asymptotic stability of the equilibrium configuration is ensured for any choice of K_P and K_D , as long as these are positive definite matrices





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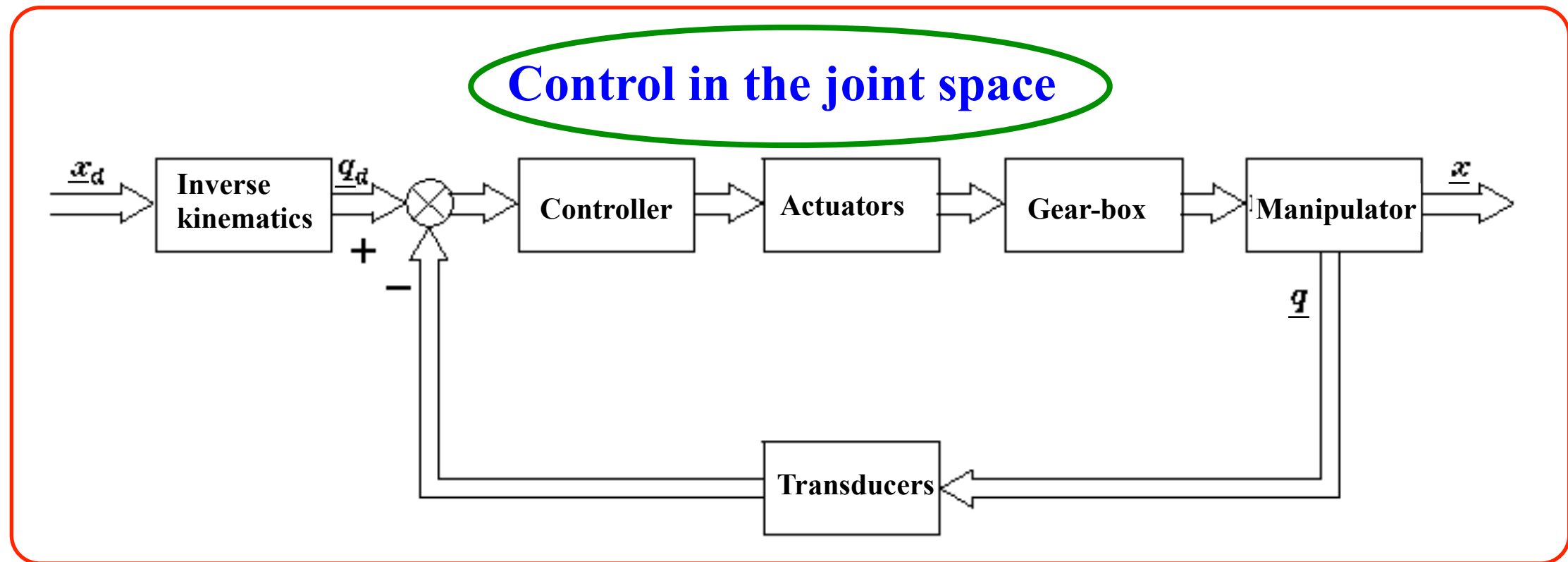
Course of Robot Control

Prof. Antonella Ferrara

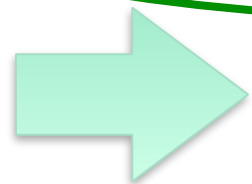


Inverse Dynamics Control

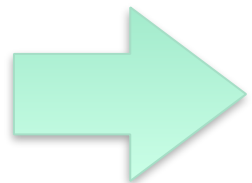
Motion Control: Control in the Joint Space



Centralized Control Approach in the Joint Space:



P and PD Control with Gravity Compensation



Inverse Dynamics Control

Motion Control: Control in the Joint Space

Centralized Approach. A Different Model Formulation

MIMO Dynamic Model of Robot Manipulator and Actuators



$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{\tau}$$

$$K_r \underline{q} = \underline{q}_m \quad \underline{\tau}_m = K_r^{-1} \underline{\tau} \quad \text{transmissions}$$

$$K_r^{-1} \underline{\tau} = K_t \dot{\underline{a}}$$

$$\underline{v}_a = R_a \dot{\underline{a}} + K_v \dot{\underline{q}}_m \quad n \text{ actuators}$$

$$\underline{v}_a = G_v \underline{v}_c.$$

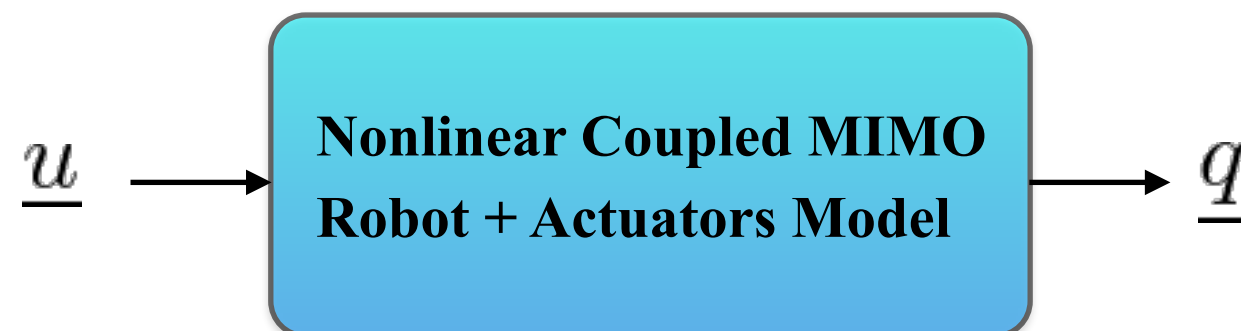
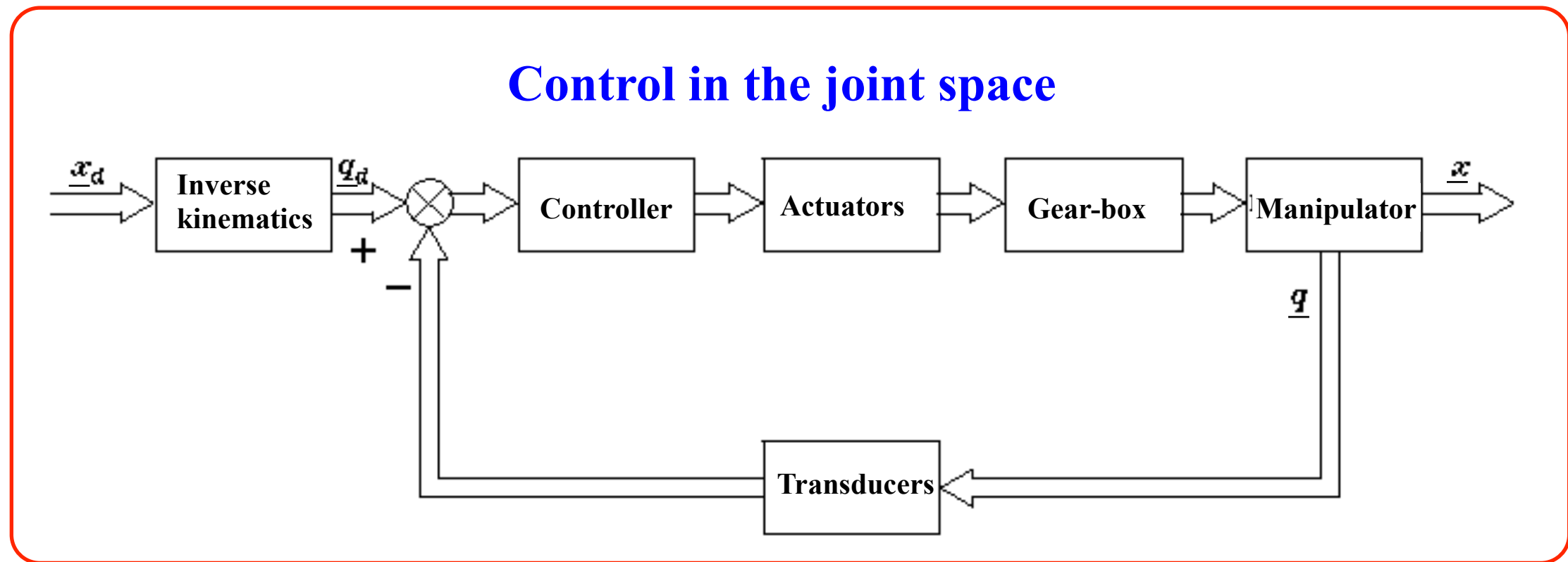
$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

Motion Control: Control in the Joint Space

Centralized Approach



Motion Control: Control in the Joint Space

Centralized Approach

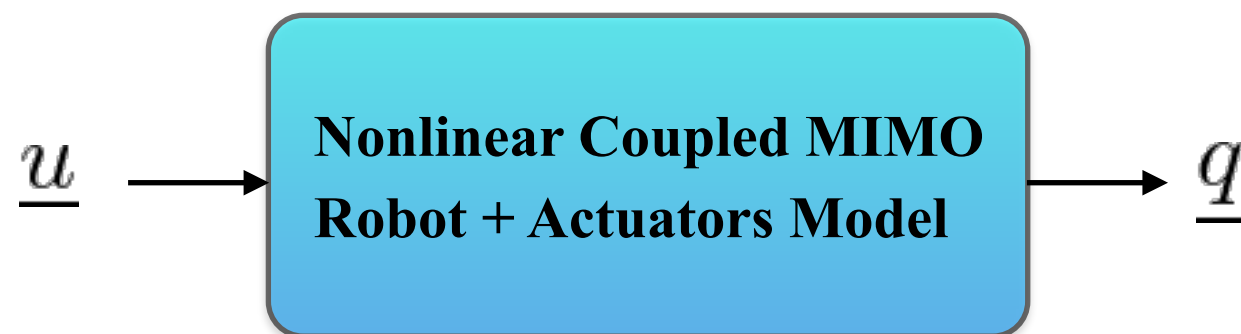
Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

$$\downarrow$$
$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$



Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

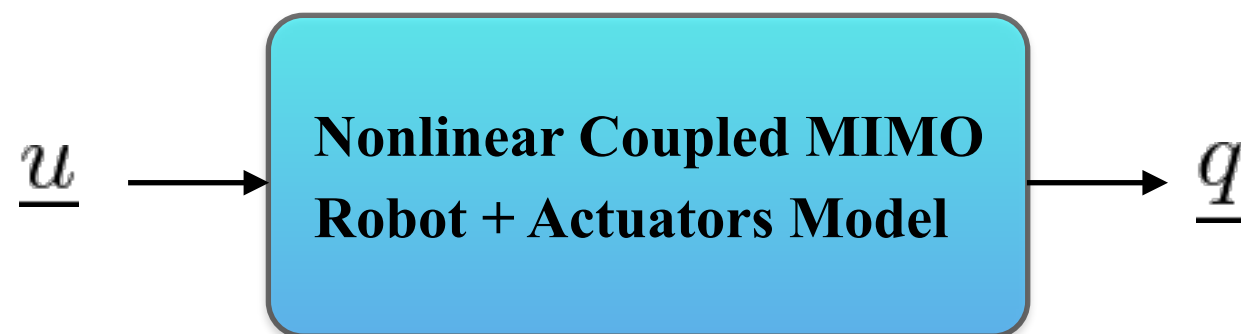
PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

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$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

find the control vector to exactly linearize the controlled system dynamics

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$



Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

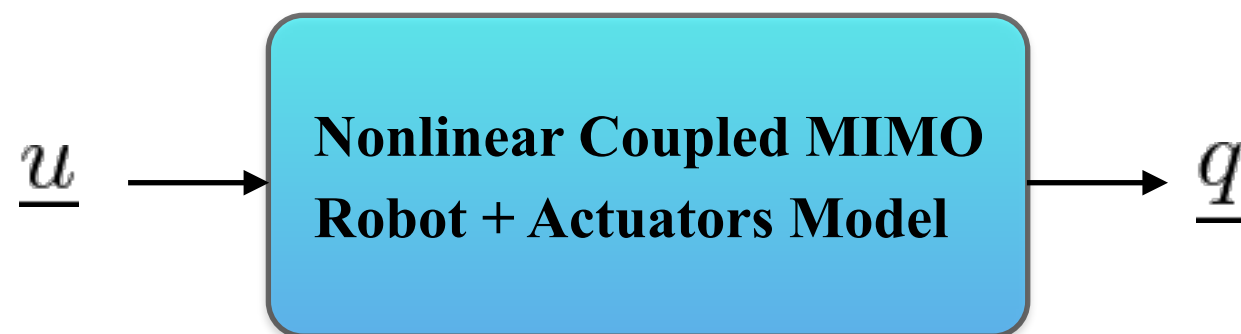
MODEL: dynamic model of an n -joint manipulator + actuators

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

find the control vector to exactly linearize the controlled system dynamics

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$

$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}})$$



$$\ddot{\underline{q}} = \underline{y}$$

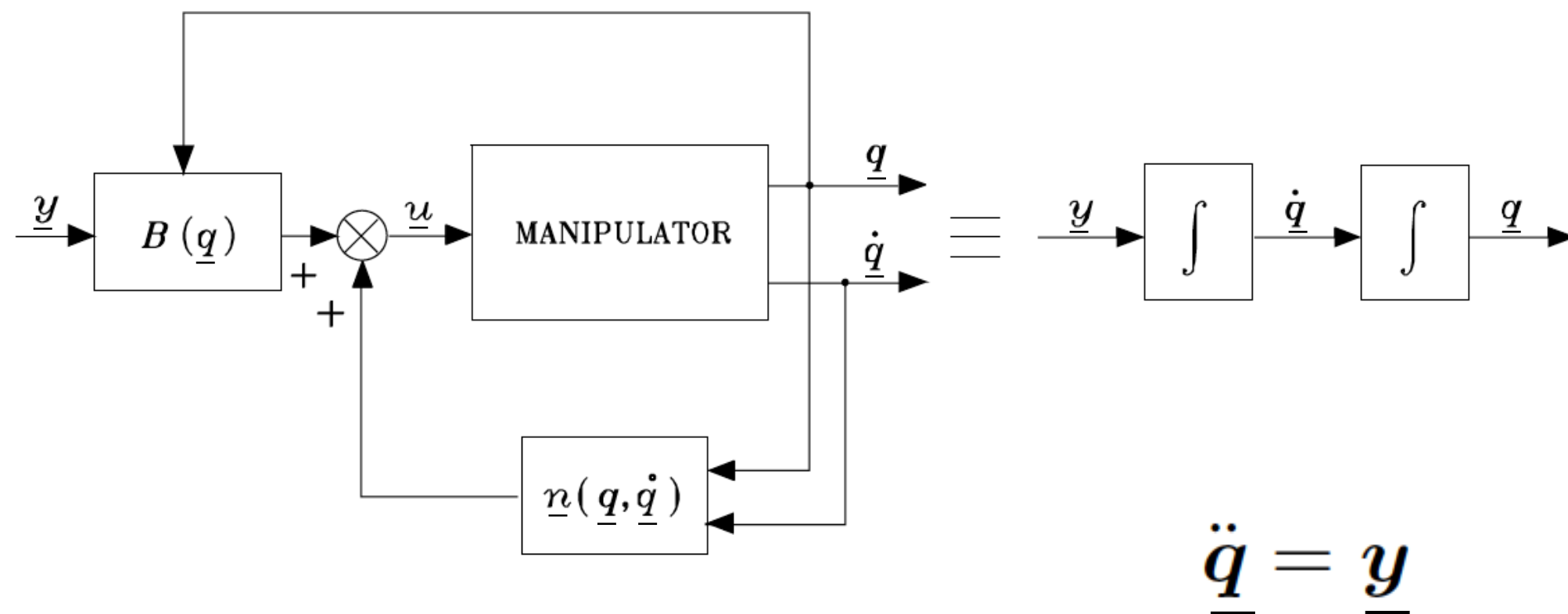
Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



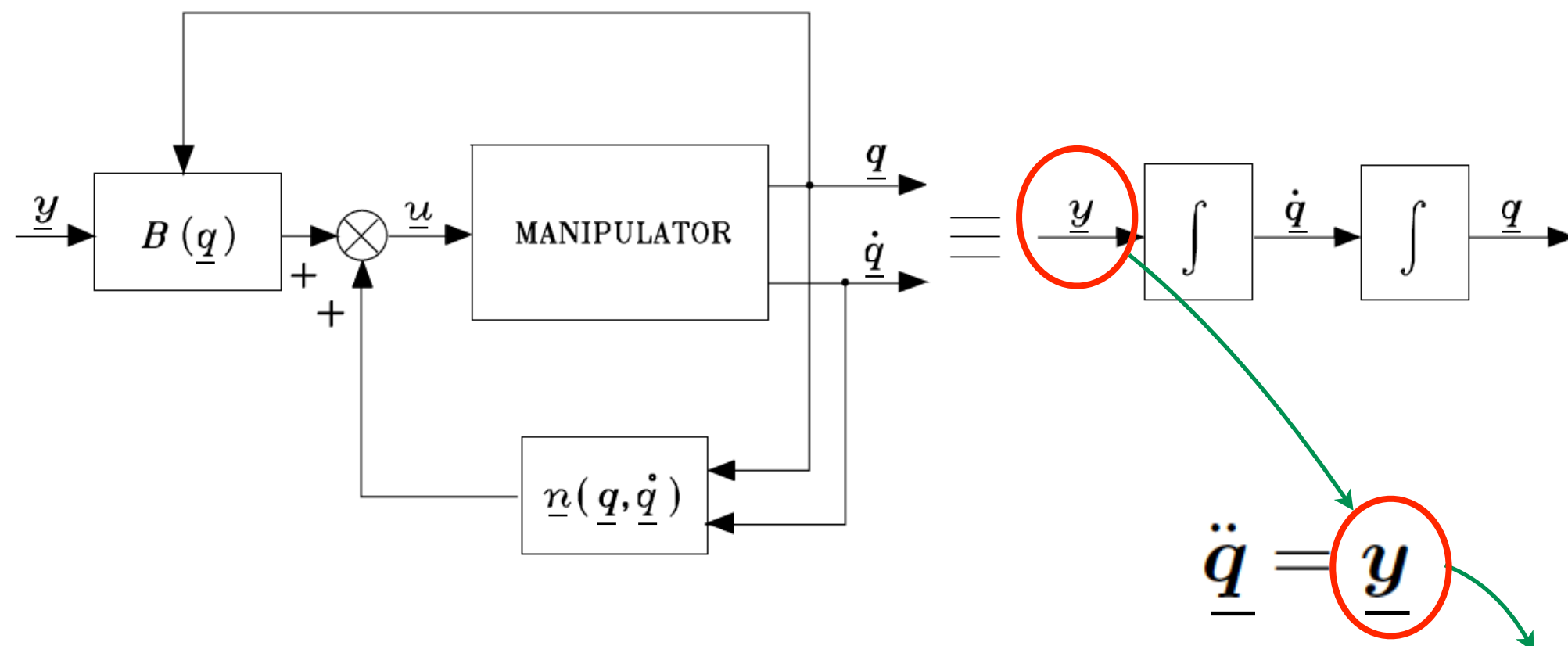
Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



auxiliary control still
to be designed

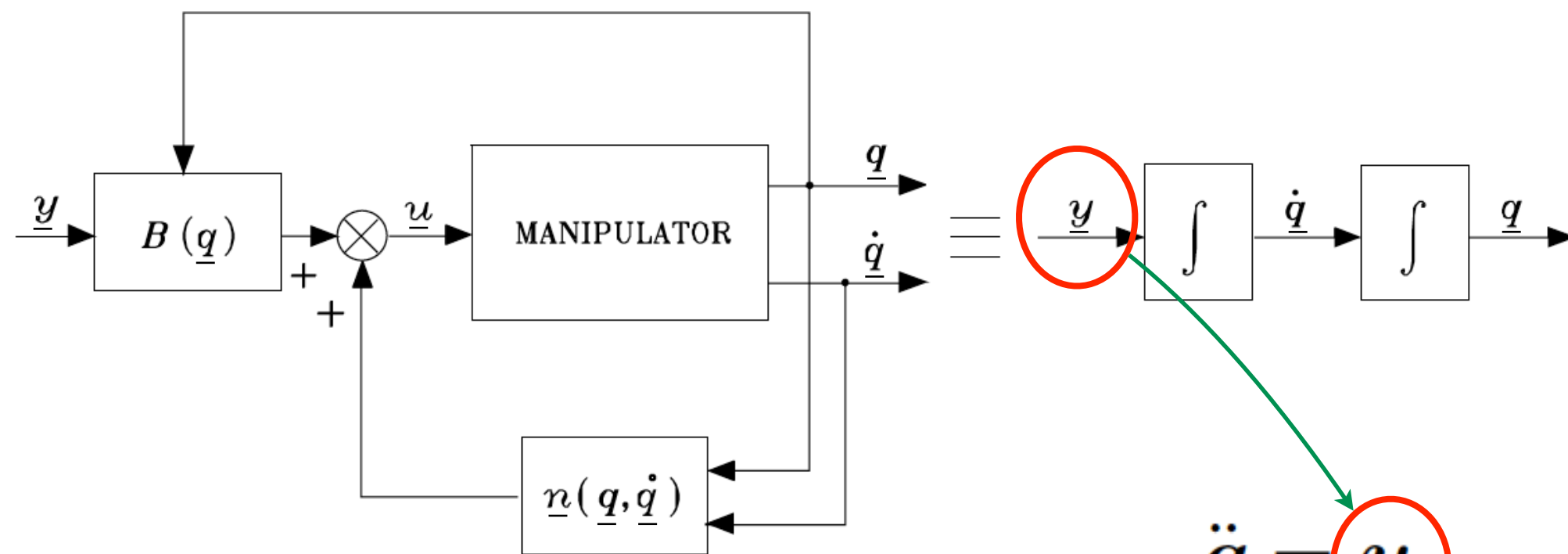
Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



$$\underline{y} = -\underline{K}_P \underline{q} - \underline{K}_D \underline{\dot{q}} + \underline{r}$$

auxiliary control still
to be designed

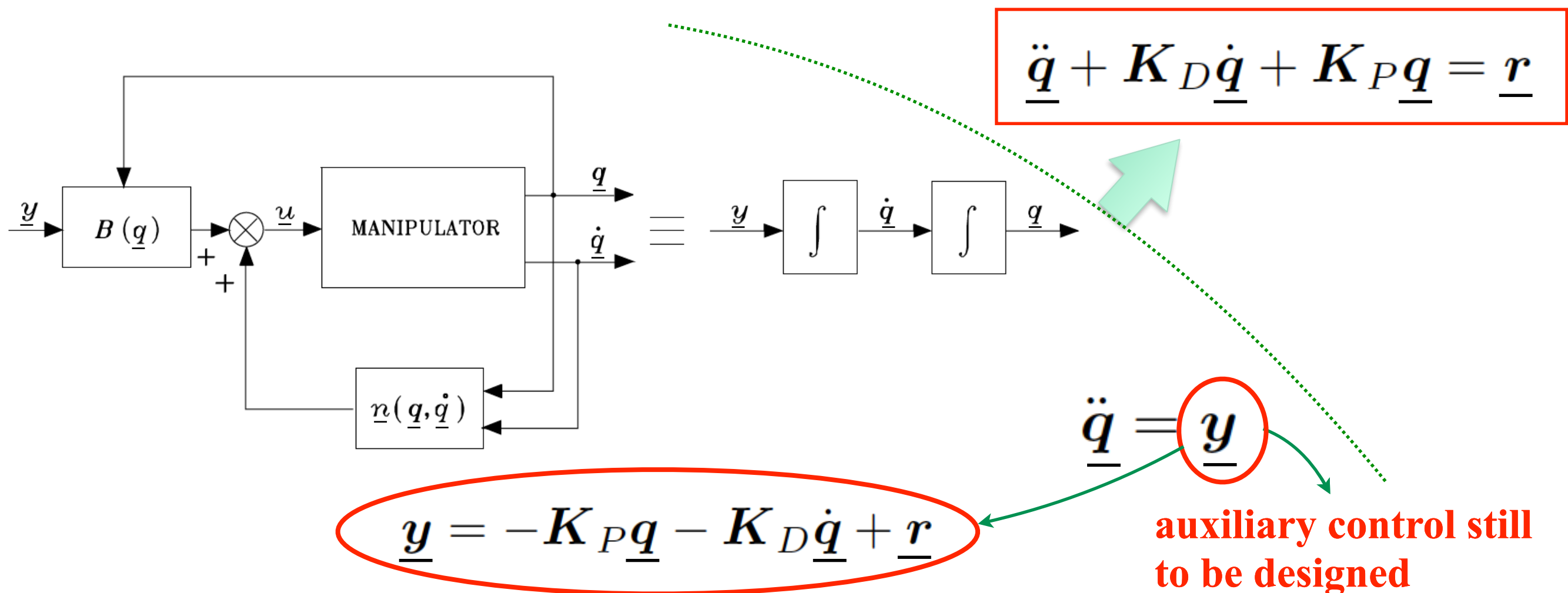
Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\underline{\ddot{q}} = \underline{y}$$

$$\underline{y} = -\underline{K}_P \underline{q} - \underline{K}_D \underline{\dot{q}} + \underline{r}$$

$$\underline{\ddot{q}} + \underline{K}_D \underline{\dot{q}} + \underline{K}_P \underline{q} = \underline{r}$$

$$\underline{\ddot{\tilde{q}}} + \underline{K}_D \underline{\dot{\tilde{q}}} + \underline{K}_P \underline{\tilde{q}} = \underline{0}$$



$$\underline{K}_P = \text{diag}\{\omega_{n1}^2, \dots, \omega_{nn}^2\} \quad \underline{K}_D = \text{diag}\{2\zeta_1\omega_{n1}, \dots, 2\zeta_n\omega_{nn}\}$$

$$\underline{r} = \underline{\ddot{q}}_d + \underline{K}_D \underline{\dot{q}}_d + \underline{K}_P \underline{q}_d$$

Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\ddot{\underline{q}} = \underline{y}$$

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$$\ddot{\underline{q}} + \underline{K}_D \dot{\underline{q}} + \underline{K}_P \underline{q} = \underline{r}$$

$$\underline{K}_P = \text{diag}\{\omega_{n1}^2, \dots, \omega_{nn}^2\} \quad \underline{K}_D = \text{diag}\{2\zeta_1\omega_{n1}, \dots, 2\zeta_n\omega_{nn}\}$$

$$\underline{r} = \ddot{\underline{q}}_d + \underline{K}_D \dot{\underline{q}}_d + \underline{K}_P \underline{q}_d$$

$$\ddot{\underline{\tilde{q}}} + \underline{K}_D \dot{\underline{\tilde{q}}} + \underline{K}_P \underline{\tilde{q}} = \underline{0}$$

Positive definite
diagonal matrices

Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\underline{\ddot{q}} = \underline{y}$$

$$\underline{y} = -\underline{K}_P \underline{q} - \underline{K}_D \underline{\dot{q}} + \underline{r}$$

$$\underline{\ddot{q}} + \underline{K}_D \underline{\dot{q}} + \underline{K}_P \underline{q} = \underline{r}$$

$$\underline{\ddot{\tilde{q}}} + \underline{K}_D \underline{\dot{\tilde{q}}} + \underline{K}_P \underline{\tilde{q}} = \underline{0}$$

$$\underline{\tilde{q}} = \underline{q}_d(t) - \underline{q}$$

$$\underline{K}_P = \text{diag}\{\omega_{n1}^2, \dots, \omega_{nn}^2\} \quad \underline{K}_D = \text{diag}\{2\zeta_1\omega_{n1}, \dots, 2\zeta_n\omega_{nn}\}$$

$$\underline{r} = \underline{\ddot{q}}_d + \underline{K}_D \underline{\dot{q}}_d + \underline{K}_P \underline{q}_d$$

Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\ddot{\underline{\tilde{q}}} + \underline{K}_D \dot{\underline{\tilde{q}}} + \underline{K}_P \underline{\tilde{q}} = \underline{0}$$



$$\lim_{t \rightarrow \infty} \underline{\tilde{q}} = \lim_{t \rightarrow \infty} (\underline{q}_d(t) - \underline{q}) = 0$$

Motion Control: Control in the Joint Space

Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\ddot{\underline{\tilde{q}}} + \underline{K}_D \dot{\underline{\tilde{q}}} + \underline{K}_P \underline{\tilde{q}} = \underline{0}$$



$$\lim_{t \rightarrow \infty} \underline{\tilde{q}} = \lim_{t \rightarrow \infty} (\underline{q}_d(t) - \underline{q}) = 0$$

Motion Control: Control in the Joint Space

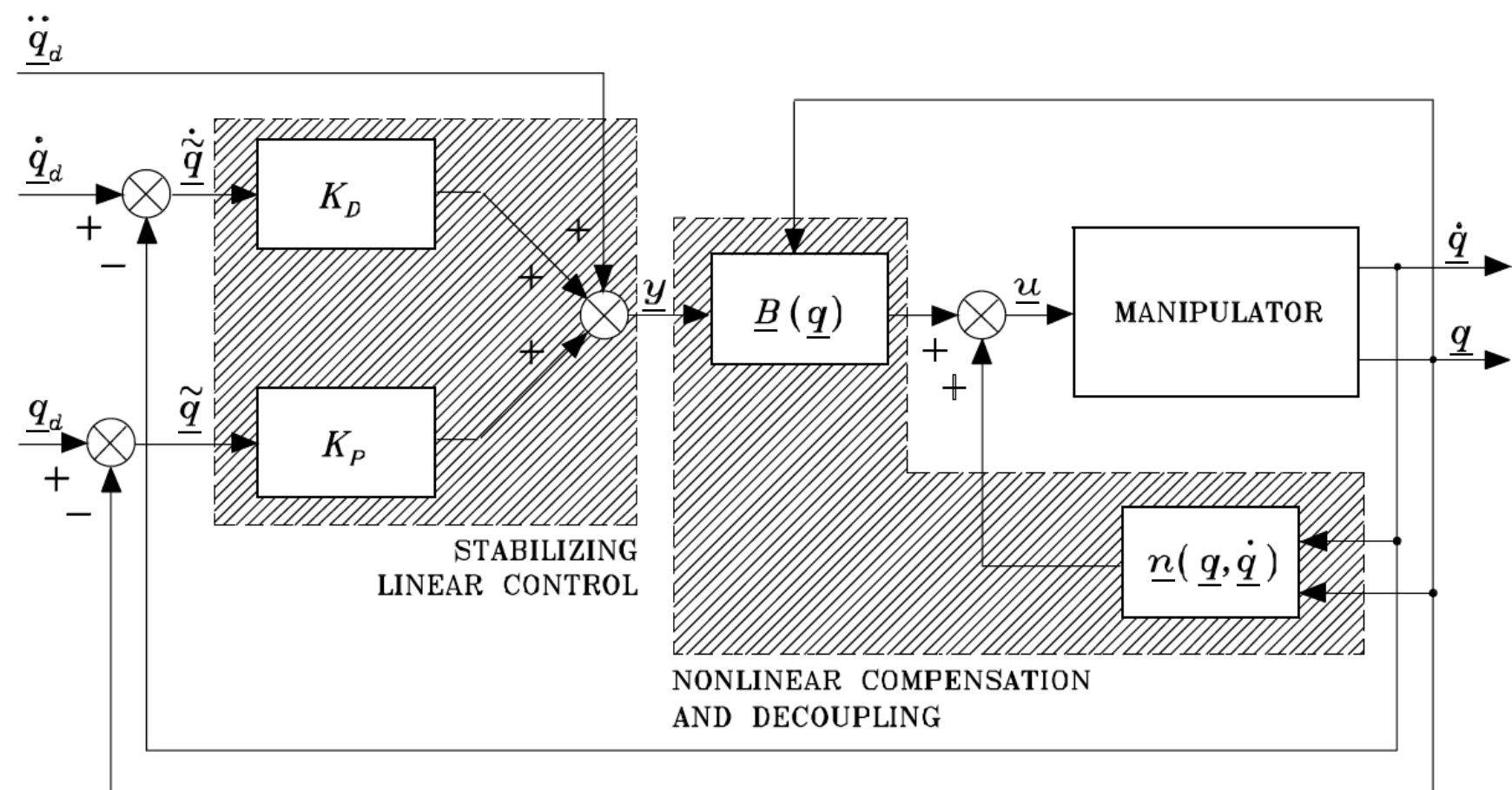
Centralized Approach

Inverse Dynamics Control

PROBLEM: tracking a joint space trajectory $\underline{q}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

FINAL CONTROL SCHEME:





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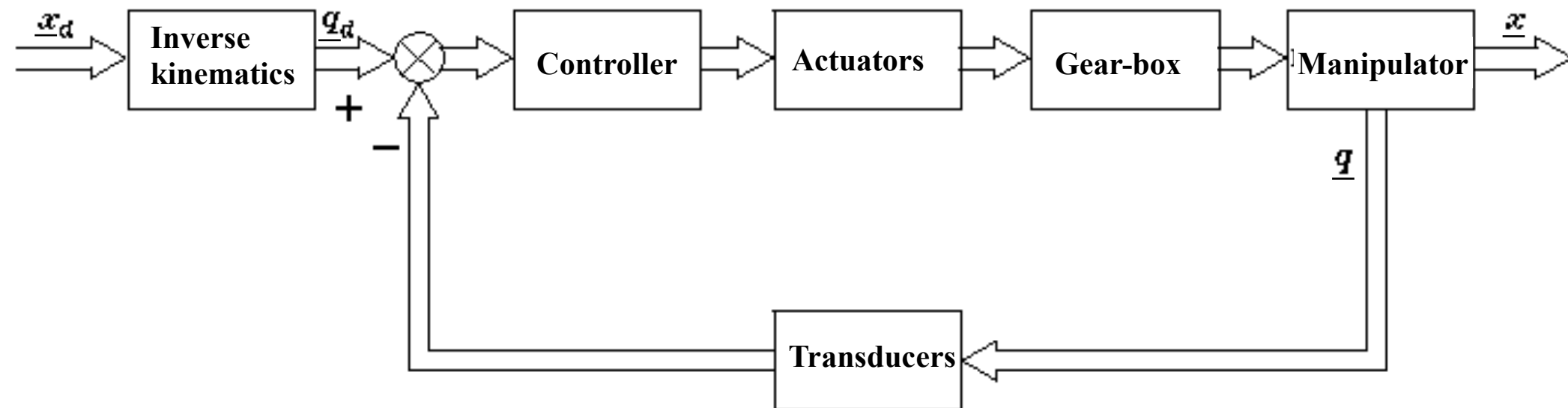


Control in the Operational Space

Motion Control: Control in the Joint Space

In previous lectures...

Control in the joint space



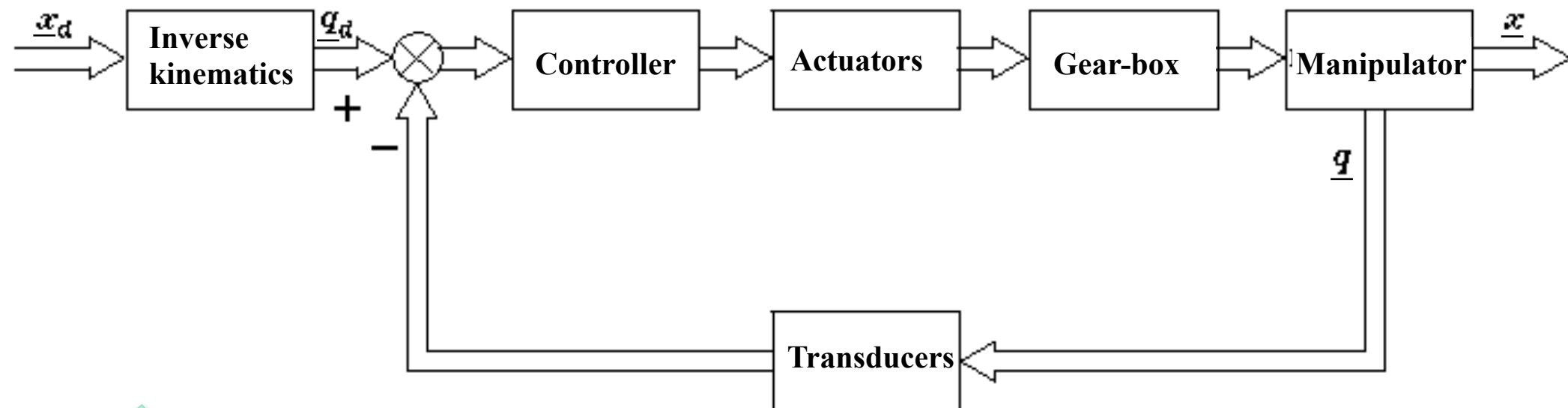
➔ **decentralized control (independent control of the joints)**

- *decentralized control with decentralized feedforward compensation*
- *decentralized control with centralized computed torque based feedforward compensation*

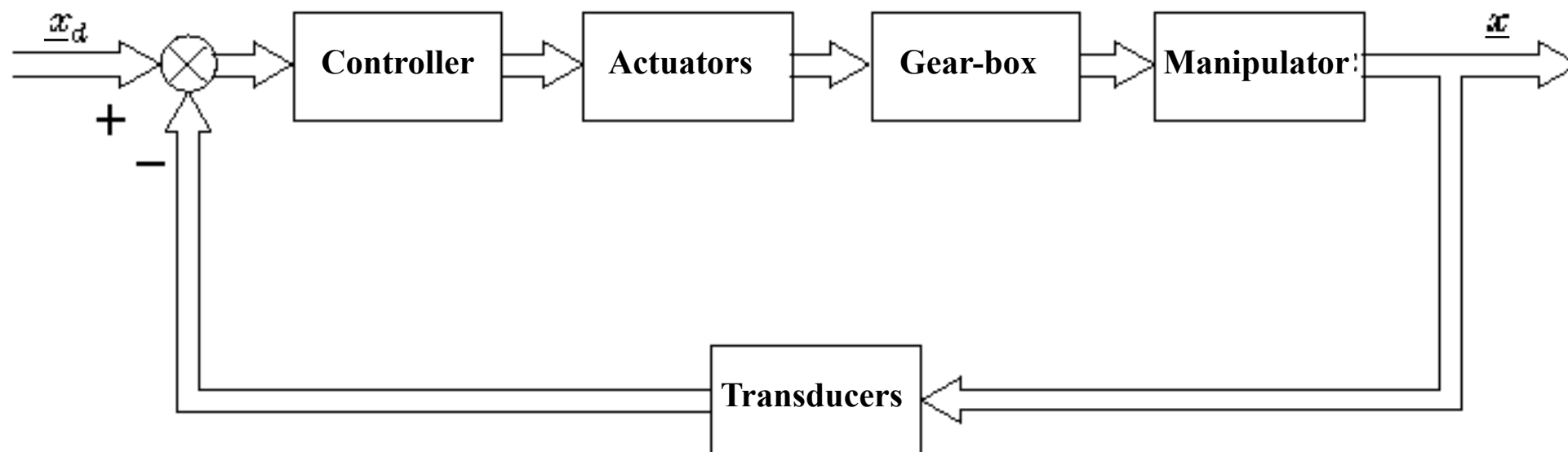
➔ **centralized control of the joints**

Motion Control of Robot Manipulators

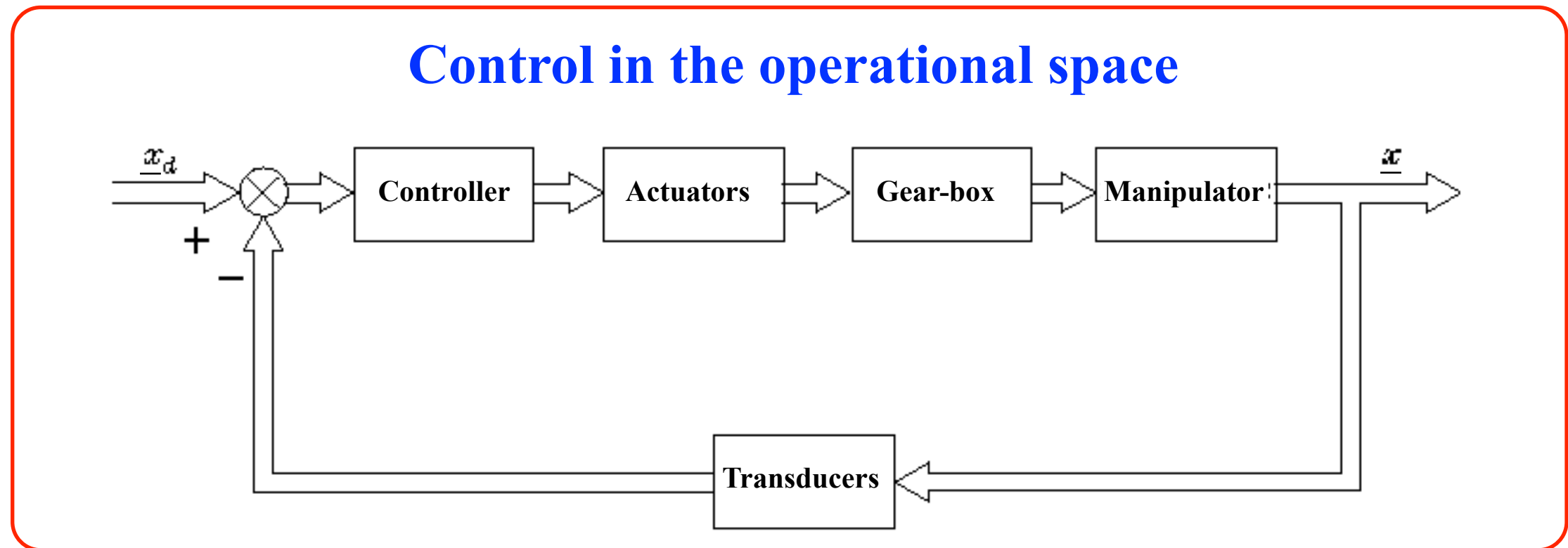
Control in the joint space



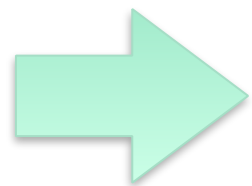
Control in the operational space



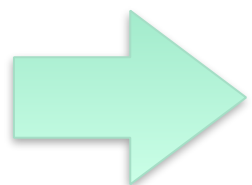
Motion Control of Robot Manipulators



Centralized Control Approach in the Operational Space:

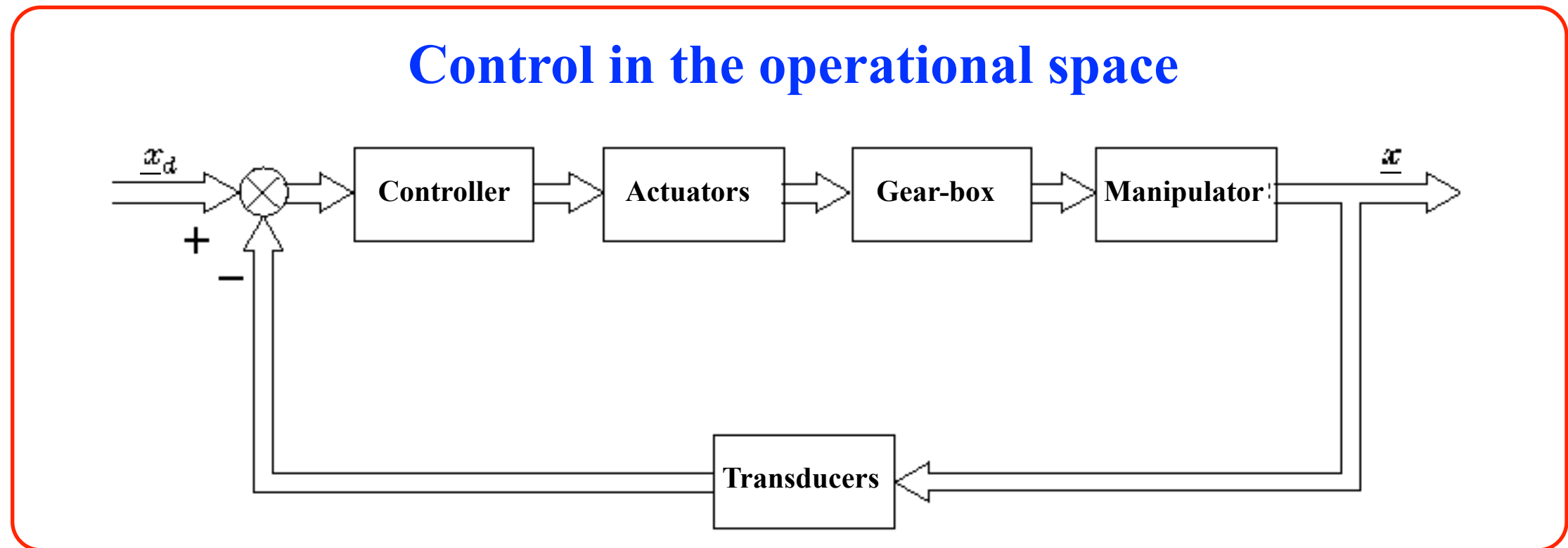


PD Control with Gravity Compensation in the Operational Space

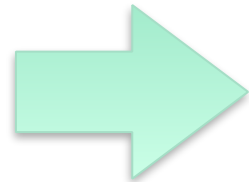


Inverse Dynamics Control in the Operational Space

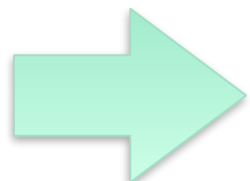
Motion Control of Robot Manipulators



Centralized Control Approach in the Operational Space:



PD Control with Gravity Compensation in the Operational Space



Inverse Dynamics Control in the Operational Space



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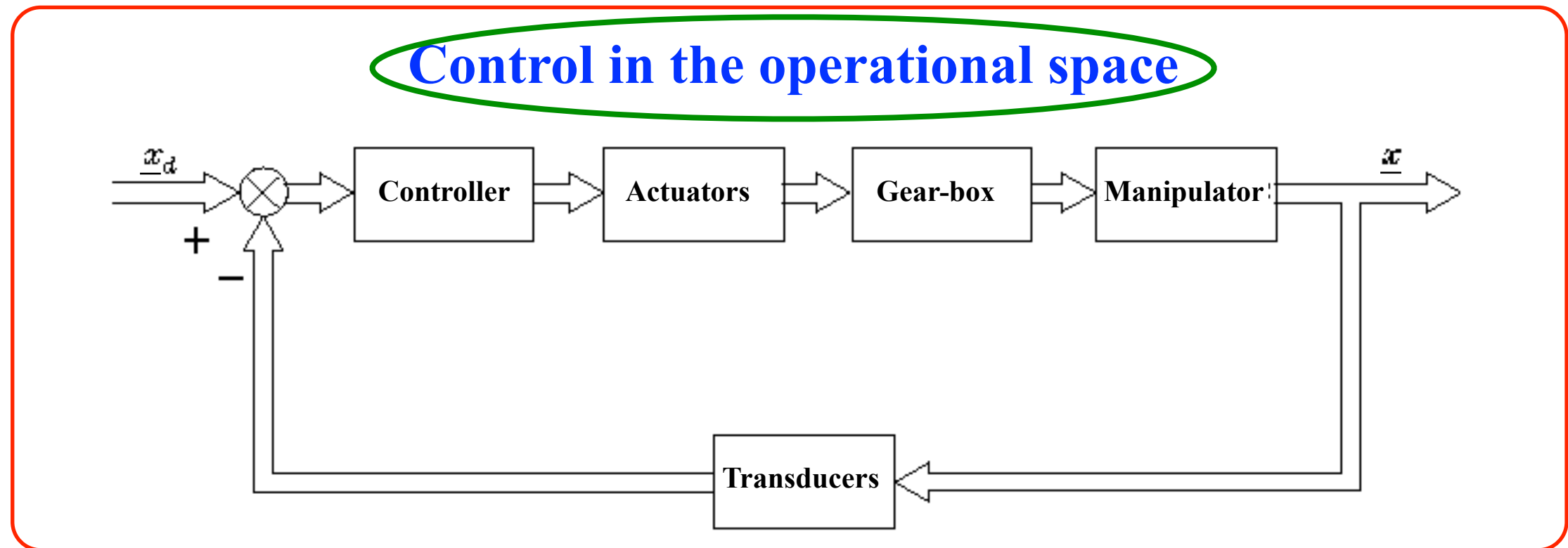
Prof. Antonella Ferrara



PD Control with Gravity Compensation in the Operational Space

Motion Control of Robot Manipulators

In this Lecture...



Centralized Control Approach in the Operational Space:

PD Control with Gravity Compensation in the Operational Space

Inverse Dynamics Control in the Operational Space

Motion Control in the Operational Space

Centralized Approach

MIMO Dynamic Model of Robot Manipulator and Actuators



$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{\tau}$$

$$K_r \underline{q} = \underline{q}_m \quad \underline{\tau}_m = K_r^{-1} \underline{\tau} \quad \text{transmissions}$$

$$K_r^{-1} \underline{\tau} = K_t \dot{\underline{a}}$$

$$\underline{v}_a = R_a \dot{\underline{a}} + K_v \dot{\underline{q}}_m \quad n \text{ actuators}$$

$$\underline{v}_a = G_v \underline{v}_c$$

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

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$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

Motion Control in the Operational Space

Centralized Approach

MIMO Dynamic Model of Robot Manipulator and Actuators



$$\underline{B}(\underline{q})\ddot{\underline{q}} + \underline{C}(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + \underline{F}\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

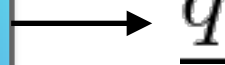
$$\underline{F} = \underline{F}_v + \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{K}_v \underline{K}_r$$

$$\underline{u} = \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{G}_v \underline{v}_c$$

$\underline{u}$



**Nonlinear Coupled MIMO
Robot + Actuators Model**



$\underline{q}$

Motion Control in the Operational Space

Centralized Approach

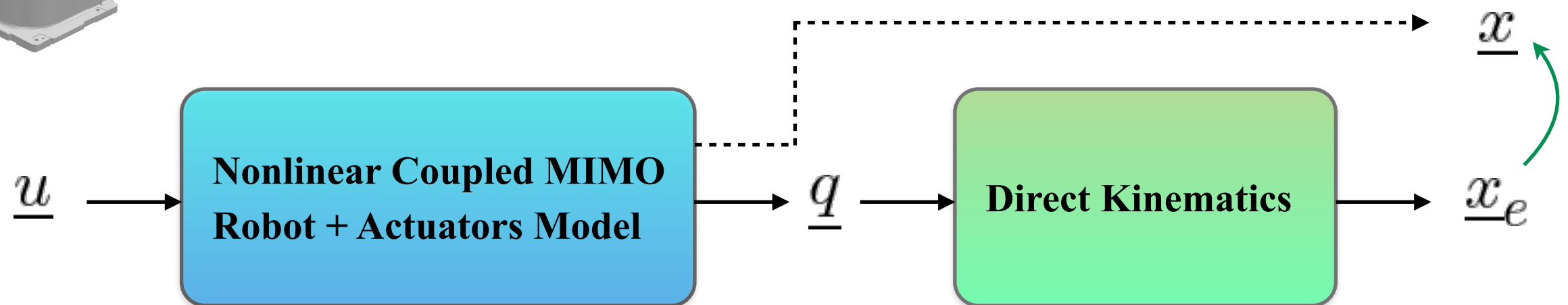
MIMO Dynamic Model of Robot Manipulator and Actuators



$$\underline{B}(\underline{q})\ddot{\underline{q}} + \underline{C}(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + \underline{F}\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$\underline{F} = \underline{F}_v + \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{K}_v \underline{K}_r$$

$$\underline{u} = \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{G}_v \underline{v}_c$$



Motion Control in the Operational Space

Centralized Approach

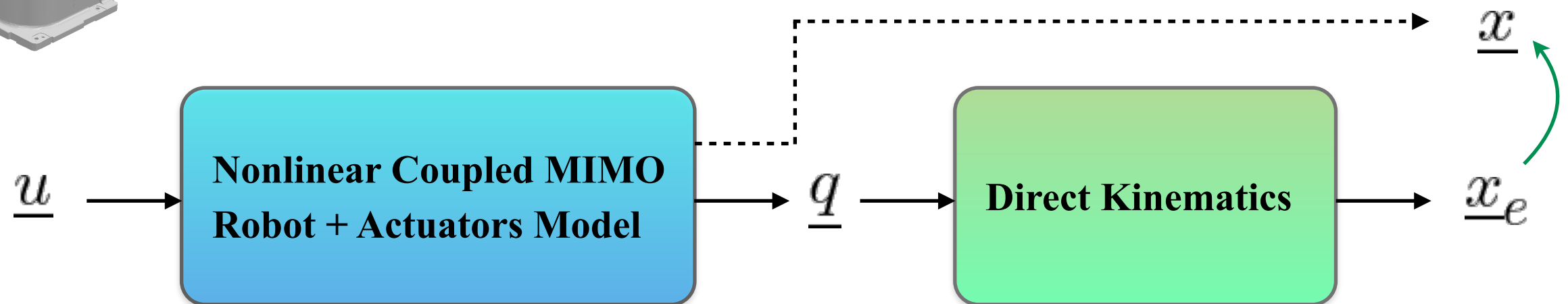
MIMO Dynamic Model of Robot Manipulator and Actuators



$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$



Note that having a direct measure of vector $\underline{x}$ is quite infrequent

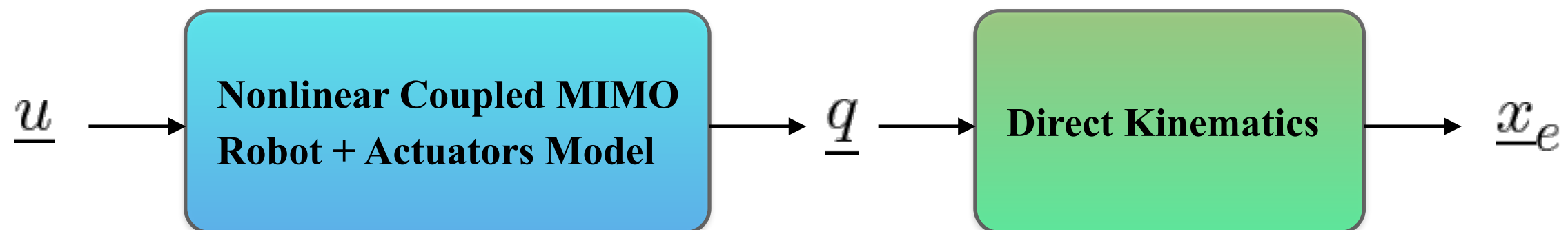
Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$

$\underline{x}_d$ is circled in red, with a green arrow pointing to the text "constant equilibrium pose".



Motion Control in the Operational Space

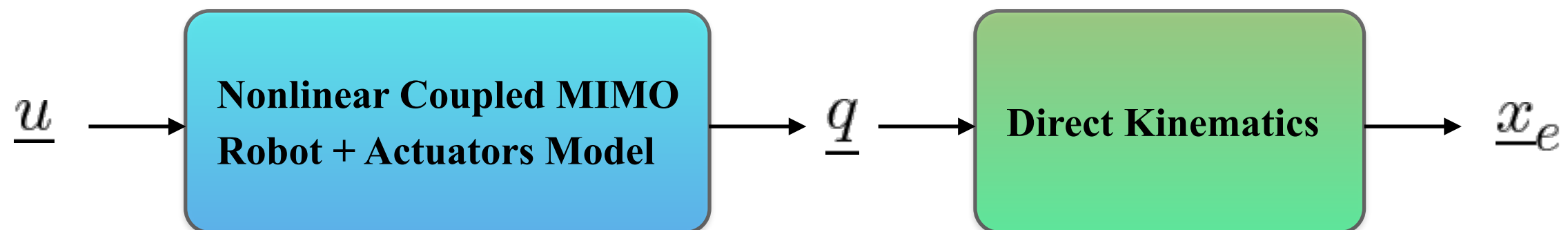
Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$

constant equilibrium pose

PROBLEM: find the structure of the controller which guarantees the global asymptotic stability of the equilibrium pose



Motion Control in the Operational Space

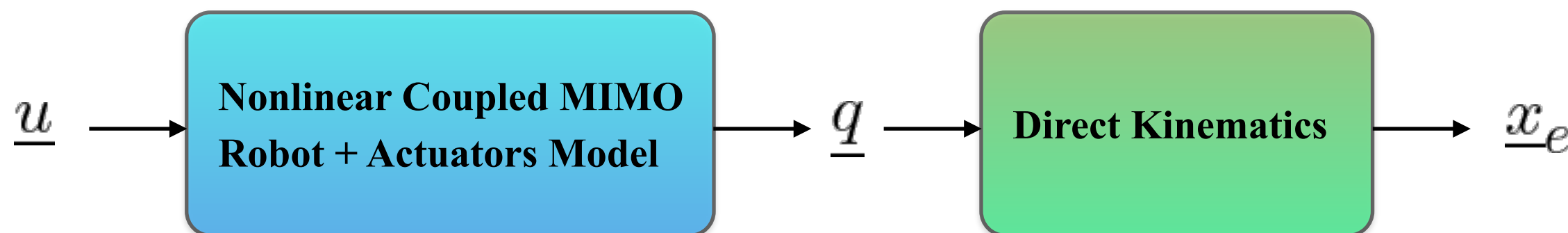
Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

Lyapunov function candidate:

$$V(\underline{\dot{q}}, \underline{\tilde{x}}) = \frac{1}{2} \underline{\dot{q}}^T B(\underline{q}) \underline{\dot{q}} + \frac{1}{2} \underline{\tilde{x}}^T K_P \underline{\tilde{x}} > 0 \quad \forall \underline{\dot{q}}, \underline{\tilde{x}} \neq \underline{0},$$



Motion Control in the Operational Space

Centralized Approach

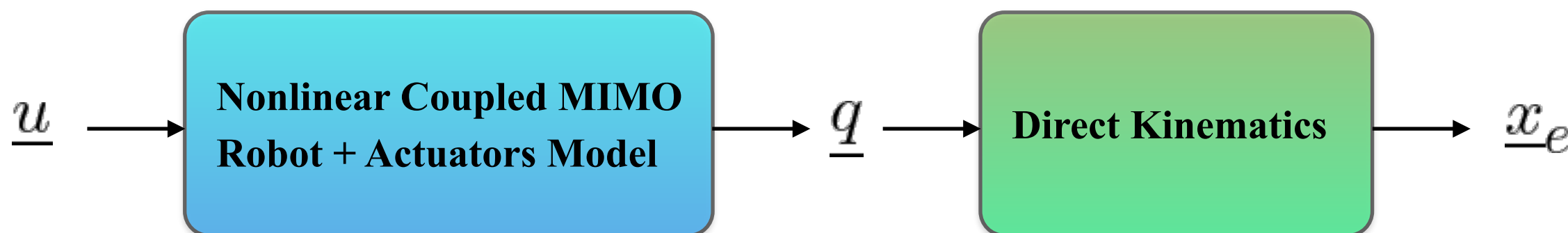
PD Control with Gravity Compensation in the Operational Space

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$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} + \underline{\dot{\tilde{x}}}^T K_P \underline{\tilde{x}}$$



Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

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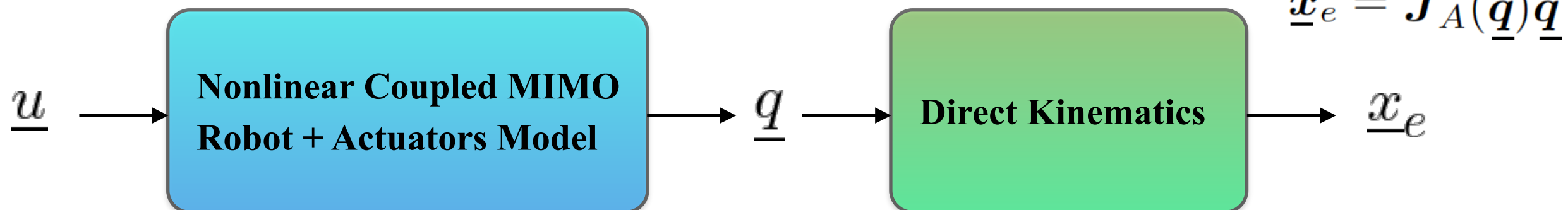
$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} + \underline{\dot{\tilde{x}}}^T K_P \underline{\tilde{x}}$$

?

since

$$\underline{\dot{x}}_d = \underline{0}$$

$$\underline{\dot{x}}_e = J_A(\underline{q}) \underline{\dot{q}}$$



Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

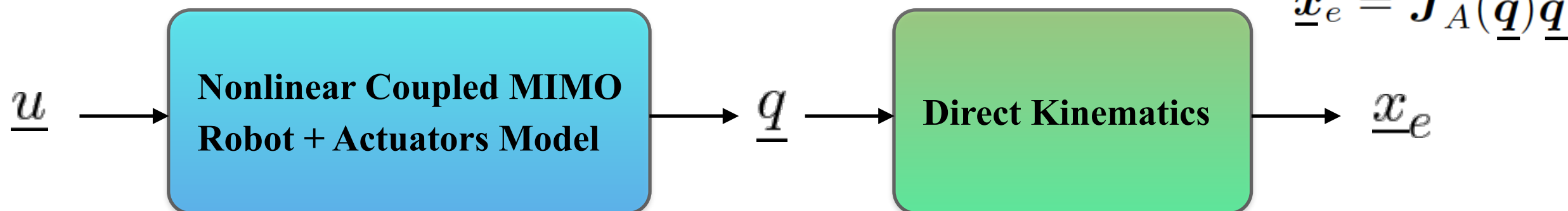
System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

Lyapunov function candidate:

$$V(\underline{\dot{q}}, \underline{\tilde{x}}) = \frac{1}{2} \underline{\dot{q}}^T B(\underline{q}) \underline{\dot{q}} + \frac{1}{2} \underline{\tilde{x}}^T K_P \underline{\tilde{x}} > 0 \quad \forall \underline{\dot{q}}, \underline{\tilde{x}} \neq \underline{0},$$

$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} + \underline{\dot{\tilde{x}}}^T K_P \underline{\tilde{x}}$$

$\underline{\dot{\tilde{x}}} = -J_A(\underline{q}) \underline{\dot{q}}$ since $\underline{\dot{x}}_d = \underline{0}$
 $\underline{\dot{x}}_e = J_A(\underline{q}) \underline{\dot{q}}$



Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

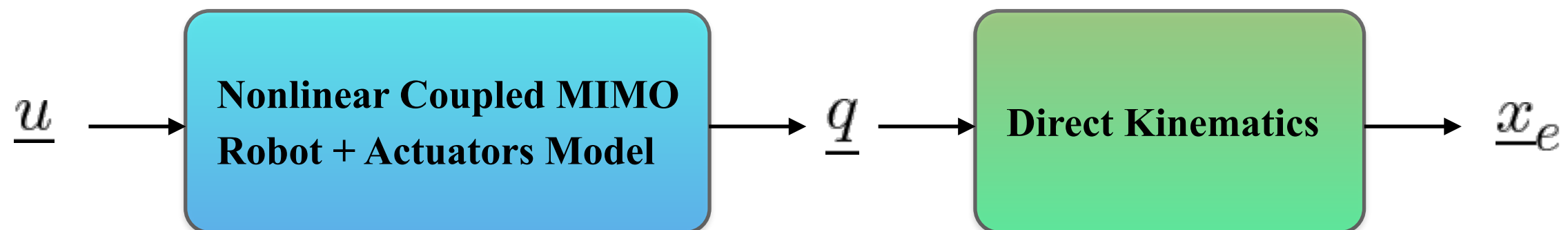
System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

Lyapunov function candidate:

$$V(\underline{\dot{q}}, \underline{\tilde{x}}) = \frac{1}{2} \underline{\dot{q}}^T B(\underline{q}) \underline{\dot{q}} + \frac{1}{2} \underline{\tilde{x}}^T K_P \underline{\tilde{x}} > 0 \quad \forall \underline{\dot{q}}, \underline{\tilde{x}} \neq \underline{0},$$

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$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

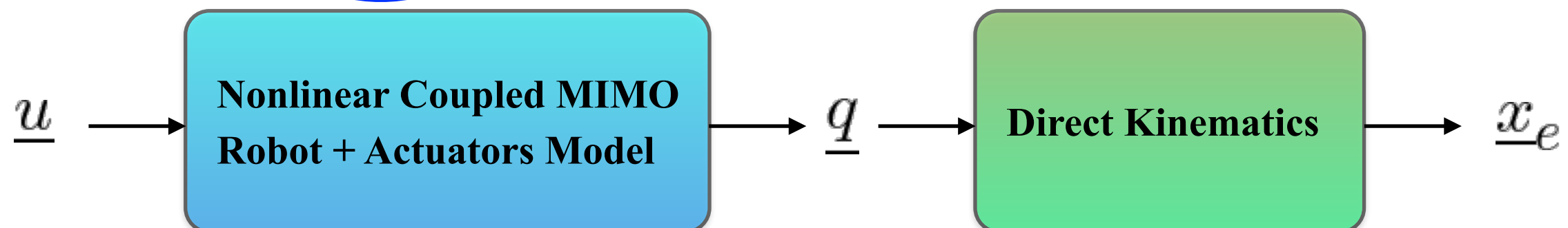
System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

$$B(\underline{q})\underline{\ddot{q}} + C(\underline{q}, \underline{\dot{q}})\underline{\dot{q}} + F\underline{\dot{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

$$\dot{V} = \underline{\dot{q}}^T B(\underline{q})\underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q})\underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

Skew symmetry property:

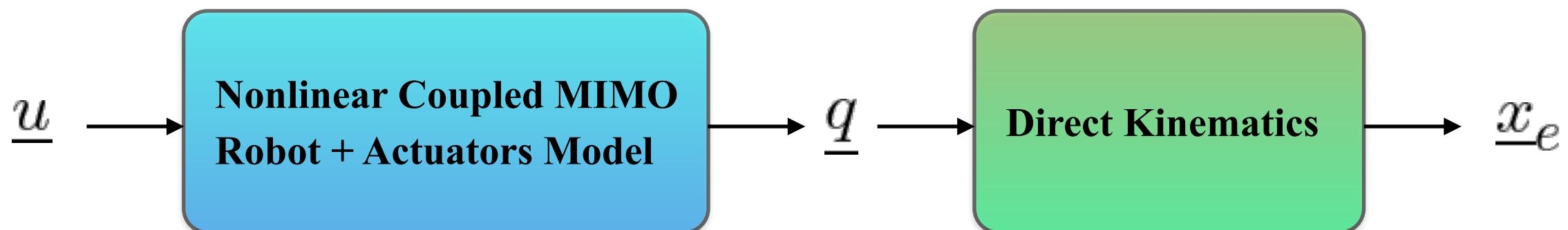
$$N(\underline{q}, \underline{\dot{q}}) = \dot{B}(\underline{q}) - 2C(\underline{q}, \underline{\dot{q}}) \quad \text{s.t.} \quad \underline{w}^T N(\underline{q}, \underline{\dot{q}}) \underline{w} = 0$$

Direct implication:

$$\underline{w} = \underline{\dot{q}}$$

$$\underline{\dot{q}}^T N(\underline{q}, \underline{\dot{q}}) \underline{\dot{q}} = 0$$

$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



Motion Control in the Operational Space

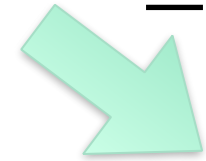
Centralized Approach

PD Control with Gravity Compensation in the Operational Space

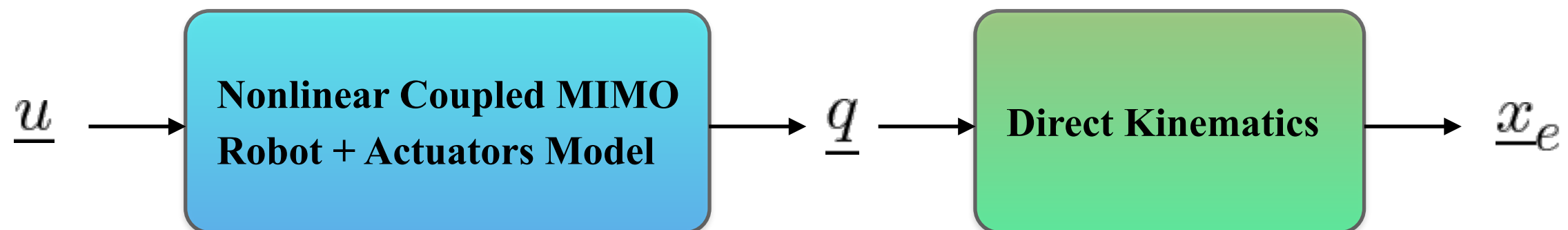
System state: $[\underline{\tilde{x}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$

constant equilibrium pose

$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



$$\dot{V} = -\underline{\dot{q}}^T F \underline{\dot{q}} + \underline{\dot{q}}^T (\underline{u} - \underline{g}(\underline{q}) - J_A^T(\underline{q}) K_P \underline{\tilde{x}})$$



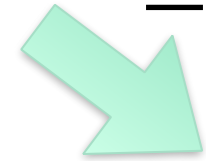
Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $[\underline{\tilde{x}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

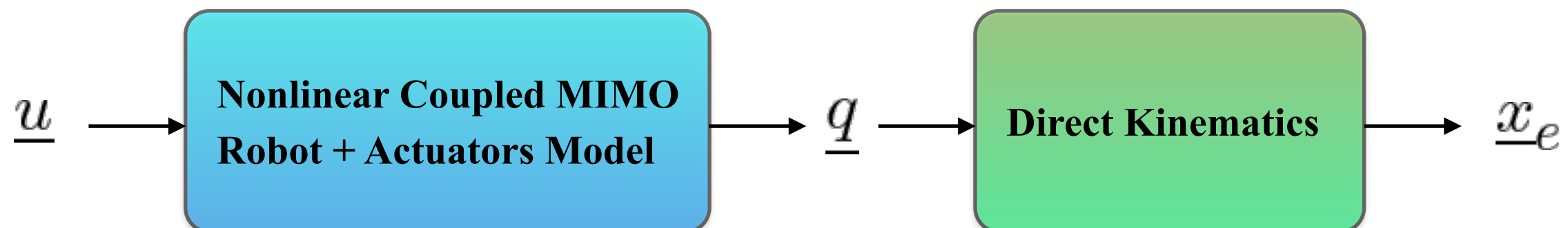
$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



$$\dot{V} = -\underline{\dot{q}}^T F \underline{\dot{q}} + \underline{\dot{q}}^T (\underline{u} - \underline{g}(\underline{q}) - J_A^T(\underline{q}) K_P \underline{\tilde{x}})$$



$$\underline{u} = \underline{g}(\underline{q}) + J_A^T(\underline{q}) K_P \underline{\tilde{x}} - J_A^T(\underline{q}) K_D J_A(\underline{q}) \underline{\dot{q}}$$



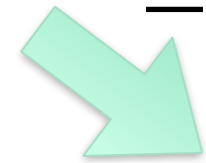
Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $[\underline{\tilde{x}}^T \quad \underline{\dot{q}}^T]^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

$$\dot{V} = \underline{\dot{q}}^T B(\underline{q}) \underline{\ddot{q}} + \frac{1}{2} \underline{\dot{q}}^T \dot{B}(\underline{q}) \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_P \underline{\tilde{x}}$$



$$\dot{V} = -\underline{\dot{q}}^T F \underline{\dot{q}} + \underline{\dot{q}}^T (\underline{u} - \underline{g}(\underline{q}) - J_A^T(\underline{q}) K_P \underline{\tilde{x}})$$



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$$\dot{V} = -\underline{\dot{q}}^T F \underline{\dot{q}} - \underline{\dot{q}}^T J_A^T(\underline{q}) K_D J_A(\underline{q}) \underline{\dot{q}}$$

Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \tilde{\underline{x}}^T & \dot{\underline{q}}^T \end{bmatrix}^T$ with $\tilde{\underline{x}} = \underline{x}_d - \underline{x}_e$

constant equilibrium pose

$$\underline{u} = \underline{g}(\underline{q}) + \underline{J}_A^T(\underline{q}) \underline{K}_P \tilde{\underline{x}} - \underline{J}_A^T(\underline{q}) \underline{K}_D \underline{J}_A(\underline{q}) \dot{\underline{q}}$$

$$\underline{B}(\underline{q}) \ddot{\underline{q}} + \underline{C}(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} + \underline{F} \dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$\underline{F} = \underline{F}_v + \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{K}_v \underline{K}_r$$

$$\underline{u} = \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{G}_v \underline{v}_c$$

Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$

constant equilibrium pose

$$\underline{u} = \underline{g}(\underline{q}) + \underline{J}_A^T(\underline{q}) \underline{K}_P \underline{\tilde{x}} - \underline{J}_A^T(\underline{q}) \underline{K}_D \underline{J}_A(\underline{q}) \underline{\dot{q}}$$

zero at equilibrium

$$\underline{B}(\underline{q}) \underline{\ddot{q}} + \underline{C}(\underline{q}, \underline{\dot{q}}) \underline{\dot{q}} + \underline{F} \underline{\dot{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$\underline{F} = \underline{F}_v + \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{K}_v \underline{K}_r$$

$$\underline{u} = \underline{K}_r \underline{K}_t \underline{R}_a^{-1} \underline{G}_v \underline{v}_c$$

Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \tilde{\underline{x}}^T & \dot{\underline{q}}^T \end{bmatrix}^T$ with $\tilde{\underline{x}} = \underline{x}_d - \underline{x}_e$

constant equilibrium pose

$$\underline{u} = \underline{g}(\underline{q}) + J_A^T(\underline{q}) K_P \tilde{\underline{x}} - J_A^T(\underline{q}) K_D J_A(\underline{q}) \dot{\underline{q}}$$

$$\cancel{B(\underline{q})\ddot{\underline{q}}} + \cancel{C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}} + \cancel{F\dot{\underline{q}}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

gravity term is compensated

Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \underline{\tilde{x}}^T & \underline{\dot{q}}^T \end{bmatrix}^T$ with $\underline{\tilde{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

$$\underline{u} = \underline{g}(\underline{q}) + J_A^T(\underline{q}) K_P \underline{\tilde{x}} - J_A^T(\underline{q}) K_D J_A(\underline{q}) \underline{\dot{q}}$$

$$\cancel{B(\underline{q})\underline{\ddot{q}}} + \cancel{C(\underline{q}, \underline{\dot{q}})\underline{\dot{q}}} + \cancel{F\underline{\dot{q}}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

$$J_A^T(\underline{q}) K_P \underline{\tilde{x}} = \underline{0}$$

the manipulator reaches an equilibrium pose

Motion Control in the Operational Space

Centralized Approach

PD Control with Gravity Compensation in the Operational Space

System state: $\begin{bmatrix} \tilde{\underline{x}}^T & \dot{\underline{q}}^T \end{bmatrix}^T$ with $\tilde{\underline{x}} = \underline{x}_d - \underline{x}_e$ constant equilibrium pose

$$\underline{u} = \underline{g}(\underline{q}) + J_A^T(\underline{q}) K_P \tilde{\underline{x}} - J_A^T(\underline{q}) K_D J_A(\underline{q}) \dot{\underline{q}}$$

$$\cancel{B(\underline{q})} \ddot{\underline{q}} + \cancel{C(\underline{q}, \dot{\underline{q}})} \dot{\underline{q}} + \cancel{F \dot{\underline{q}}} + \cancel{\underline{g}(\underline{q})} = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$

if the Jacobian
is full-rank

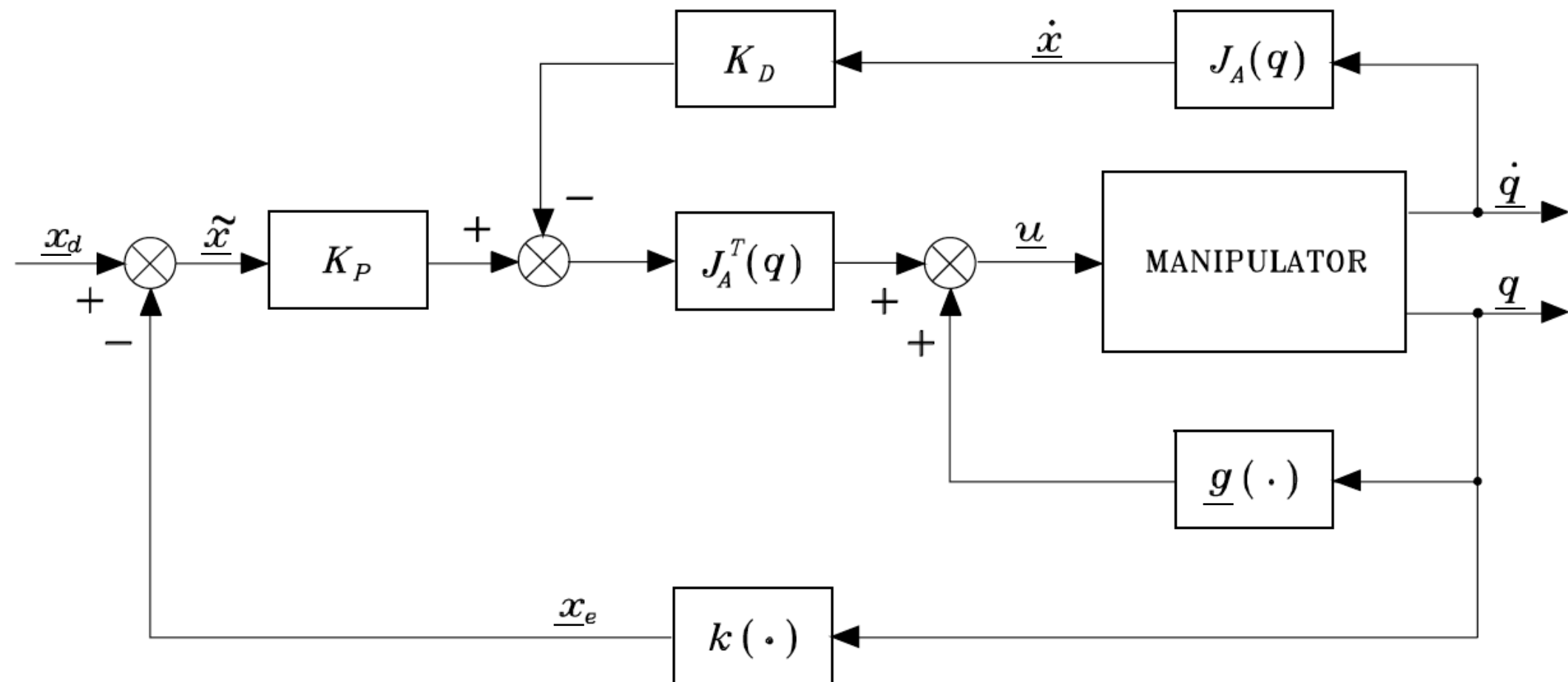
$$J_A^T(\underline{q}) K_P \tilde{\underline{x}} = \underline{0}$$

$$\tilde{\underline{x}} = \underline{x}_d - \underline{x}_e = \underline{0}$$

Motion Control in the Operational Space

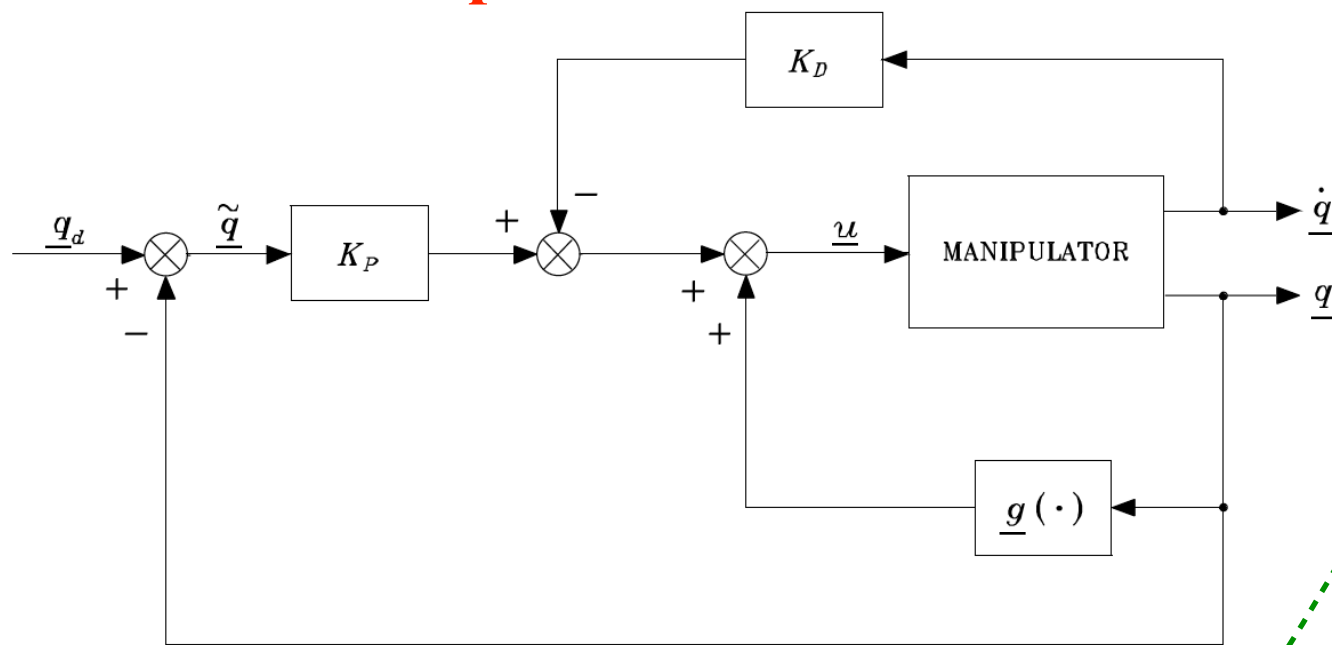
Centralized Approach

PD Control with Gravity Compensation in the Operational Space

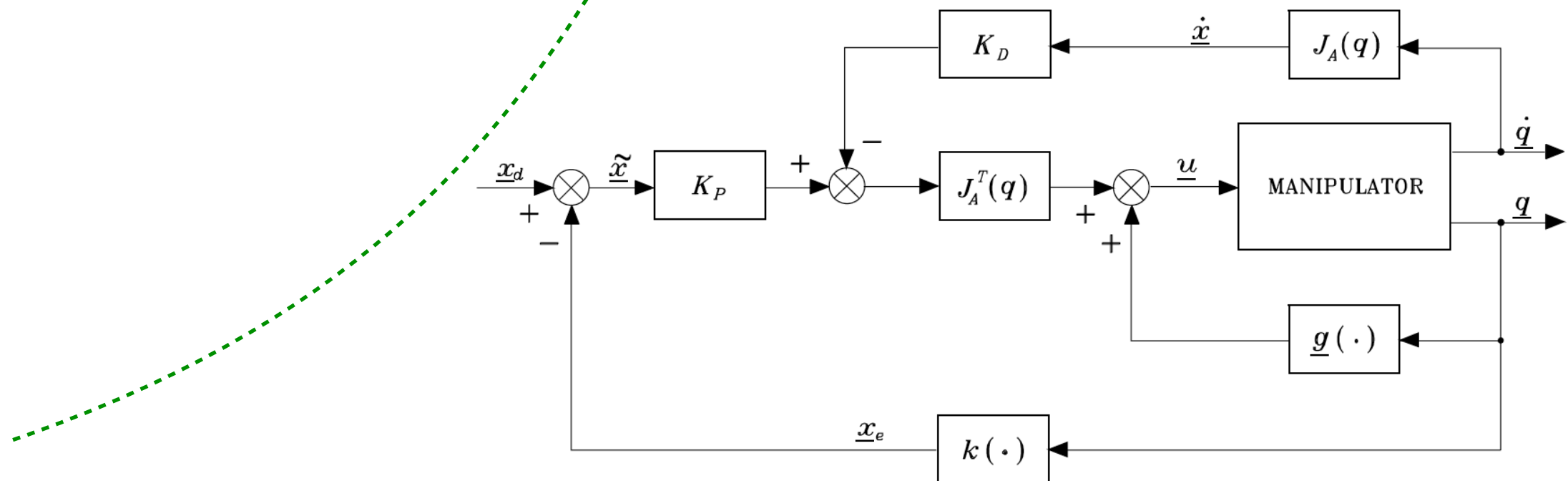


Comparison: Centralized Motion Control in the Joint Space vs in the Operational Space

PD Control with Gravity Compensation in the Joint Space



PD Control with Gravity Compensation in the Operational Space





UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

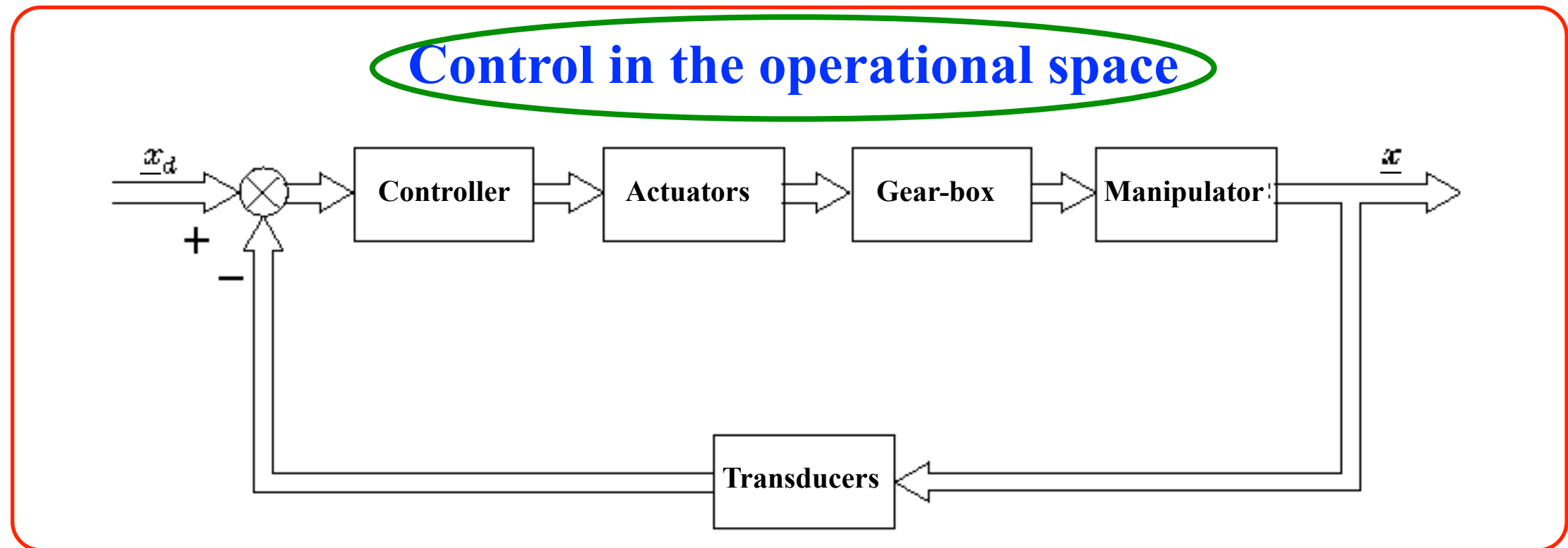
Prof. Antonella Ferrara



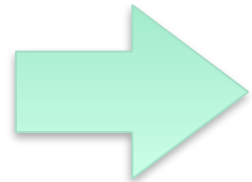
Inverse Dynamics Control in the Operational Space

Motion Control of Robot Manipulators

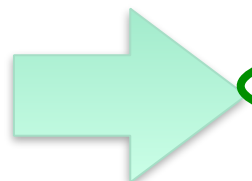
In this Lecture...



Centralized Control Approach in the Operational Space:



PD Control with Gravity Compensation in the Operational Space



Inverse Dynamics Control in the Operational Space

Motion Control in the Operational Space

Centralized Approach

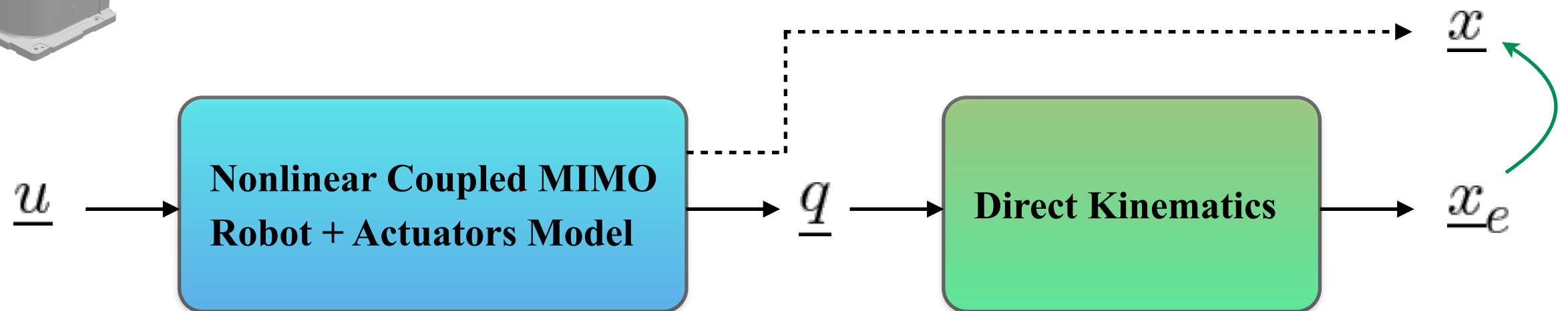
MIMO Dynamic Model of Robot Manipulator and Actuators



$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u}$$

$$F = F_v + K_r K_t R_a^{-1} K_v K_r$$

$$\underline{u} = K_r K_t R_a^{-1} G_v \underline{v}_c$$



Motion Control in the Operational Space

Centralized Approach

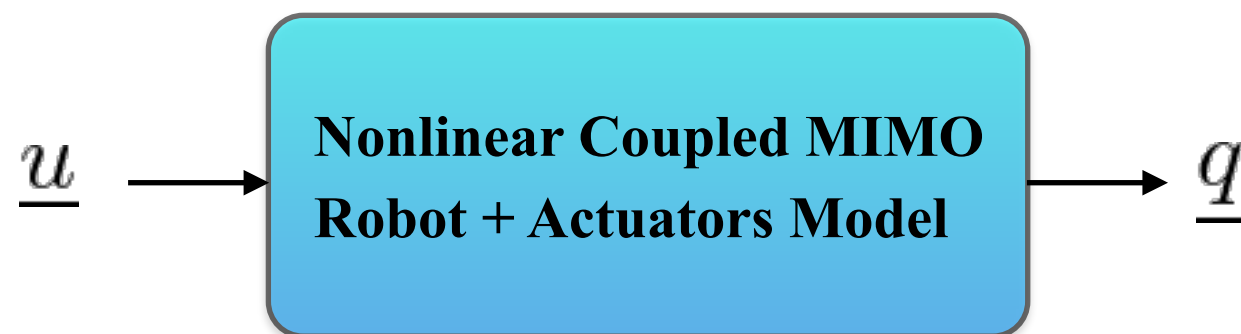
Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

$$\downarrow$$
$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$



Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

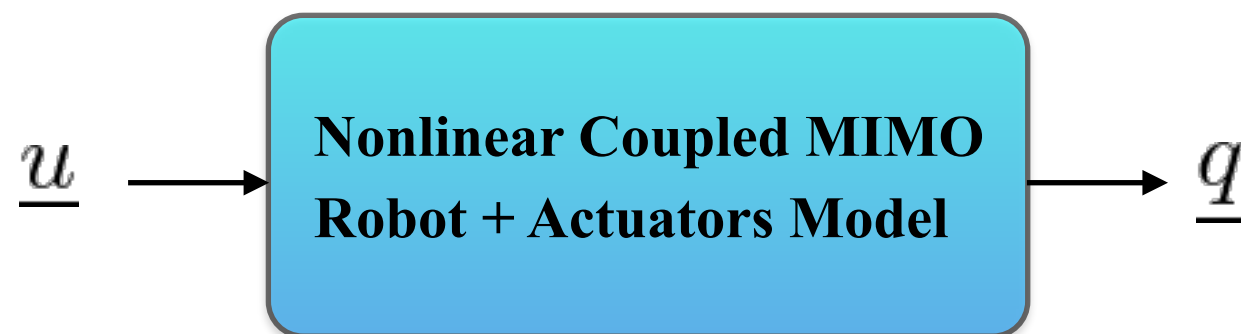
MODEL: dynamic model of an n -joint manipulator + actuators

$$\underline{B}(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

to exactly linearize the
controlled system dynamics

$\downarrow$

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{C}(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + \underline{F}\dot{\underline{q}} + \underline{g}(\underline{q})$$



Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

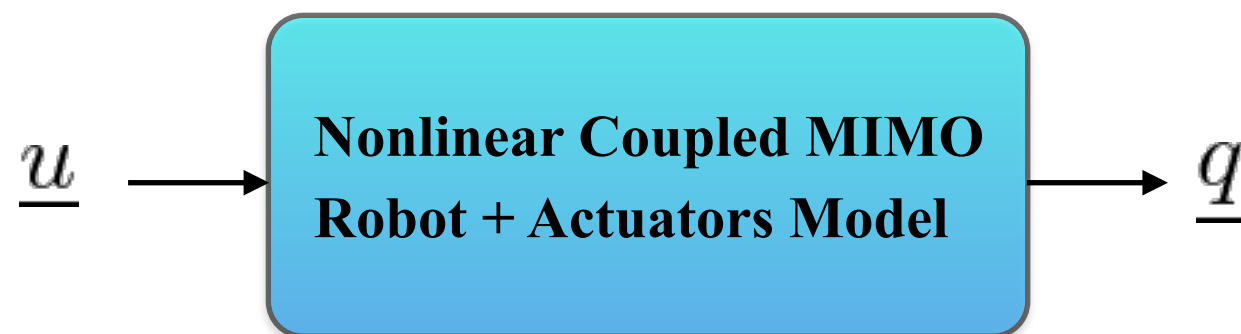
MODEL: dynamic model of an n -joint manipulator + actuators

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

to exactly linearize the controlled system dynamics

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$

$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}})$$



$$\ddot{\underline{q}} = \underline{y}$$

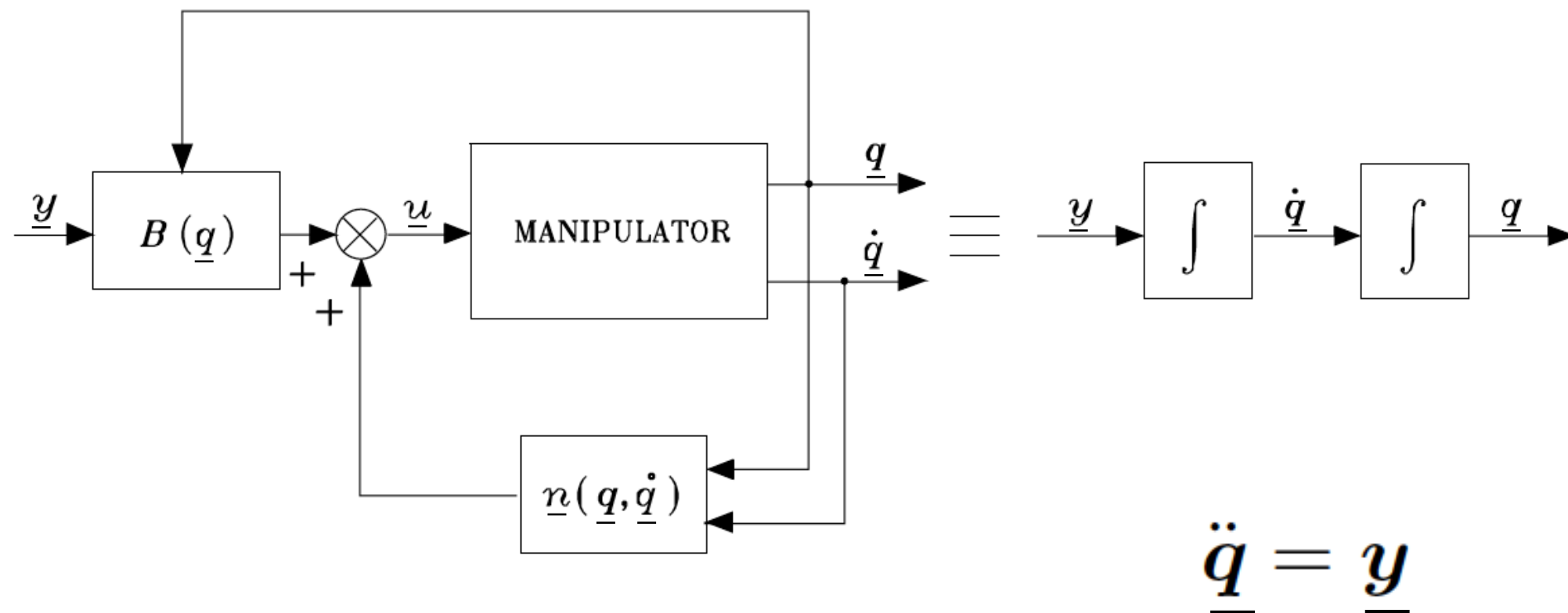
Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



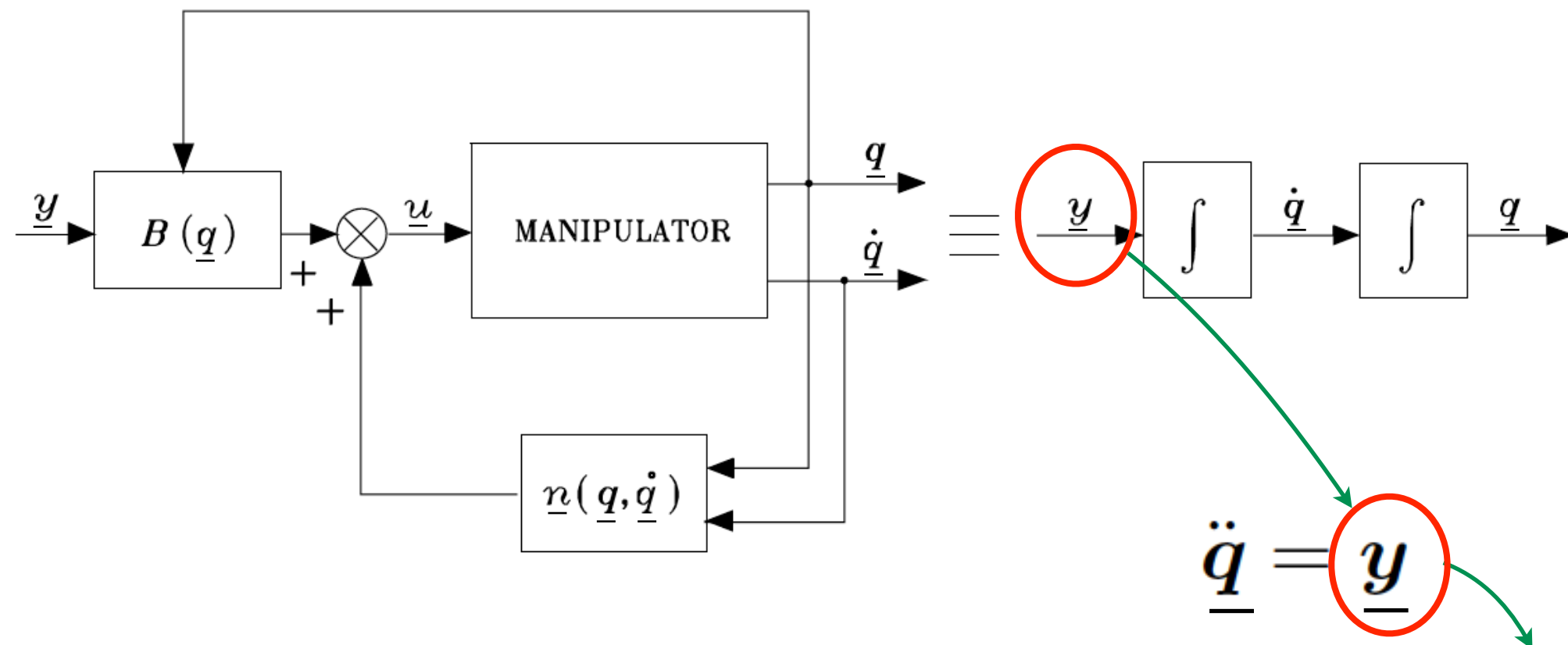
Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



auxiliary control still
to be designed

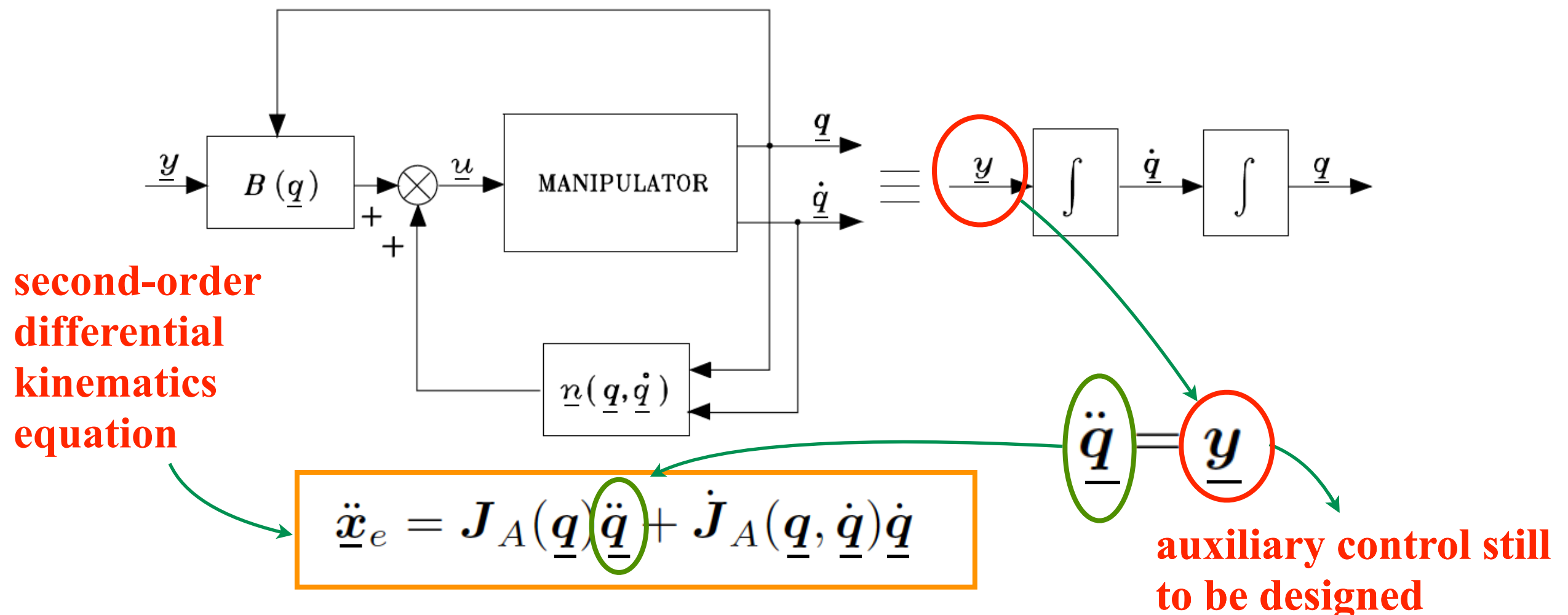
Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



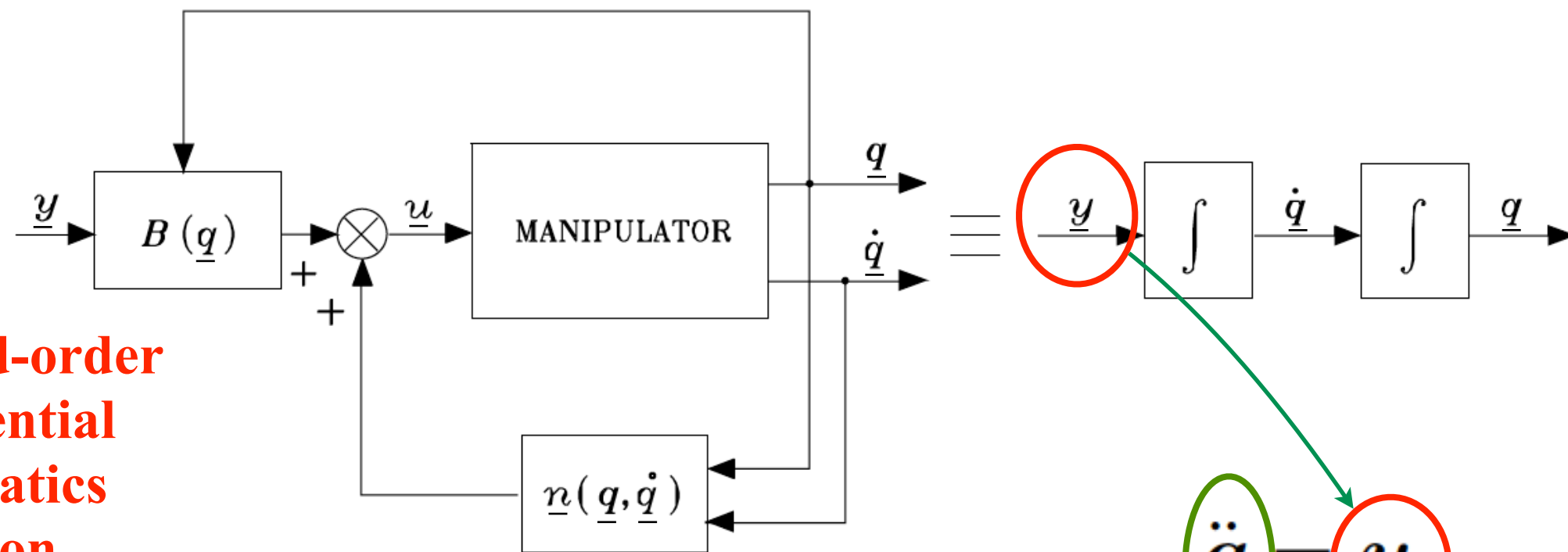
Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators



second-order
differential
kinematics
equation

$$\ddot{\underline{x}}_e = \underline{J}_A(\underline{q})\ddot{\underline{q}} + \dot{\underline{J}}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}$$

$$\dot{\underline{x}}_e = \underline{J}_A(\underline{q})\dot{\underline{q}}$$

auxiliary control still
to be designed

Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

second-order
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kinematics
equation

$$\ddot{\underline{x}}_e = \underline{J}_A(\underline{q})\ddot{\underline{q}} + \dot{\underline{J}}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}$$

auxiliary control still
to be designed

$$\ddot{\underline{q}} = \underline{y}$$

$$\dot{\underline{x}}_e = \underline{J}_A(\underline{q})\dot{\underline{q}}$$

$$\underline{y} = \underline{J}_A^{-1}(\underline{q})\left(\ddot{\underline{x}}_d + \underline{K}_D\dot{\underline{x}} + \underline{K}_P\tilde{\underline{x}} - \dot{\underline{J}}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}\right)$$

Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

second-order
differential
kinematics
equation

$$\ddot{\underline{x}}_e = \underline{J}_A(\underline{q})\ddot{\underline{q}} + \dot{\underline{J}}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}$$

auxiliary control still
to be designed

$$\ddot{\underline{q}} = \underline{y}$$

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$$\underline{y} = \underline{J}_A^{-1}(\underline{q})\left(\ddot{\underline{x}}_d + \underline{K}_D\dot{\underline{\tilde{x}}} + \underline{K}_P\tilde{\underline{x}} - \dot{\underline{J}}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}\right)$$

$$\ddot{\underline{\tilde{x}}} + \underline{K}_D\dot{\underline{\tilde{x}}} + \underline{K}_P\tilde{\underline{x}} = \underline{0}$$

Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

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auxiliary control still
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$$\ddot{\underline{q}} = \underline{y}$$

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$$\ddot{\tilde{\underline{x}}} + \underline{K}_D\dot{\tilde{\underline{x}}} + \underline{K}_P\tilde{\underline{x}} = \underline{0}$$

m (at most 6) second order
LTI SISO systems with 2
negative real eigenvalues

Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space

PROBLEM: tracking a pose trajectory $\underline{x}_d(t)$

MODEL: dynamic model of an n -joint manipulator + actuators

$$\ddot{\underline{\tilde{x}}} + \underline{K}_D \dot{\underline{\tilde{x}}} + \underline{K}_P \underline{\tilde{x}} = \underline{0}$$

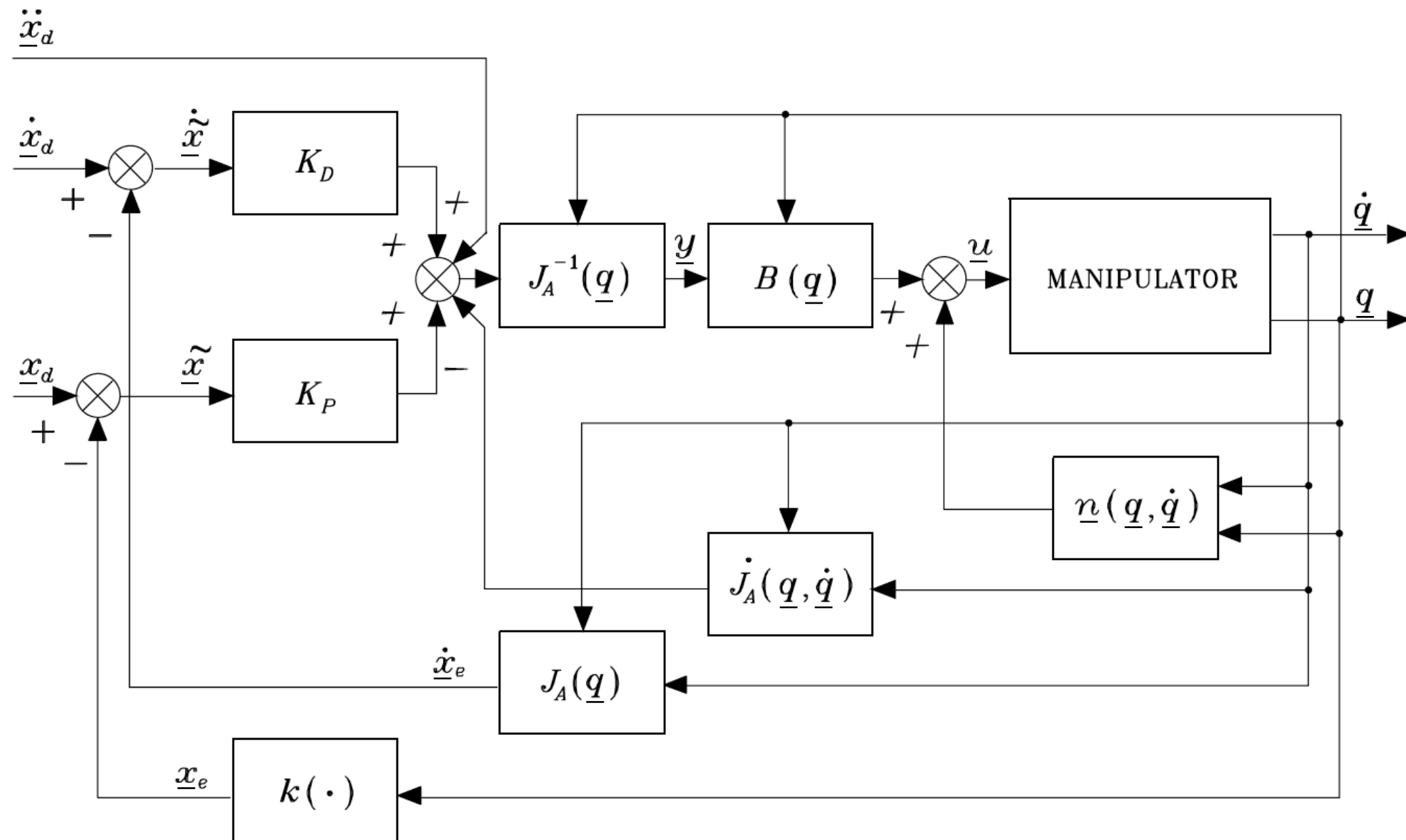


$$\lim_{t \rightarrow \infty} \underline{\tilde{x}} = \lim_{t \rightarrow \infty} (\underline{x}_d(t) - \underline{x}_e) = \underline{0}$$

Motion Control in the Operational Space

Centralized Approach

Inverse Dynamics Control in the Operational space



Final Comment on Motion Control Schemes and Supplementary Material

A. Ferrara, G.P. Incremona, M. Cucuzzella, “**Advanced and Optimization Based Sliding Mode Control**”, SIAM, 2019

Ferrara A., Incremona G.P., Sangiovanni B., “Tracking Control via Switched Integral Sliding Mode with Application to Robot Manipulators”, **Control Engineering Practice**, Volume 90, Pages 257-266 September 2019

G. P. Incremona, A. Ferrara and L. Magni, “MPC for Robot Manipulators with Integral Sliding Modes Generation”, **IEEE/ASME Transactions on Mechatronics**, Volume: 22, Issue 3, Pages 1299-1307, June 2017

A. Ferrara and G. P. Incremona, “Design of an Integral Suboptimal Second-Order Sliding Mode Controller for the Robust Motion Control of Robot Manipulators”, **IEEE Transactions on Control Systems Technology**, Volume 23, Issue 6, Pages 2316-2325, Nov. 2015

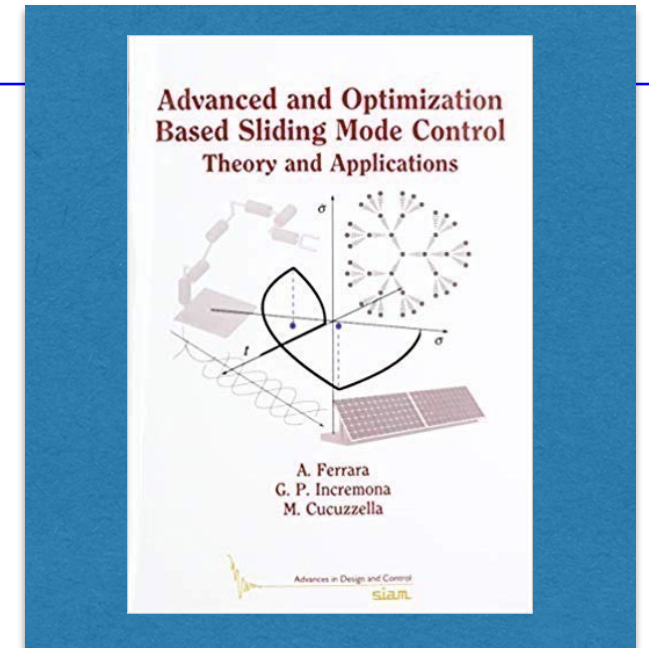
Gian Paolo Incremona, Gianluca De Felici, Antonella Ferrara and Ezio Bassi, A Supervisory Sliding Mode Control Approach for Cooperative Robotic Systems of Systems, **IEEE Systems Journal**, vol. 9, no. 1, pp. 263-272, March 2015

Capisani L., Facchinetti T., Ferrara A., Martinelli A., Obstacle Modelling Oriented to Safe Motion Planning and Control for Planar Rigid Robot Manipulators, **Journal of Intelligent and Robotic Systems**, Volume 71, Issue 2, pp. 159–178, August 2013

Capisani L., Ferreira de Loza A., Ferrara A., Fridman L., Manipulators Fault Diagnosis via Higher Order Sliding Mode Observers, **IEEE Transactions on Industrial Electronics**, Vol. 59, No. 10, 3979 - 3986, 2012

Capisani L., Facchinetti T., Ferrara A., Real-time networked control of an industrial robot manipulator via discrete-time second order sliding modes, **International Journal of Control**, Vol. 83, No. 8, pp. 1595-1611, Aug., 2010

Capisani L., Ferrara A., Magnani L., Design and experimental validation of a second order sliding mode motion controller for robot manipulators, **International Journal of Control**, Volume 82 Issue 2, pp. 365-377, 2008





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e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara

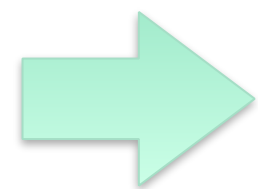
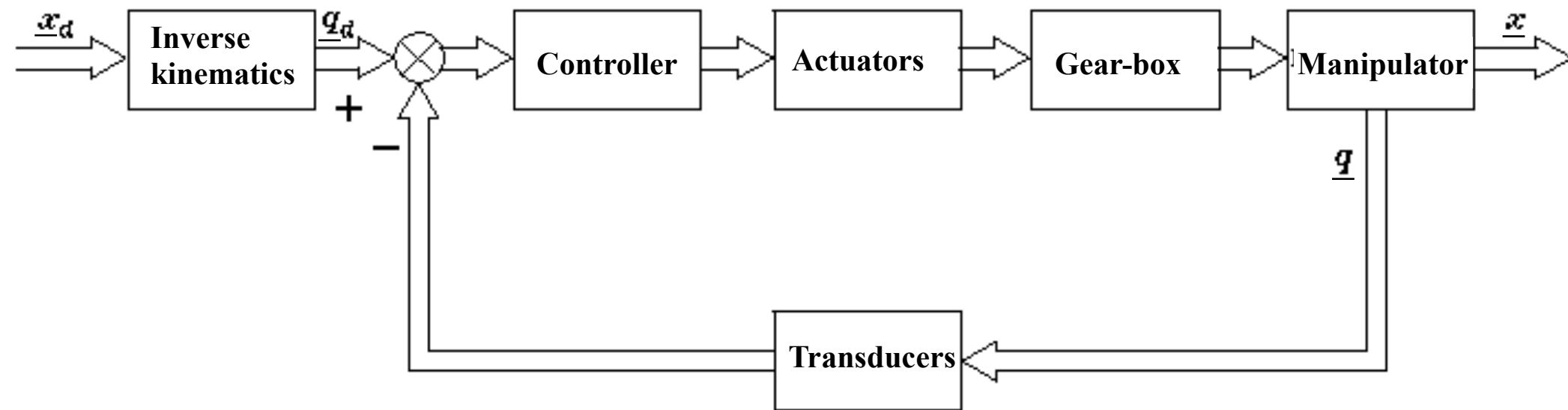


Interaction Control of Robot Manipulators

Motion Control: Control in the Joint Space

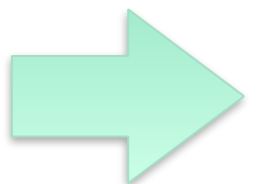
In previous lectures...

Control in the joint space



decentralized control (independent control of the joints)

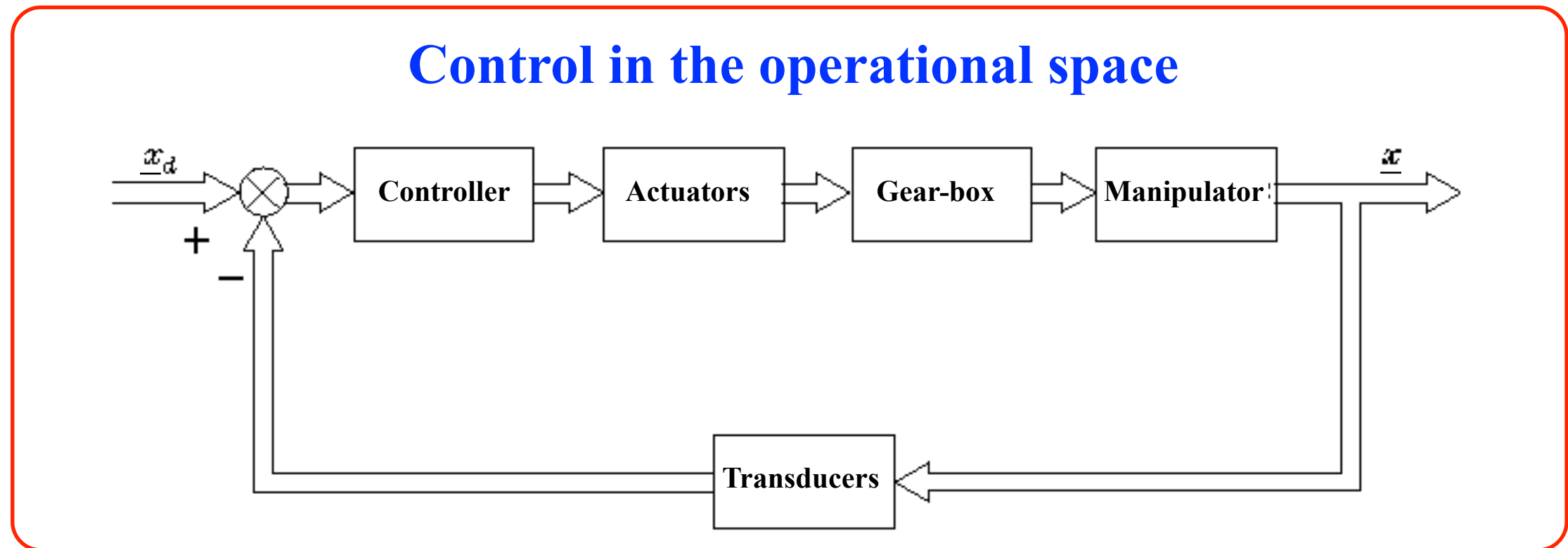
- *decentralized control with decentralized feedforward compensation*
- *decentralized control with centralized computed torque based feedforward compensation*



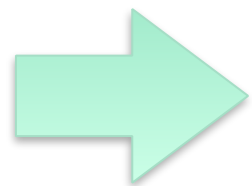
centralized control of the joints

Motion Control of Robot Manipulators

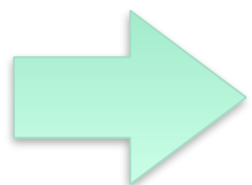
In previous lectures...



Centralized Control Approach in the Operational Space:



PD Control with Gravity Compensation in the Operational Space

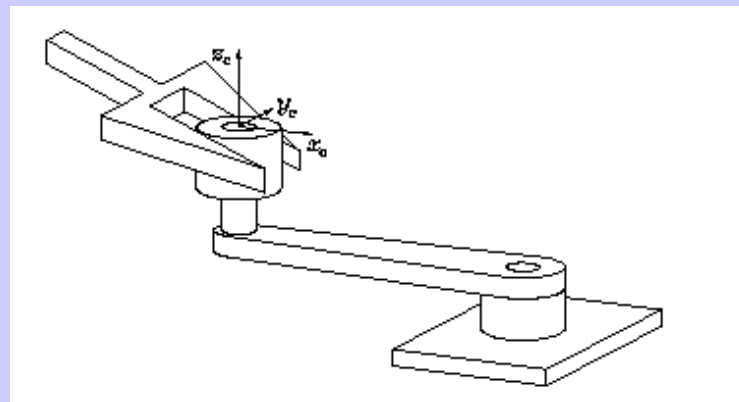
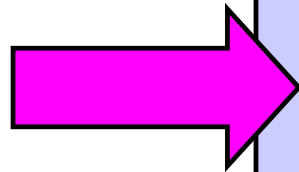


Inverse Dynamics Control in the Operational Space

Interaction Control of Robot Manipulators

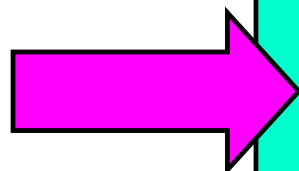
In this Lecture...

When a task requires...

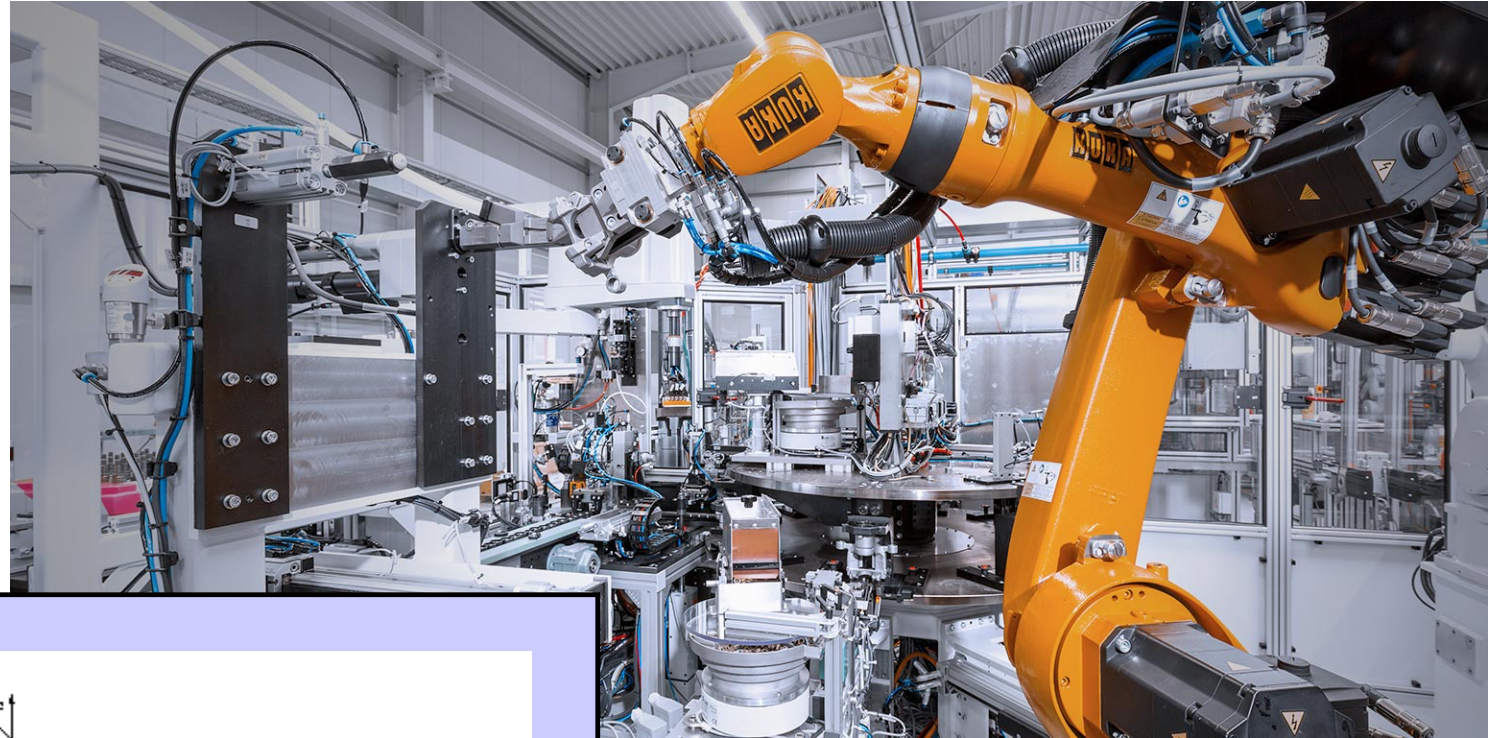


... the interaction with the environment

The robot motion becomes...



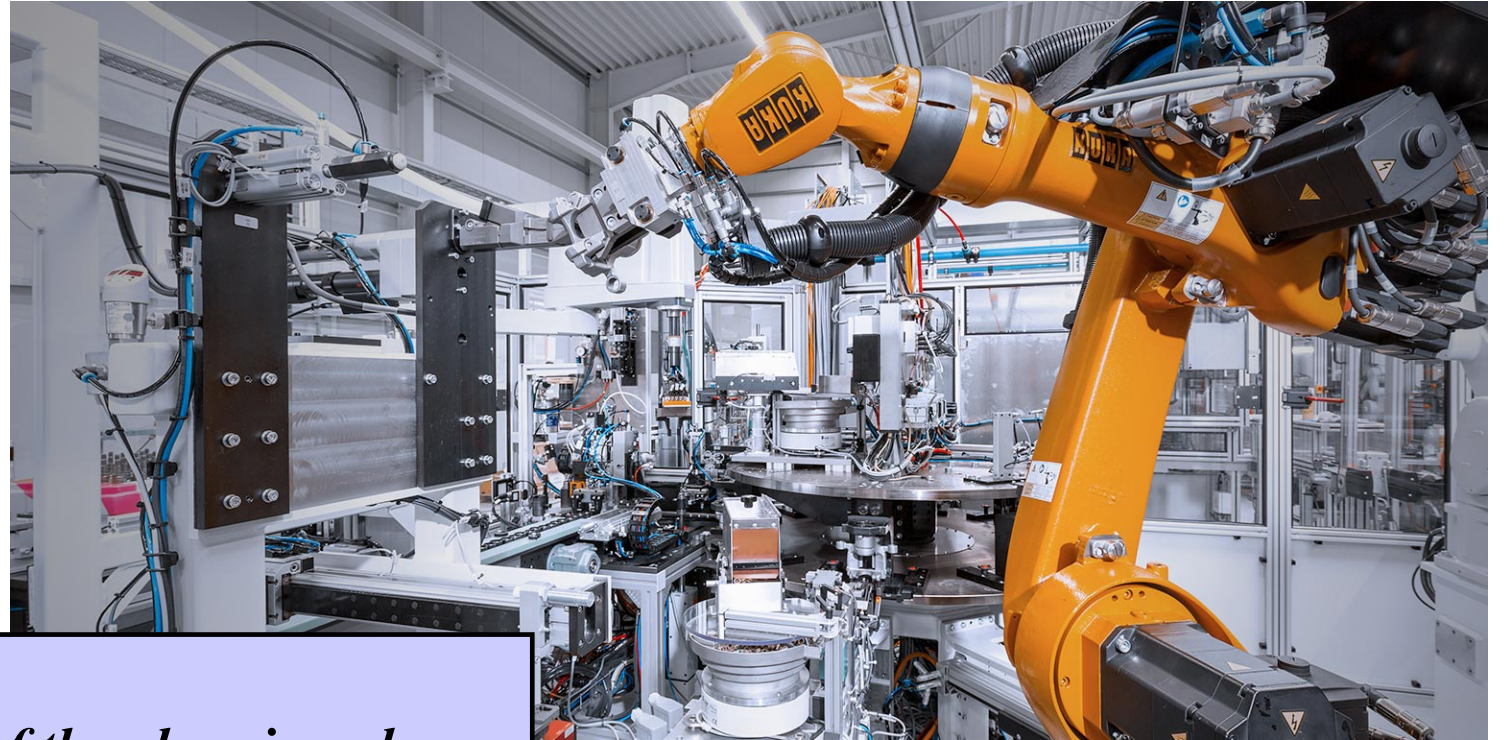
... a constrained motion



Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control of Robot Manipulators

In this Lecture...



Requirements...

*Accuracy of the planning phase
and high performance of the
control system in terms of
precision and quality of tracking*

**Accurate knowledge of the kinematical and
dynamical model of the robot as well as of the
geometric and mechanical features
of the environment**

Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control of Robot Manipulators

In this Lecture...

Interaction Control

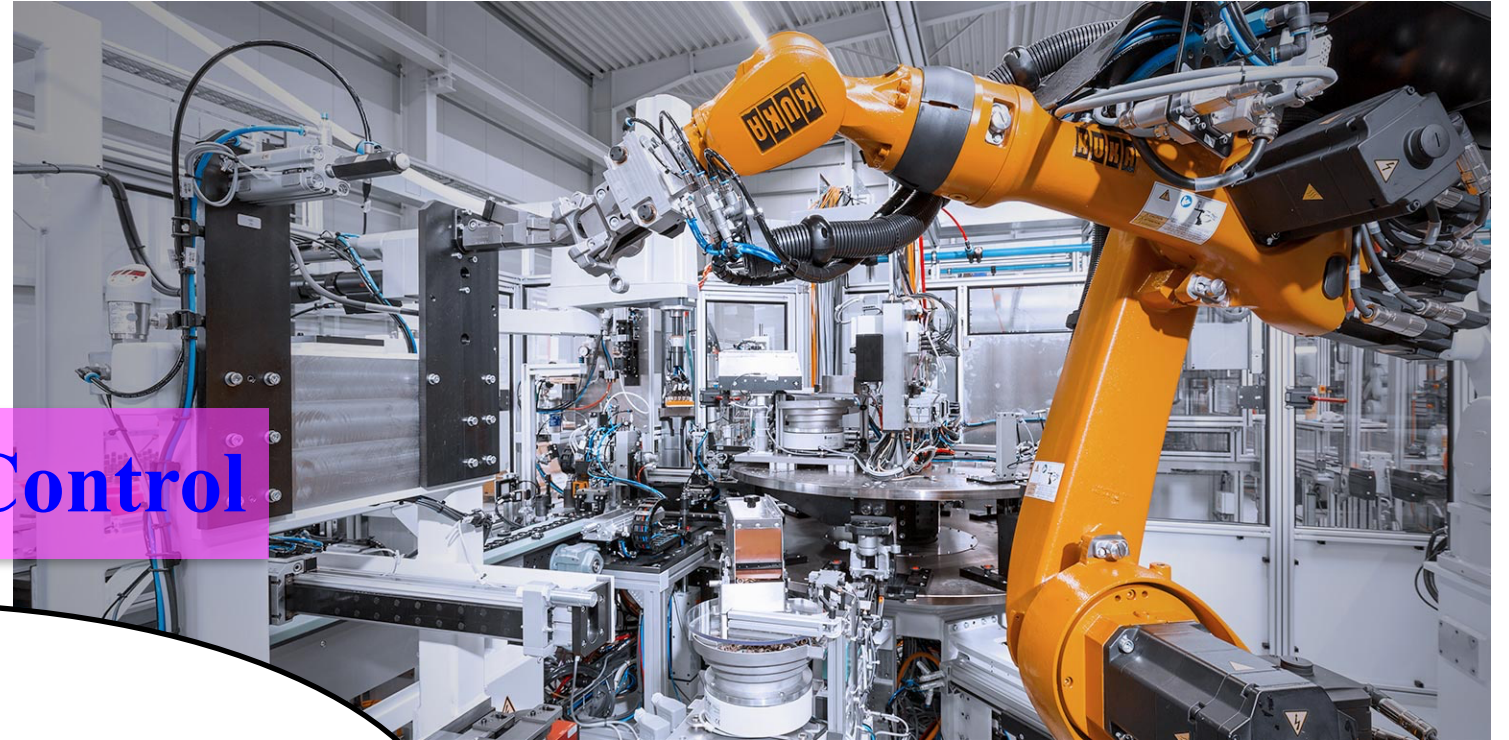
Strategies to respect the constraints on the *contact force*

INDIRECT APPROACH:

By means of position control laws

DIRECT APPROACH:

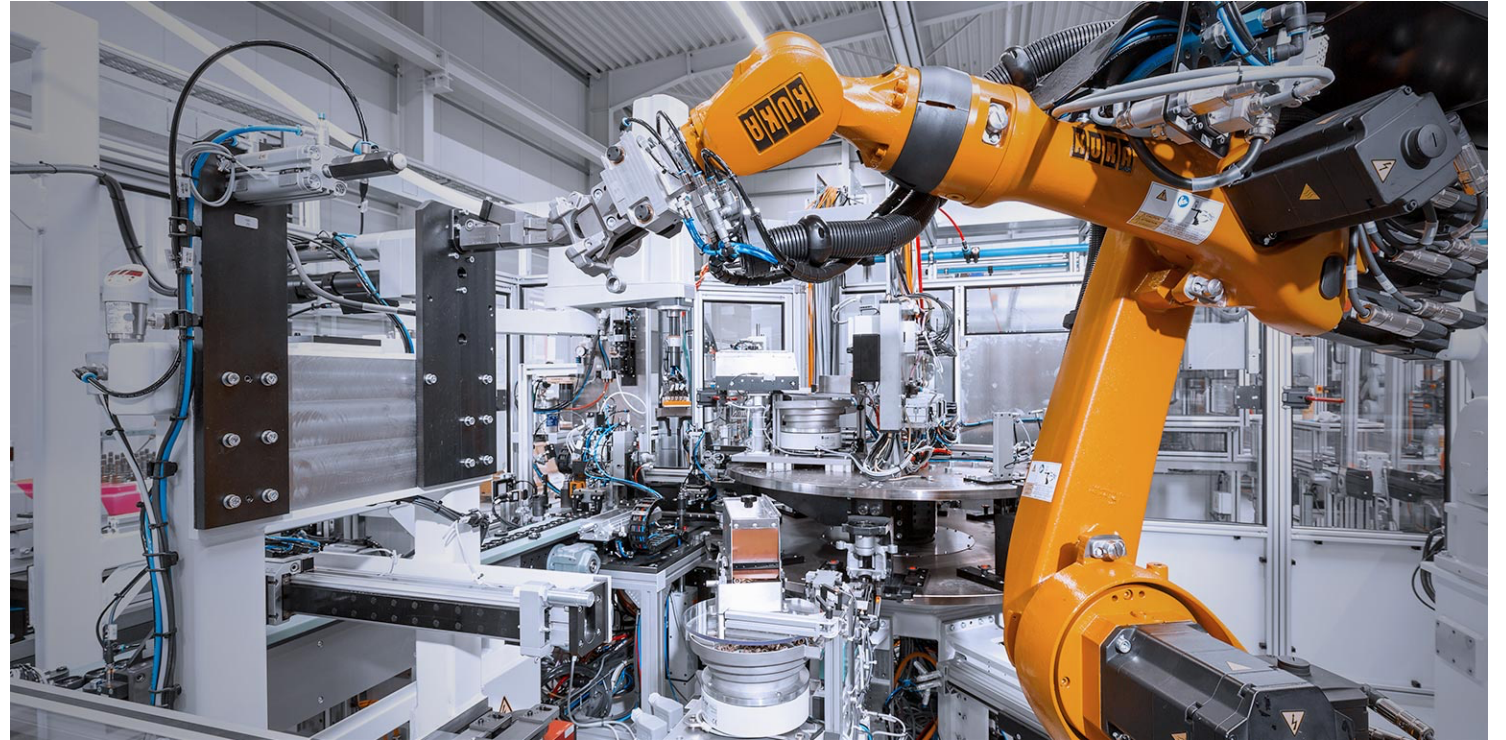
By means of force control laws



Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

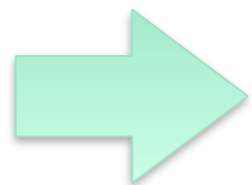
Interaction Control of Robot Manipulators

In this Lecture...



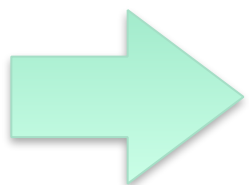
Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control: Indirect Approach



Impedance control

Interaction Control: Direct Approach



Force control



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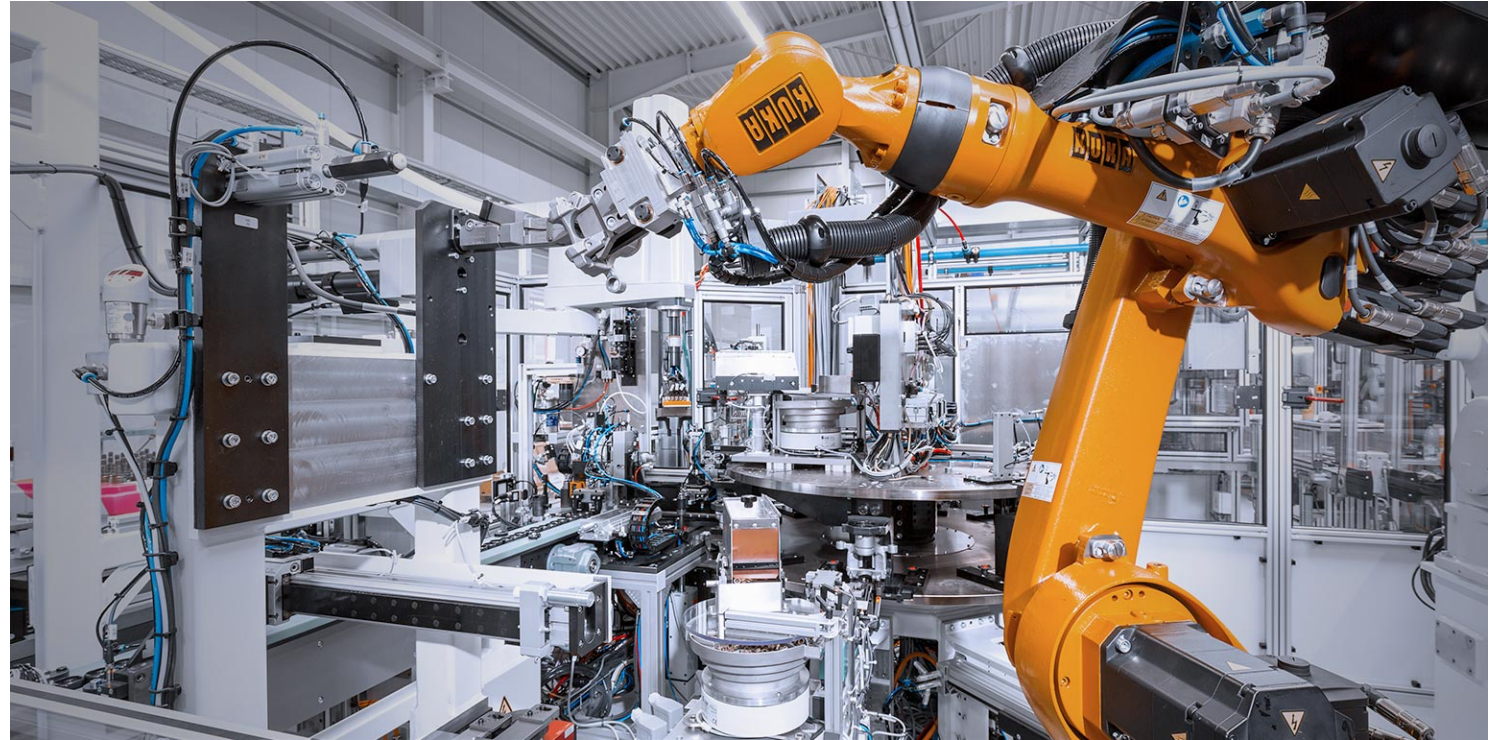
Course of Robot Control

Prof. Antonella Ferrara



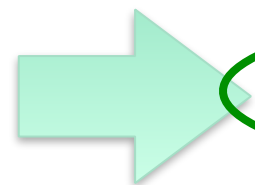
Impedance control

Interaction Control of Robot Manipulators



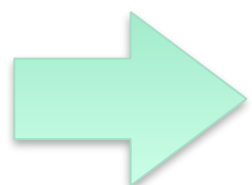
Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control: Indirect Approach



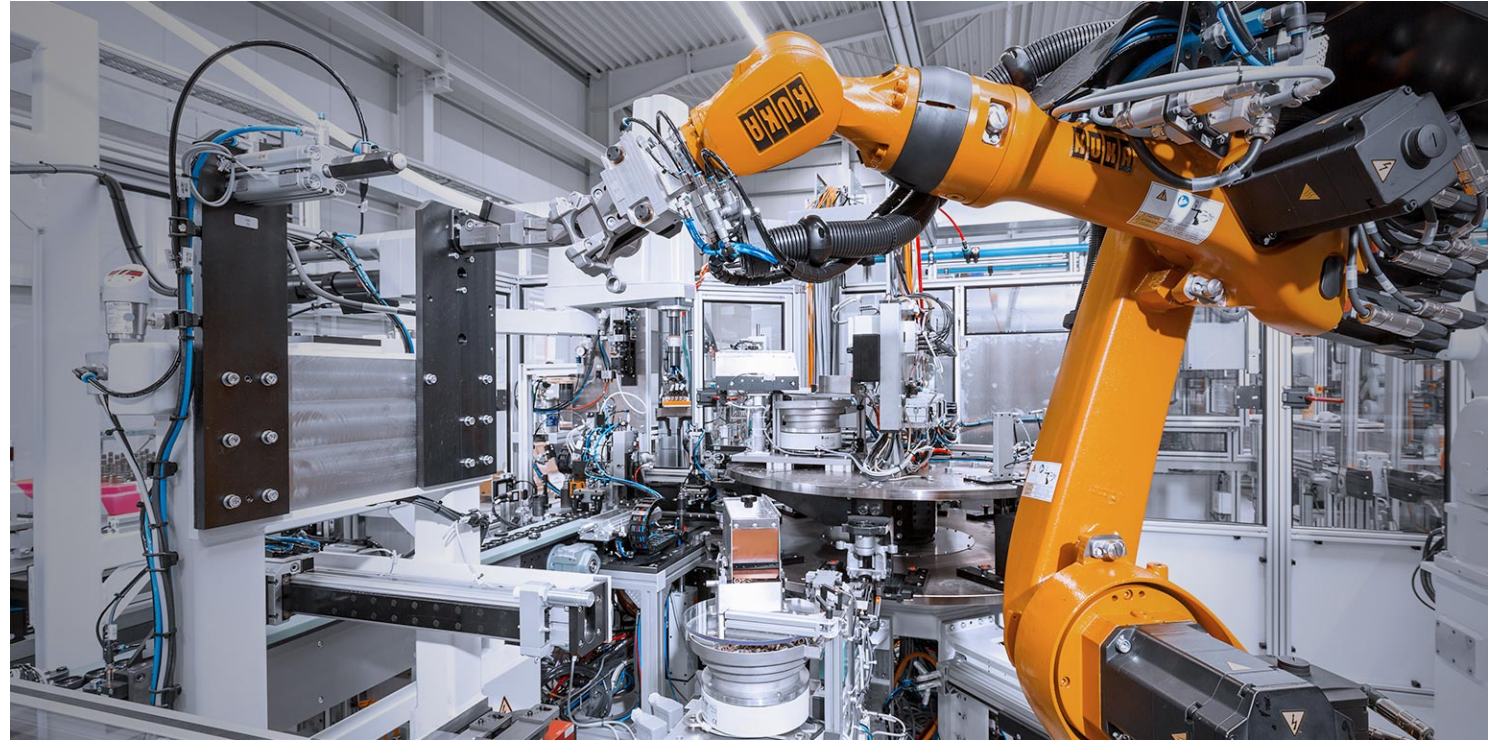
Impedance control

Interaction Control: Direct Approach



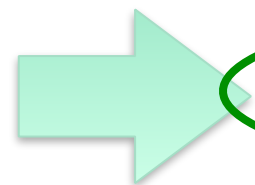
Force control

Interaction Control of Robot Manipulators



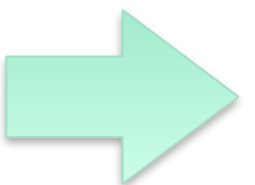
Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control: Indirect Approach



Impedance control

Interaction Control: Direct Approach



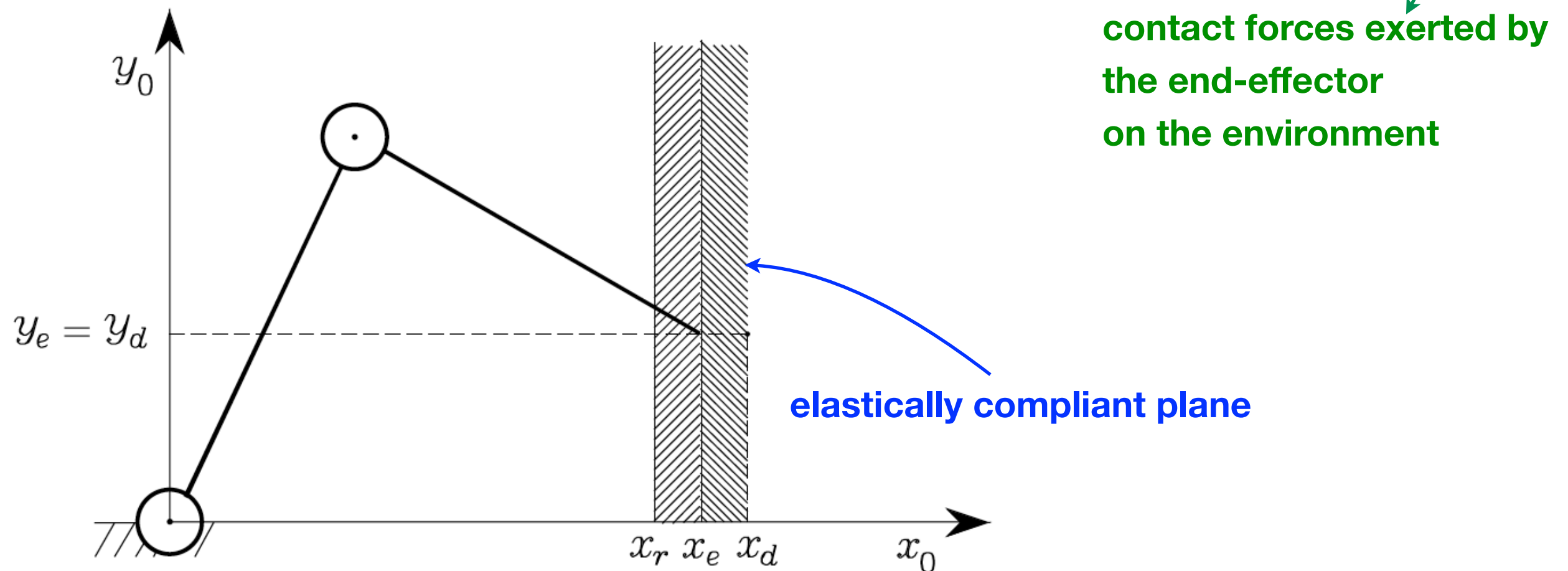
Force control

**Control in the
Operational Space**

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$



Source: Siciliano et al., Robotics, Springer:

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

to exactly linearize the
controlled system dynamics

contact forces exerted by
the end-effector
on the environment

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$

$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}})$$

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q})J^T(\underline{q})\underline{h}_e$$

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

to ~~exactly~~ linearize the
controlled system dynamics

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$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$

$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}})$$

there is still a NL
coupling term
due to the
contact
forces/torques

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q})J^T(\underline{q})\underline{h}_e$$

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

contact forces exerted by
the end-effector
on the environment

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q})J^T(\underline{q})\underline{h}_e$$

$$\ddot{\underline{x}}_e = J_A(\underline{q})\ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}$$

second-order
differential
kinematics
equation

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

contact forces exerted by
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$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q})J^T(\underline{q})\underline{h}_e$$

$$\ddot{\underline{x}}_e = J_A(\underline{q})\ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}}$$

second-order
differential
kinematics
equation

$$\underline{y} = J_A^{-1}(\underline{q})M_d^{-1}(M_d\ddot{\underline{x}}_d + K_D\dot{\underline{x}} + K_P\underline{x} - M_d\dot{J}_A(\underline{q}, \dot{\underline{q}})\dot{\underline{q}})$$

Interaction Control of Robot Manipulators: Impedance Control

$$M_d \ddot{\underline{\tilde{x}}} + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} = M_d B_A^{-1}(\underline{q}) \underline{h}_A$$

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q}) J^T(\underline{q}) \underline{h}_e$$

$$\ddot{\underline{x}}_e = J_A(\underline{q}) \ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

**second-order
differential
kinematics
equation**

$$\underline{y} = J_A^{-1}(\underline{q}) M_d^{-1} (M_d \ddot{\underline{x}}_d + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} - M_d \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}})$$

Interaction Control of Robot Manipulators: Impedance Control

$$M_d \ddot{\underline{\tilde{x}}} + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} = M_d B_A^{-1}(\underline{q}) \underline{h}_A$$

inertia matrix of the
manipulator in the
operational space

$$B_A(\underline{q}) = J_A^{-T}(\underline{q}) B(\underline{q}) J_A^{-1}(\underline{q})$$

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q}) J^T(\underline{q}) \underline{h}_e$$

$$\ddot{\underline{x}}_e = J_A(\underline{q}) \ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

second-order
differential
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$$\underline{y} = J_A^{-1}(\underline{q}) M_d^{-1} (M_d \ddot{\underline{x}}_d + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} - M_d \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}})$$

Interaction Control of Robot Manipulators: Impedance Control

$$M_d \ddot{\underline{\tilde{x}}} + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} = M_d B_A^{-1}(\underline{q}) \underline{h}_A$$

inertia matrix of the
manipulator in the
operational space

$$B_A(\underline{q}) = J_A^{-T}(\underline{q}) B(\underline{q}) J_A^{-1}(\underline{q})$$

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q}) J^T(\underline{q}) \underline{h}_e$$

$$T_A^T(\underline{x}_e) \underline{h}_e = \underline{h}_A$$

$$\ddot{\underline{x}}_e = J_A(\underline{q}) \ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

second-order
differential
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$$\underline{y} = J_A^{-1}(\underline{q}) M_d^{-1} (M_d \ddot{\underline{x}}_d + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} - M_d \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}})$$

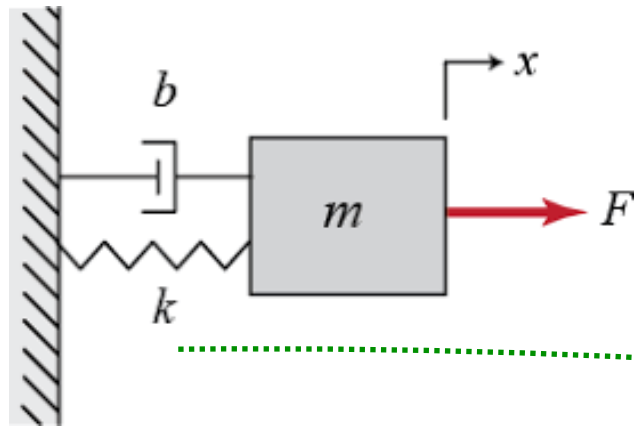
Interaction Control of Robot Manipulators: Impedance Control

The controlled system becomes a generalized mechanical impedance

$$M_d \ddot{\underline{\tilde{x}}} + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} = M_d B_A^{-1}(\underline{q}) \underline{h}_A$$

inertia matrix of the
manipulator in the
operational space

$$B_A(\underline{q}) = J_A^{-T}(\underline{q}) B(\underline{q}) J_A^{-1}(\underline{q})$$



$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q}) J^T(\underline{q}) \underline{h}_e$$

$$T_A^T(\underline{x}_e) \underline{h}_e = \underline{h}_A$$

$$\ddot{\underline{x}}_e = J_A(\underline{q}) \ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

second-order
differential
kinematics
equation

$$\underline{y} = J_A^{-1}(\underline{q}) M_d^{-1} (M_d \ddot{\underline{x}}_d + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} - M_d \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}})$$

Interaction Control of Robot Manipulators: Impedance Control

this matrix makes the system coupled

$$M_d \ddot{\underline{\tilde{x}}} + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} = M_d B_A^{-1}(\underline{q}) \underline{h}_A$$

inertia matrix of the
manipulator in the
operational space

$$B_A(\underline{q}) = J_A^{-T}(\underline{q}) B(\underline{q}) J_A^{-1}(\underline{q})$$

$$T_A^T(x_e) h_e = h_A$$

$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q}) J^T(\underline{q}) \underline{h}_e$$

$$\ddot{\underline{x}}_e = J_A(\underline{q}) \ddot{\underline{q}} + \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

second-order
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kinematics
equation

$$\underline{y} = J_A^{-1}(\underline{q}) M_d^{-1} (M_d \ddot{\underline{x}}_d + K_D \dot{\underline{\tilde{x}}} + K_P \underline{\tilde{x}} - M_d \dot{J}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}})$$

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

to ~~exactly~~ linearize the
controlled system dynamics

contact forces exerted by
the end-effector
on the environment

$$B(\underline{q})\ddot{\underline{q}} + \underline{n}(\underline{q}, \dot{\underline{q}}) = \underline{u}$$

$$\underline{n}(\underline{q}, \dot{\underline{q}}) = C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q})$$

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$$\ddot{\underline{q}} = \underline{y} - B^{-1}(\underline{q})J^T(\underline{q})\underline{h}_e$$

Interaction Control of Robot Manipulators: Impedance Control

MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

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to exactly linearize the
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$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}})$$

$$\underline{u} = B(\underline{q})\underline{y} + \underline{n}(\underline{q}, \dot{\underline{q}}) + J^T(\underline{q})\underline{h}_e$$

$$\ddot{\underline{q}} = \underline{y}$$

Interaction Control of Robot Manipulators: Impedance Control

Before (no measure of the generalized contact forces)

$$\mathbf{M}_d \ddot{\underline{\tilde{x}}} + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} = \mathbf{M}_d \mathbf{B}_A^{-1}(\underline{q}) \underline{h}_A$$

Now (with measure of the generalized contact forces)

$$\mathbf{M}_d \ddot{\underline{\tilde{x}}} + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} = \underline{h}_A$$

second-order
differential
kinematics
equation

$$\ddot{\underline{x}}_e = \mathbf{J}_A(\underline{q}) \ddot{\underline{q}} + \dot{\mathbf{J}}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

$$\mathbf{T}_A^T(\underline{x}_e) \underline{h}_e = \underline{h}_A$$

$$\underline{y} = \mathbf{J}_A^{-1}(\underline{q}) \mathbf{M}_d^{-1} (\mathbf{M}_d \ddot{\underline{x}}_d + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} - \mathbf{M}_d \dot{\mathbf{J}}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} - \underline{h}_A)$$

Interaction Control of Robot Manipulators: Impedance Control

Before (no measure of the generalized contact forces)

$$\mathbf{M}_d \ddot{\underline{\tilde{x}}} + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} = \mathbf{M}_d \mathbf{B}_A^{-1}(\underline{q}) \underline{h}_A$$

Now (with measure of the generalized contact forces)

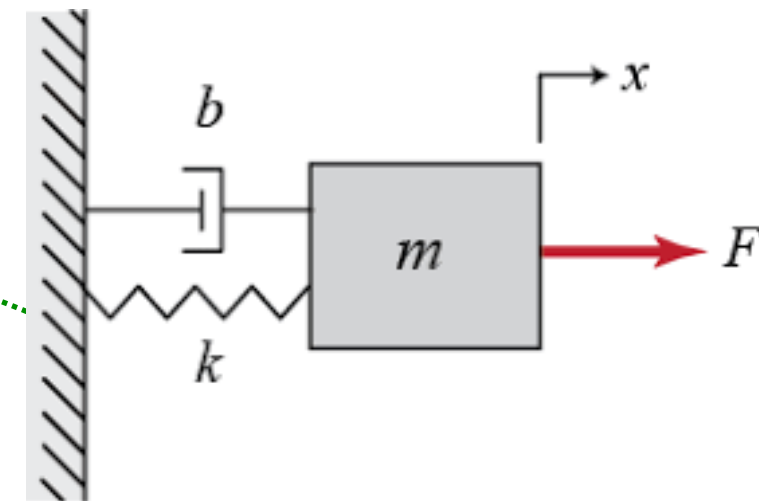
$$\mathbf{M}_d \ddot{\underline{\tilde{x}}} + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} = \underline{h}_A$$

$$\ddot{\underline{q}} = \underline{y}$$

second-order
differential
kinematics
equation

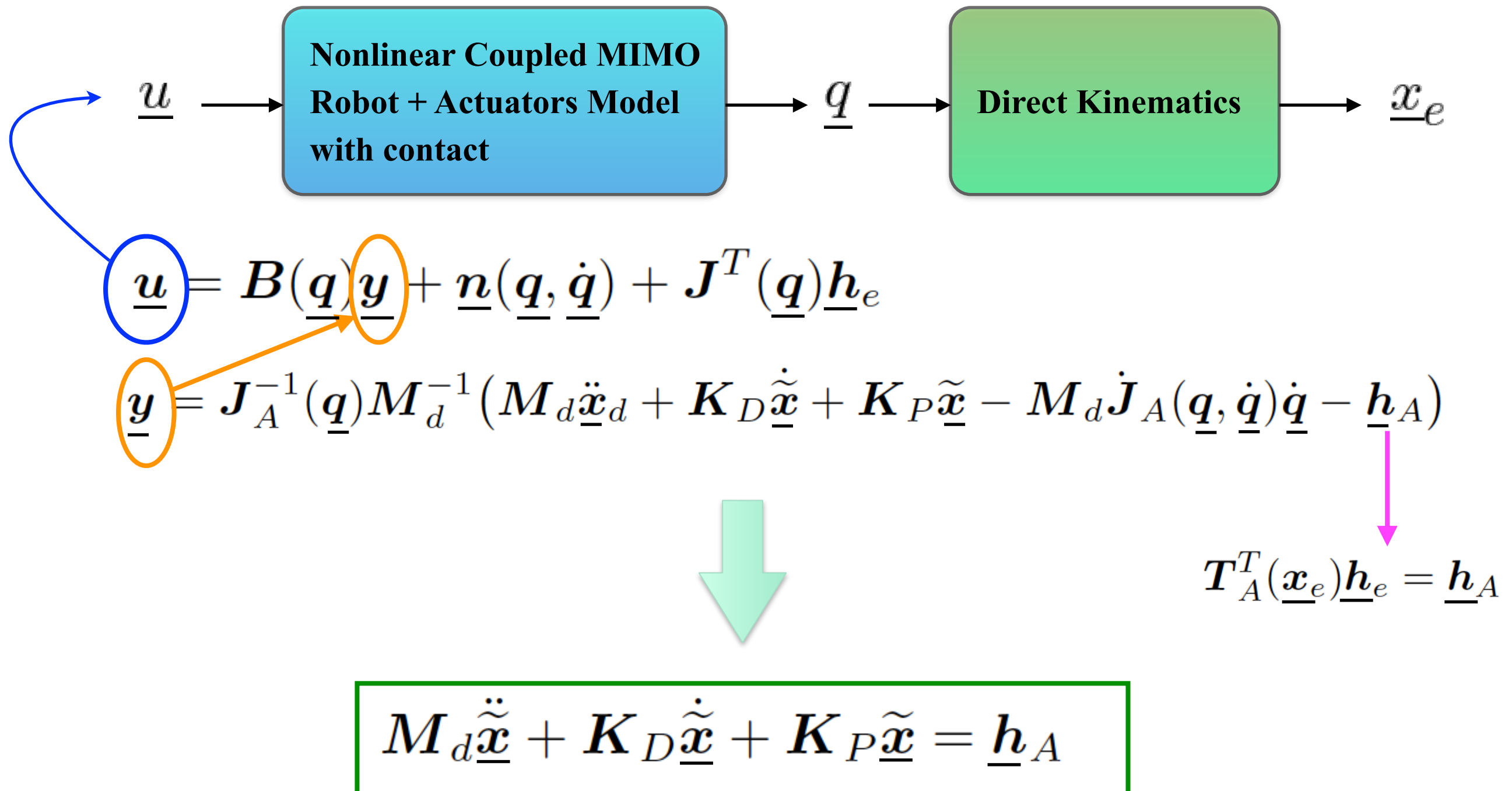
$$\ddot{\underline{x}}_e = \mathbf{J}_A(\underline{q}) \ddot{\underline{q}} + \dot{\mathbf{J}}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}}$$

$$\underline{y} = \mathbf{J}_A^{-1}(\underline{q}) \mathbf{M}_d^{-1} (\mathbf{M}_d \ddot{\underline{x}}_d + \mathbf{K}_D \dot{\underline{\tilde{x}}} + \mathbf{K}_P \underline{\tilde{x}} - \mathbf{M}_d \dot{\mathbf{J}}_A(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} - \underline{h}_A)$$

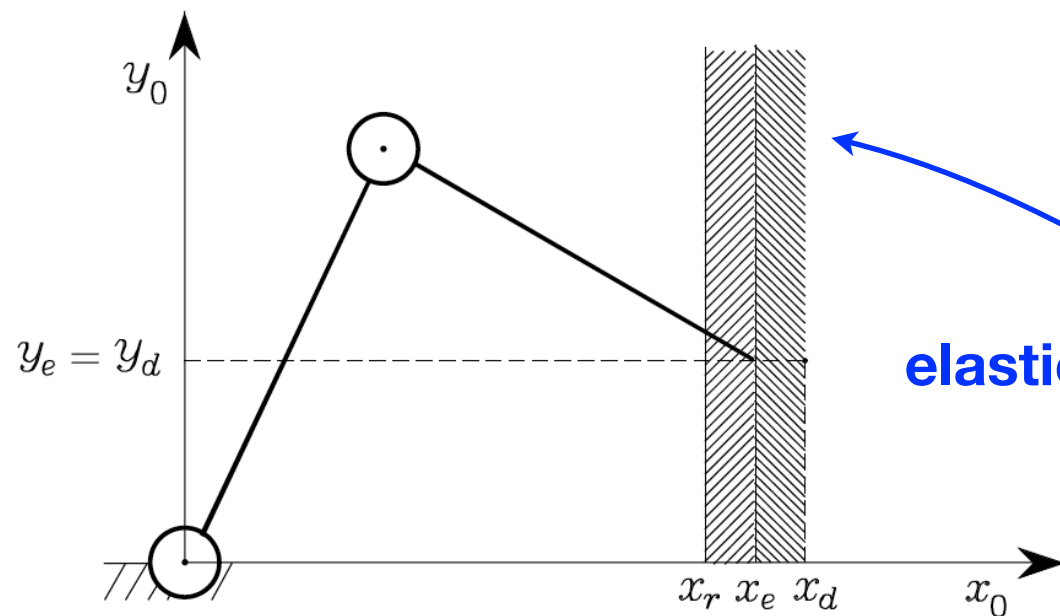


$$\mathbf{T}_A^T(\underline{x}_e) \underline{h}_e = \underline{h}_A$$

Interaction Control of Robot Manipulators: Impedance Control



Interaction Control of Robot Manipulators: Impedance Control



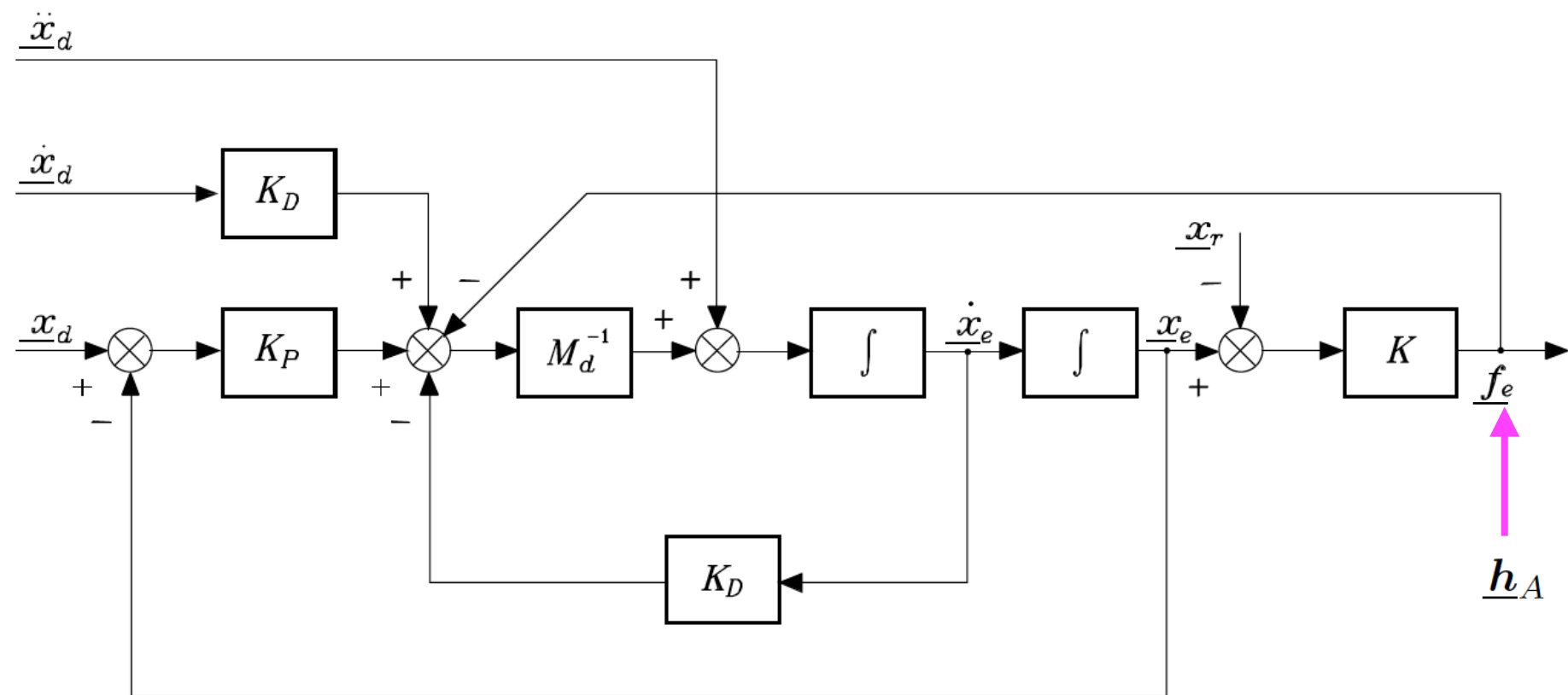
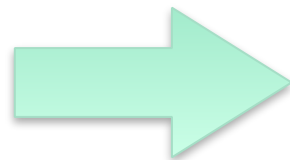
Source: Siciliano et al., Robotics, Springer:

Model of the environment
(with the assumption that only
forces arise at the contact)

$$\underline{f}_e = \underline{K}(\underline{x}_e - \underline{x}_r)$$

elastically compliant plane

$$\underline{M}_d \ddot{\underline{x}} + \underline{K}_D \dot{\underline{x}} + \underline{K}_P \underline{x} = \underline{h}_A$$





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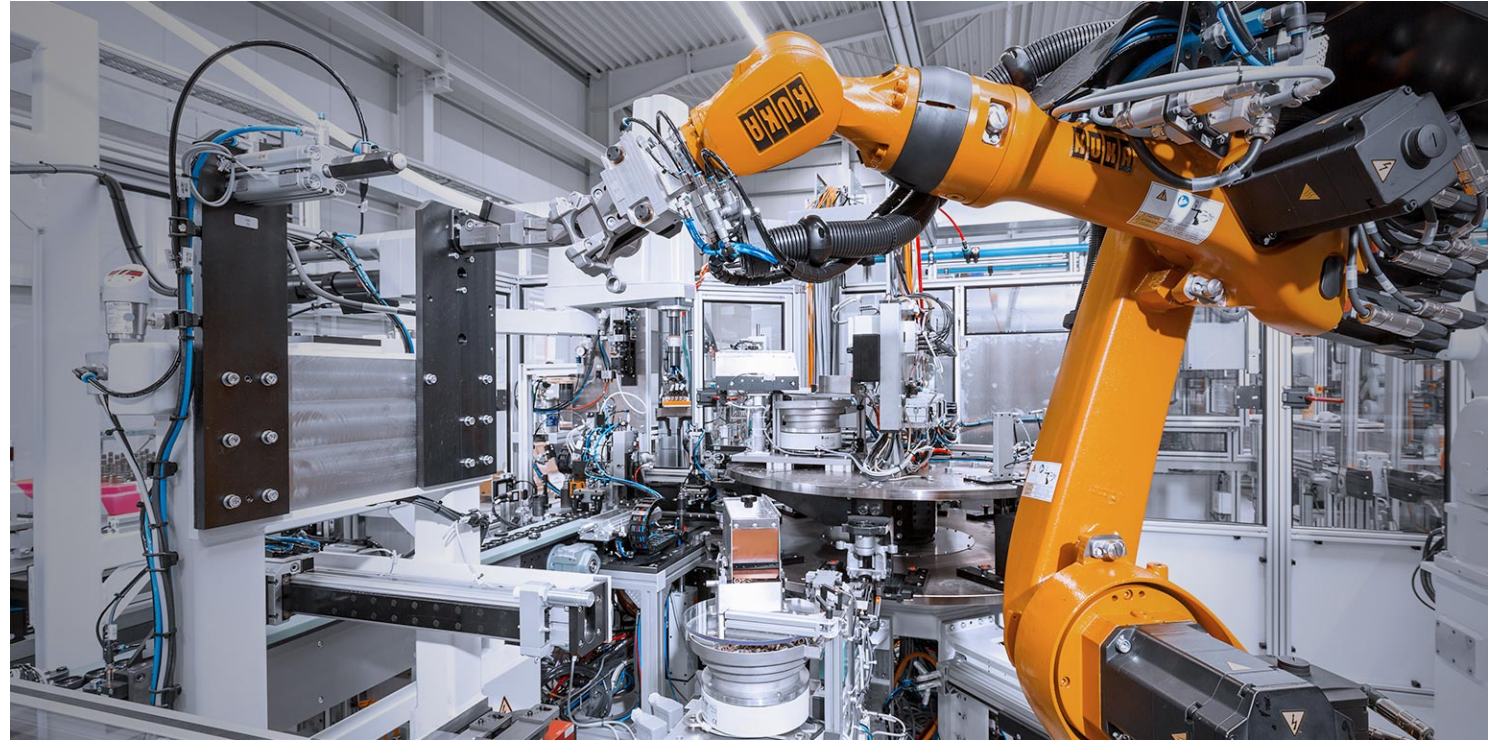
Course of Robot Control

Prof. Antonella Ferrara



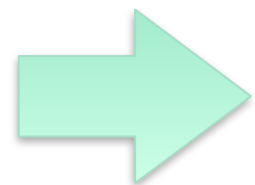
Force control

Interaction Control of Robot Manipulators



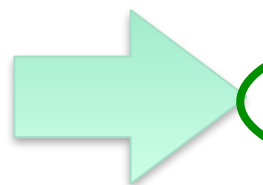
Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control: Indirect Approach



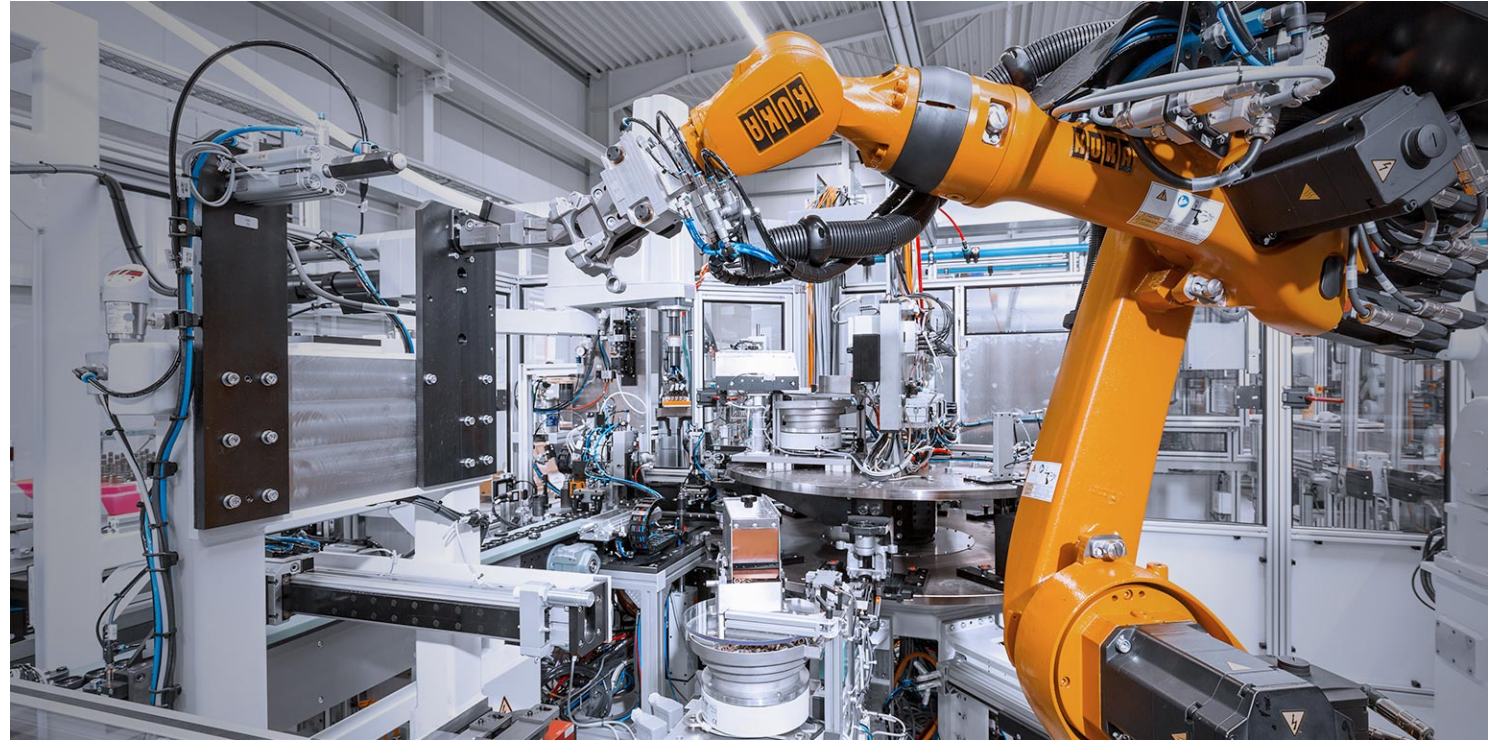
Impedance control

Interaction Control: Direct Approach



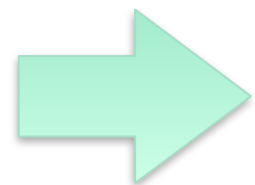
Force control

Interaction Control of Robot Manipulators



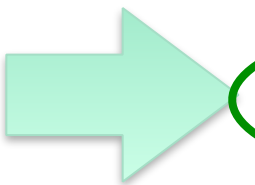
Source: <https://www.aumann.com/en/services/machines/assembly-and-handling-technology/robot-cells/>

Interaction Control: Indirect Approach



Impedance control

Interaction Control: Direct Approach



Force control

**Control in the
Operational Space**

Interaction Control of Robot Manipulators: Force Control

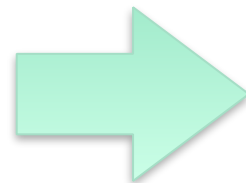
MIMO Dynamic Model of Robot Manipulator and Actuators in case of contact with the environment

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F\dot{\underline{q}} + \underline{g}(\underline{q}) = \underline{u} - J^T(\underline{q})\underline{h}_e$$

contact forces exerted by
the end-effector
on the environment

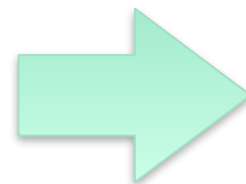
Typical structure of a force control scheme:

Inner loop



Inverse dynamics motion control

Outer loop



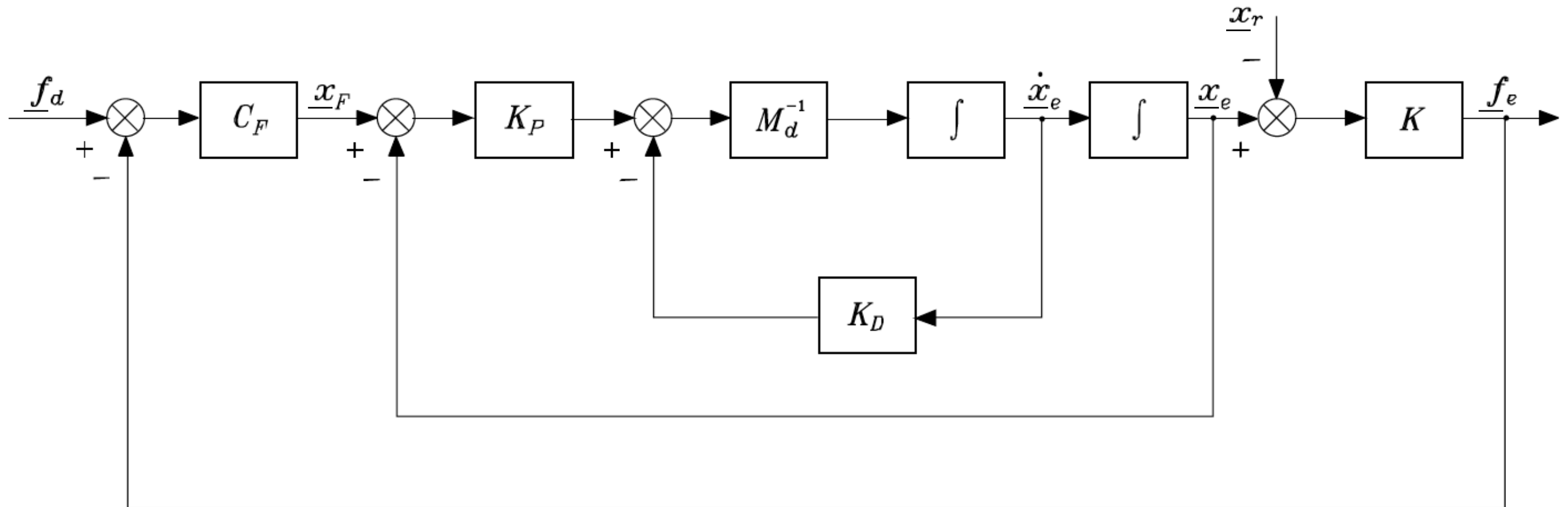
Force regulation feedback loop

Typical model of the environment: $\underline{f}_e = K(\underline{x}_e - \underline{x}_r)$

(with the assumption that only forces arise at the contact)

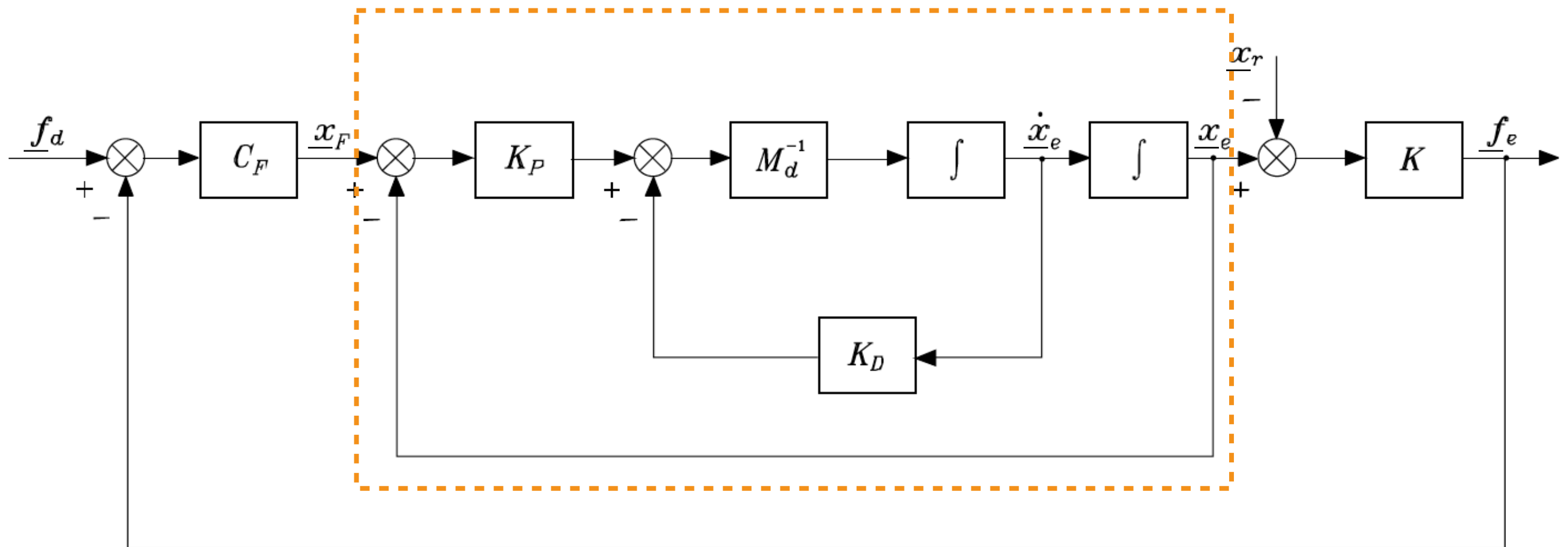
Interaction Control of Robot Manipulators: Force Control

Force control with inner position loop



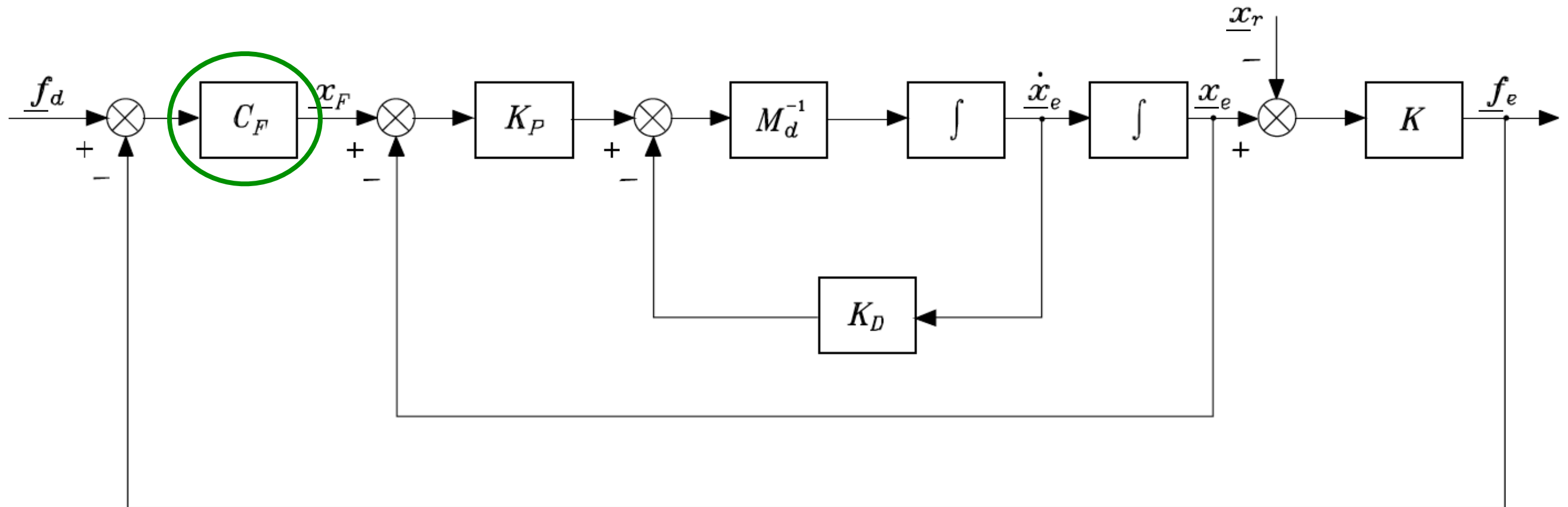
Interaction Control of Robot Manipulators: Force Control

Force control with inner position loop



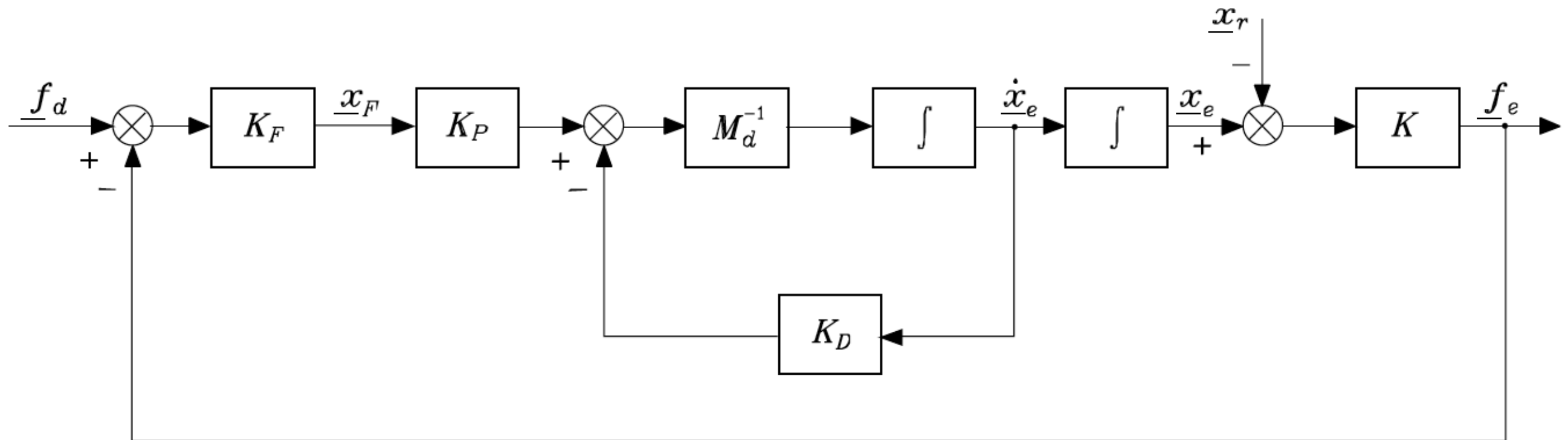
Interaction Control of Robot Manipulators: Force Control

Force control with inner position loop



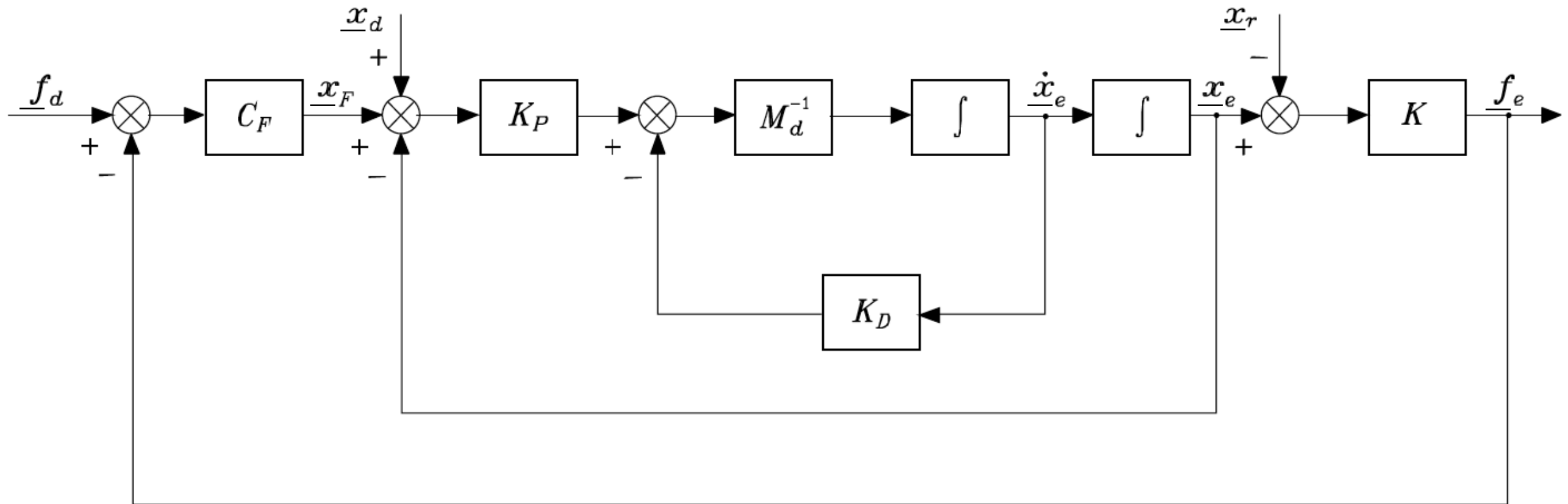
Interaction Control of Robot Manipulators: Force Control

Force control with inner velocity loop



Interaction Control of Robot Manipulators: Force Control

Parallel Force/Position Control



End-effector equilibrium position: $\underline{x}_e = \underline{x}_d + C_F (K(\underline{x}_r - \underline{x}_e) + \underline{f}_d)$

Interaction Control Schemes

Supplementary Material

A. Ferrara, G.P. Incremona, M. Cucuzzella, “**Advanced and Optimization Based Sliding Mode Control**”, SIAM, 2019

Ferrara A., Lombardi C., Interaction Control of Robotic Manipulators Via Second Order Sliding Modes, **International Journal of Adaptive Control and Signal Processing**, Vol. 21, Pages 708-730, No. 8-9, 2007.

Capisani L., Ferrara A., Trajectory Planning and Second Order Sliding Mode Motion/Interaction Control for Robot Manipulators in Unknown Environments, **IEEE Transactions on Industrial Electronics**, (Regular), Year: 2012, Volume: 59, Issue: 8, Pages: 3189 - 3198

Gian Paolo Incremona, Gianluca De Felici, Antonella Ferrara and Ezio Bassi, A Supervisory Sliding Mode Control Approach for Cooperative Robotic Systems of Systems, **IEEE Systems Journal**, vol. 9, no. 1, pp. 263-272, March 2015

Bassi E., Benzi F., Capisani L. M., Cuppone D., Ferrara A., “Characterization of the Dynamical Model of a Force Sensor for Robot Manipulators”, Robot Motion and Control 2009, Krzysztof Kozłowski ed., (LNCiS) Lecture Notes in Control and Information Sciences n. 396, chapter 22, pp 243-253, Springer Verlag Berlin Heidelberg, 2009.

Bassi E., Benzi F., Capisani L., Cuppone D., Ferrara A., Hybrid Position/Force Sliding Mode Control of a Class of Robotic Manipulators, Proc. 48th IEEE Conference on Decision and Control and 28th Chinese Control Conference, December 2009, Shanghai, P.R. China.

